

UC Berkeley

UC Berkeley Electronic Theses and Dissertations

Title

Automatic Grading and Feedback for Students' Short Written Responses

Permalink

<https://escholarship.org/uc/item/4f54k5tf>

Author

Condor, Aubrey

Publication Date

2024

Peer reviewed|Thesis/dissertation

Automatic Grading and Feedback for Students' Short Written Responses

By

Aubrey Condor

A dissertation submitted in partial satisfaction of the

requirements for the degree of

Doctor of Philosophy

in

Education

in the

Graduate Division

of the

University of California, Berkeley

Committee in charge:

Professor Zachary Pardos, Chair

Professor Mark Wilson

Professor Marcia Linn

Spring 2024

Abstract

Automatic Grading and Feedback for Students' Short Written Responses

by

Aubrey Condor

Doctor of Philosophy in Education

University of California, Berkeley

Professor Zachary Pardos, Chair

This work focuses on automatic grading and revision feedback for open-ended (OE) student responses relating to mathematics and science topics. The dissertation will be divided into three closely related chapters, where each build upon the work of the previous chapter(s). In the first chapter, we focus on improving the performance of an Automatic Short Answer Grading (ASAG) model, where performance is measured by how closely the machine ratings match those of a human rater. We experiment with incorporating domain-related text into the model with sources such as the item text, context, and rubric, and investigate how this might affect the generalizability of the model by testing the ASAG model on out-of-training-sample questions. Further, we examine if representing not only the text, but also the ordinal structure of a scoring rubric is useful by using a graph sampling method prior to incorporating the text into the ASAG model.

In the second chapter, we address the lack of explainability in typical ASAG models. We train a deep reinforcement learning (RL) agent to revise a student's response by inserting key phrases, or deleting portions of the response, to achieve a high grade from an ASAG model in the least number of revisions. The intent is that the agent's revisions may provide some explanation as to what was missing or unnecessarily included in a student's response. We also examine the intelligibility of a Neural Additive Model (NAM) using the RL agent's key phrases as input features. NAMs provide the explainability of additive models with the ability to approximate high dimensional functions and inputs like a neural network. Further, we use the RL agent to expose shortcomings in the ASAG model by finding revisions that achieve a high grade from the automatic grader but may not be considered good responses by a human rater.

In the third chapter, we focus on whether the RL agent's revisions can provide feedback to encourage a student to revise their own short response. We collected data with a randomized controlled design from students interacting with an online tutoring system. Students who receive an intervention from the RL agent's revisions will be compared with both students who receive an intervention from a popular open-source chatbot, ChatGPT, and students who received a generic intervention. We examine student perceptions of the feedback, how prior knowledge interacts with the interventions, and the accuracy of ChatGPT's feedback.

Table of Contents

List of Tables

List of Figures

Chapter 1

- 1.1 Introduction
- 1.2 Related Work
 - 1.2.1 Automatic Short Answer Grading
 - 1.2.2 Natural Language Processing for Education
- 1.3 Chapter 1, Study 1: Using Domain-related Text for Generalizability and Model Performance
 - 1.3.1 The Dataset
 - 1.3.2 Methods
 - 1.3.2.1 Input Representations
 - 1.3.2.1.1 Sentence-BERT
 - 1.3.2.1.2 Word2Vec
 - 1.3.2.1.3 Bag of Words
 - 1.3.2.2 Input Content
 - 1.3.2.2.1 Question Text
 - 1.3.2.2.2 Question Context
 - 1.3.2.2.3 Rubric Text
 - 1.3.2.3 Classification Models
 - 1.3.2.4 Evaluation and Model Comparison
 - 1.3.2.4.1 Experiments
 - 1.3.2.4.2 Leave-one-question-out Evaluation
 - 1.3.2.4.3 Evaluation Metrics
 - 1.3.2.4.4 Empirical Monte Carlo Confidence Intervals
 - 1.3.3 Results
 - 1.3.4 Discussion
 - 1.3.4.1 Next Steps
- 1.4 Chapter 1, Study 2: Utilizing a Scoring Rubric to Increase Model Performance
 - 1.4.1 Background
 - 1.4.1.1 BERT
 - 1.4.1.2 Node2Vec
 - 1.4.2 Methods
 - 1.4.2.1 The Dataset
 - 1.4.2.2 Architecture Modification
 - 1.4.2.4 Fine-tuning Experiments
 - 1.4.3 Results
 - 1.4.4 Discussion

Chapter 2

- 2.1 Introduction
- 2.2 Related Work
 - 2.2.1 Explainable AI and ASAG
 - 2.2.2 Reinforcement Learning for Education

2.2.3	Applications of Neural Additive Models	
2.3	The Dataset	
2.4	Chapter 2, Study 1: Neural Additive Models for Explainable ASAG	
2.4.1	Background	
2.4.1.1	Neural Additive Models	
2.4.1.2	Logistic Regression	
2.4.1.3	DeBERTa'	
2.4.2	Methods	
2.4.2.1	Feature Engineering	
2.4.2.2	Evaluation	
2.4.3	Results	
2.4.4	Discussion	
2.5	Chapter 2, Study 2: Training an RL Agent to Revise a Short Response	
2.5.1	Background	
2.5.1.1	Deep Reinforcement Learning	
2.5.1.1.1	Offline (batch) Learning	
2.5.1.1.2	Proximal Policy Optimization	
2.5.1.2	The ASAG Model	
2.5.2	The RL Formulation	
2.5.2.1	The Algorithm	
2.5.2.2	The State	
2.5.2.3	The Action Space	
2.5.2.4	The Policy	
2.5.2.5	The Reward	
2.6	Chapter 2, Study 3: Auditing an ASAG model with deep RL	
2.6.1	The Experiments	
2.6.2	Results	
2.6.2.1	Experiment 1: Helpful Phrases	
2.6.2.2	Experiment 2: Unhelpful Phrases	
2.6.2.3	Experiment 3: All Phrases	
2.6.3	Discussion	
Chapter 3		
3.1	Abstract	
3.2	Introduction	
3.3	Related Work	
3.3.1	Actionable Automated Feedback	
3.3.2	AI-assisted, and Automated Feedback	
3.3.3	Prior Knowledge and Feedback	
3.3.4	ChatGPT for Education	
3.4	Feedback Design	
3.5	Methods	
3.5.1	ChatGPT	
3.5.2	The Data Set and Student Model	
3.5.3	ChatGPT Prompt Design	
3.6	Evaluation of the RL Agent and Machine Revisions	
3.6.1	Quantitative Evaluation	

3.6.1.1	Quantitative Evaluation Results	
3.6.2	Qualitative Evaluation	
3.6.2.1	Qualitative Evaluation Results	
3.7	Data Collection: Processing and Results	
3.7.1	Data Processing	
3.7.2	Summary Statistics	
3.7.3	Comparing Treatment Groups for Revision Efficacy	
3.7.3.1	Assumptions of ANCOVA	
3.7.3.2	ANCOVA Results	
3.7.3.3	Heterogeneity of Treatment Effect	
3.7.4	Examining Student Perceptions	
3.7.4.1	Engaging in Revision	
3.7.4.2	Similarity Analysis of Machine and Student Revisions	
3.7.4.3	Analysis of Student Critiques	
3.7.5	Analysis of ChatGPT Machine Revisions	
3.8	Discussion and Limitations	
	References	
	Appendix: Screenshots of the Survey in OATutor	

List of Tables

Table

- 1.1 Experiment Results: Multi-label Accuracy
- 1.2 Experimental Results: 95% Monte Carlo Confidence Intervals
- 1.3 Test Set Weighted F1 Score and Cohen's Kappa
- 1.4 5x2 Cross Validation Paired One-tailed t-test
- 2.1 KI Soundwave Scoring Rubric Sample
- 2.2 5x2 Cross Validation Paired One-tailed t-test and Cohen's D Effect Sizes
- 2.3 5x2 Cross Validation Averages
- 2.4 Pseudocode for the RL Algorithm
- 3.1 The IMR Construct Map
- 3.2 Scoring Guide for the OE Item
- 3.3 Summary Statistics for Number of Revisions
- 3.4 Summary Statistics for Construct Level Increase
- 3.5 Coded Likert Scale Averages and Standard Deviations
- 3.6 Counts of Participants Partitioned by Treatment and Demographic Groups
- 3.7 Summary Statistics of OE and Revision Scores by Group
- 3.8 ANCOVA Results
- 3.9 Results of Tukey's HSD Test
- 3.10 OLS Fit Results for the Baseline Model
- 3.11 Summary Statistics of Pretest Scores
- 3.12 OLS Fit Results for Model 2 with Prior Knowledge and Group Interaction

List of Figures

Figure

- 1.1 An Example of an Item from the Data Set
- 1.2 An Example of a Scoring Rubric Corresponding to the Item in Figure 1.1
- 1.3 2D Visuals of Input Vector Representations.
- 1.4 The Modified BERT Architecture
- 1.5 Example of a Scoring Rubric and Corresponding Graph Structure
- 2.1 KI Soundwave Item
- 2.2 NAM Mean Feature Importance
- 2.3 NAM Shape Functions (with highest and lowest rating categories)
- 2.4 NAM Shape Functions (with all rating categories)
- 2.5 A Comparison of the Smoothed Mean Episode Returns
- 2.6 Kernel Density Estimated Distribution Plots for each Experiment
- 3.1 The Four Building Blocks in the BEAR Assessment System
- 3.2 The Context and Graph for the Pretest Questions
- 3.3 The Context and Graph for the OE Item
- 3.4 Smoothed (n=200) mean Episode Returns
- 3.5 Frequency of Likert Agreement Levels for Statement 1
- 3.6 Frequency of Likert Agreement Levels for Statement 2
- 3.6 Frequency of Likert Agreement Levels for Statement 3
- 3.7 Histogram of ANCOVA Residual Values
- 3.8 Box Plot of Expected Revision Score by Treatment Group
- 3.9 Box Plot of Residuals by Treatment Group
- 3.10 Box Plot of Expected OE Score by Treatment Group
- 3.11 Scatterplot and Linear Fit for Expected OE versus Revision Score: RL Group
- 3.12 Scatterplot and Linear Fit for Expected OE versus Revision Score: Control Group
- 3.13 Scatterplot and Linear Fit for Expected OE versus Revision Score: GPT Group
- 3.14 Linear Fit of Covariate to Dependent Variable by Treatment Group

Chapter 1

Automatic Short Answer Grading: Using domain-related text for generalizability and model performance.

1.1 Introduction

Automatic Short Answer Grading (ASAG) is an emerging field of research, as the education community has started to embrace the use of machine learning to assist both students and education professionals. ASAG refers to the machine scoring of open-ended (OE) items that are shorter than essay length - often a sentence or two. It has been shown that the use of open-ended questions helps facilitate learning by self-explanation (Chi et al., 1994), or information recall (Bertsch et al., 2007). The “generation effect” has been studied extensively and findings show that subjects who generate information are able to remember the information better than material they have simply read (Bertsch et al., 2007). Additionally, open-ended items may be adapted to assess student knowledge and understandings across many content areas and have been shown to enhance learning (Hancock, 1995). However, educators are often deterred from the use of OE items because grading requires much more time than that for multiple choice items (Hancock, 1995). The use of an automated system for grading short answer questions could decrease the amount of time teachers spend grading, while allowing students to grow in their knowledge through self-explanation and information recall.

In this chapter, we focus on improving the performance of an Automatic Short Answer Grading (ASAG) model, where performance is measured by how closely the machine ratings match those of a human rater. We experiment with incorporating domain-related text into the model with sources such as the item text, context, and rubric, and investigate how this might affect the generalizability of the model by testing the ASAG model on out-of-training-sample questions. Further, we examine if representing not only the text, but also the ordinal structure of a scoring rubric is useful by using a graph sampling method prior to incorporating the text into the ASAG model.

1.2 Related Work

In this section, we give a brief background of research relating to ASAG and more broadly, applications of Natural Language Processing for educational purposes.

1.2.1 Automatic Short Answer Grading (ASAG)

A systematic review of trends in ASAG (Burrows, et al., 2015) illustrates an increasing interest in the field of automatic grading for education. Unsupervised methods have been explored such as concept mapping, semantic similarity, and clustering to assign ratings. For example, Mohler and Mihalcea (2009) compared knowledge based and corpus based semantic similarity measures for automatic grading, Klein et al. (2011) implemented a latent semantic analysis approach, and Basu et al. (2013) used clustering to provide rich feedback to groups of similar responses. In addition, many types of supervised classification methods have been utilized for ASAG. Notable examples include Hou and Tsao (2011) who incorporated POS tags and term frequency with a Support Vector Machine classifier, and Madnani et al. (2013) who made use of simple features such as a count of commonly used words and length of response with a logistic regression classifier.

More recent ASAG research exploits deep learning methods. Haller et al. (2022) provide a comprehensive analysis of recently published methods which deploy deep learning approaches for ASAG. They emphasize that the progression from engineered features of text to representation learning methods, including the use of word embeddings, sequential models and attention-based methods, has contributed to recent improvements in ASAG models. Notable work includes Zhang et al. (2016) who used a combination of feature engineering and deep belief networks, Liu et al. (2019) who employed multi-way attention networks, and Yang et al. (2018) who considered a deep autoencoder model specific to Chinese responses. Additionally, Qi et al. (2019) created a hierarchical word-sentence model with a CNN and Bi-LSTM model and Tan et al. (2020) explored the use of a graph convolutional network (GCN) to encode a graph of all student responses. Tulu et al. (Tulu et al., 2021) proposed a new approach to ASAG using a MaLSTM (a LSTM based method for creating sentence vectors).

Further, much of the newest ASAG work makes use of state-of-the-art transformer based models, including Gaddipati et al. (2020) who evaluated four different types of response embeddings, ELMo, GPT, BERT, and GPT-2 for their performance on an ASAG task, and Camus and Filighera (2020) who compared the performance of transformer models for ASAG in terms of the size of the transformer and the ability to generalize to other languages. Agarwal et al. (2022) introduced the Multi-Relational Graph Transformer (MitiGaTe) to prepare token representations that consider the structural context of a sentence and Zhu et al. (2022) propose a novel BERT-based, NN framework for ASAG by adjusting BERT embeddings with a semantic refinement layer consisting of a bidirectional-Long Short-Term Memory (LSTM) network and a Capsule network with position information.

More specifically related to our project, researchers have experimented with using extra, question-relevant information within ASAG models. Sung et al. (2019; 2019) examined the effectiveness of pre-training BERT on relevant domain texts, as a function of the size of training data and generalizability across domains, and found that the pre-trained BERT model can be improved by augmenting data on domain-specific texts. Wang et al. (2021) used an attention-based method to extract key phrases from student answers that are highly related to phrases and answers from a scoring rubric. They find that the performance of both grading and justification identification is improved through their method. Chen and Li (2021) make use of prior knowledge to enrich features of student responses by incorporating question text information within an extra forward propagation training step. They show that by incorporating this question-specific information, their model beats current state-of-the-art performance.

1.2.2 Natural Language Processing for Education

Literature addressing the general application of natural language processing (NLP) for uses in the field of education has grown in recent years. Shaik et al. (2022) provide a review of the trends and challenges of using existing NLP methodologies for educational domain applications like sentiment annotations, entity annotations, text summarization and topic modeling, and Ahadi et al. (2022) analyze the metadata of publications using text mining or NLP in education settings and report on key themes as well as current state-of-the-art methods. More specific applications include Fonseca et al. (2020) who used NLP to automatically classify the programming assignments for students within academic context, and Thaker et al. (2020) who incorporated textual similarity techniques to recommend remedial readings to students. Further, Arthurs and Alvero (2020) examined bias in word representations for college admissions essays, and Xiao et al. (2020) employed NLP and transfer learning methods for problem detection in

peer assessments. Finally, Dowell and Kovanovic (2022) describe the current landscape of NLP tools and approaches to discern patterns in conversations that take place across educational technology platforms.

1.3 Chapter 1, Study 1: Using Domain-related Text for Generalizability and Model Performance

This section contributes to the field of automatic grading in three related ways. We focus on classification performance and generalizability of the supervised grading model in terms of 1) the textual representation type, 2) the content of the input and 3) the classification model.

A key challenge with supervised automatic grading models, where supervised learning refers to a sub-category of machine learning defined by the necessity of labeled data sets to train a model, is gathering a large enough sample of labeled data for training. While some labeling tasks for supervised learning may be straightforward such as identifying an image as either a dog or a cat, others like rating student responses require careful consideration. In high-stakes assessment scenarios, two or more ratings by different experts are often necessary to form a reliable consensus rating. Thus, obtaining labeled data to train an ASAG system can be arduous. It follows that quickly introducing new questions to an existing system may not be feasible if a data collection as well as the meticulous rating of new responses is necessary. Further, a new model would have to be trained and tuned with the newly collected responses. If we create more generalizable ASAG models, educators may have the flexibility to add new questions to an existing assessment with very little effort, thus increasing the practical use of the ASAG system.

We hypothesize that the inclusion of extra question related information within the model input may improve both the classification performance, and the generalizability of the model. For the purposes of this project, we formally define the generalizability of an ASAG model as the capacity to classify responses from out-of-training-sample questions, and the classification performance represents alignment with human scores.

We compare three different representation types, including those of state-of-the-art models: Sentence-BERT (SBERT), Word2Vec (W2V), and Bag of Words (BOW). In terms of input content, we experiment with including previously untapped resources relating to the questions in the model input. Such resources include a question-bundle identifier, the question stem text, question context text, and rubric information. Extra input content is vectorized (if the source is textual) and concatenated to the response vectors to be used as input to the classification model. Finally, we compare a non-neural model (a multinomial logistic regression) and a simple neural model (a three layer feed-forward network).

In order to examine the generalizability of the model for each experiment, we use a leave-one-question-out evaluation procedure where we train the model on N-1 questions and use the one left-out question data as our test set. We repeat this method N=31 times for each question, and reported accuracies are the average of the N experiments. Additionally, we compare results of our experiments against a typical random, student level N=31-fold cross validation. We randomly partition the data into 31 splits, and use one as the test set, and the rest of the data is used for training. This is repeated N=31 times such that each Nth partition is used as the test set once, and reported results are the average of the N holdout sets. Results from a majority class classifier are included as well for a baseline comparison.

Finally, we use each of the neural network leave-one-question-out experimental results (accuracies) as an observed distribution on which to employ Monte Carlo sampling and compute empirical confidence intervals of the sampling means. To compare results over the different

vectorization methods and different variations of included question-related information, we investigate if the computed empirical confidence intervals of the sampling mean accuracies overlap between the different experiments. As detailed in the results, highly overlapping confidence intervals suggest that we do not have strong evidence to support the hypothesis that adding extra question-related information helps the auto-grader achieve higher accuracies when grading out-of-sample questions.

The novel contribution of this section includes both the focus on generalizability of the model to out-of-training-sample questions, as well as the leveraging of previously untapped, question related information as input to the model.

1.3.1 Data Set

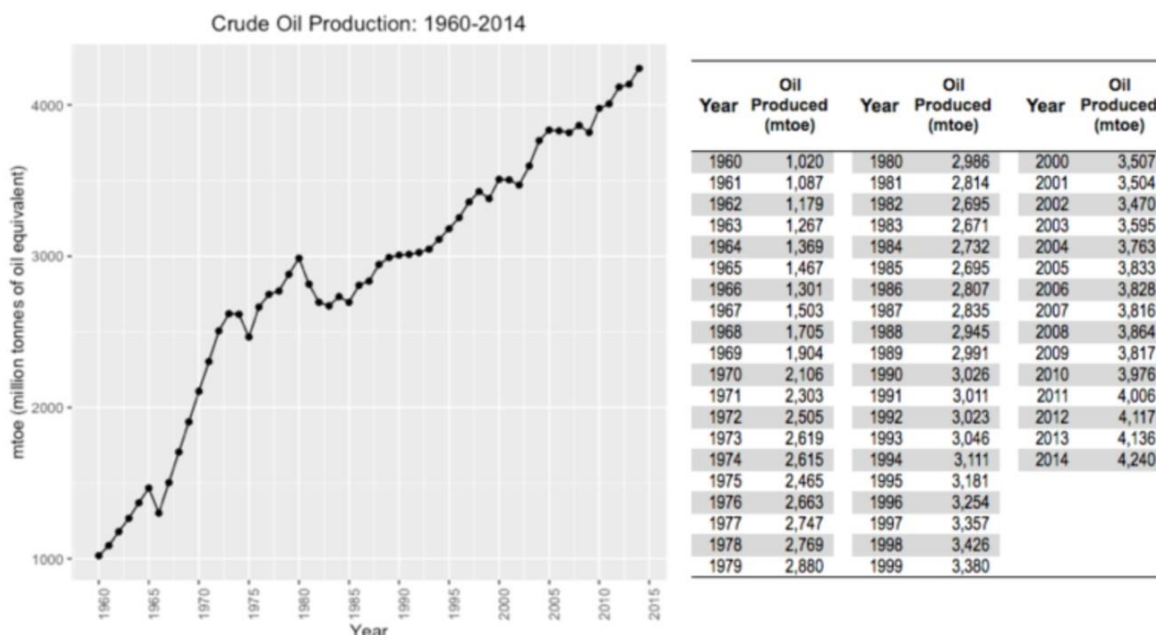
The data we used for this project was sourced from a 2019 field test of a Critical Reasoning for College Readiness (CR4CR) assessment (Arneson et al., 2019) created at the Berkeley Evaluation and Assessment Research (BEAR) center. The data consists of 5,550 student responses from 558 distinct students to 31 different items. The field test included other items that were multiple choice, but these questions were filtered out of the data for our use. The mean number of responses per question is 179 with the minimum being 128 and the maximum being 313. Most of the items belong to an item bundle - a grouping of items that are related in context and/or share a common question context. Additionally, the items were administered in four different test forms, where some items were included in multiple forms. The items all relate to four constructs about student understanding of algebra.

An example of one of the items, labeled 'Crude Oil 4ab' is included in Figure 1.1. For this item, students are presented with two images relating to oil production - one being a line graph and the other, a table. With the given context, students are presented with a choice between the graph or the table for which would be better to represent the historical patterns, or change over time of the oil production. The correct answer for this question is the graph, and students are expected to provide reasons why this is the right choice.

Figure 1.1

An Example of an Item from the Data Set: Crude Oil 4ab

The following graph and table show annual crude oil production in million tonnes of oil equivalent (mtoe) from 1960 to 2014.



[a] For a group project, you and your classmates have to present the overall historical patterns of annual oil production as a poster. Due to limited space, only one of the following representations can be included in the poster: a table, a graph, or a set of equations. Which representation should you use?

[b] Briefly explain your answer choice in [a].

An example of a student response to the Crude Oil 4ab question (shown in Figure 1) rated at the highest (most correct) score category is shown:

"The graph easily displays patterns over time whereas the table and equations require more analyzing."

In contrast, a student's response to the same question rated at the lowest (most incorrect) category is shown:

"The table is more clear, the information is seen in the table."

Responses were rated from 0 (fully incorrect) to 4 (fully correct) by multiple researchers and subject-matter experts at the BEAR center. The quality and consistency of ratings were evaluated by an inter-rater reliability score, and when a high percentage of rating mismatches between raters existed, incongruous ratings were discussed until a consensus was reached by the raters.

1.3.2 Methods

In this section, we briefly introduce the representation methods and model classes that we include in our experiments. Additionally, we describe the question related information, beyond that of the responses, that are used as inputs to the model. Further, we outline our experiments in detail including methods for evaluation and comparison.

1.3.2.1 Input Representations

We chose to include three distinct, yet commonly used in the field of natural language processing (NLP), representation types in our experiments: A count-based method that elicits the distinct vocabulary of our data (Bag of Words), a simple neural method that utilizes pre-trained word vectors (Word2Vec), and a state-of-the-art, contextual neural method (Sentence-BERT). A short description of each representation type is included.

1.3.2.1.1 Sentence-BERT

Sentence-BERT (SBERT) is a modification of the BERT network that utilizes siamese and triplet network structures to create semantically meaningful sentence embeddings (Reimers & Gurevych, 2019). SBERT re-tunes the BERT network on a combination of the SNLI dataset and the Multi-Genre NLI datasets, totaling about one million sentence pairs. Although sentence embeddings can be derived from the original BERT model using methods such as averaging the BERT output layers or using the “[CLS]” token embedding, it has been shown that such methods yield poor sentence embeddings (Pennington et al., 2014). In comparison, SBERT sentence embeddings outperformed other state-of-the-art methods such as InferSent (Conneau et al., 2017) and Universal Sentence Encoder (Cer et al., 2018) on the SentEval (Conneau et al., 2017) benchmark, which gives an idea of the quality of sentence embeddings for various task.

1.3.2.1.2 Word2Vec

Word2Vec (W2V) is a neural model that creates vector representations of words that have been shown to be semantically meaningful and useful in different NLP tasks (Mikolov et al., 2013). We use an extension of the previously introduced Skip-gram model (Mikolov et al., 2013) that incorporates sub-sampling of frequent words during training in order to speed up training, and improves accuracy of representations of less frequent words. For this project, we use the Google News corpus of pre-trained word embeddings. Vectors of size 300 are created for each word, and in order to construct response embeddings from the individual word vectors, we employ a simple yet popular method: averaging the vectors of all words in the response.

1.3.2.1.3 Bag of Words

The bag-of-words (BOW) model (Harris, 1954) represents a document as a vector, or “bag,” of length equal to the number of unique words in the entire corpus and values of the vector equal to the frequency with which its corresponding word occurred in the document. In our application, the bags are student short answers and additional question information.

1.3.2.2 Input Content

In an item design context, there are various, typically untapped sources of information relating to a particular question that may be useful to include as input to a classification model. We explore the use of four different sources of information, outside that of the response itself. A brief description of such sources are included below.

1.3.2.2.1 Question Text

The question text consists of the direct question stem. As in the example item provided in Figure 1.1, the question text would be: *“Briefly explain your answer choice in [a].”*

1.3.2.2.2 Question Context

The question context includes any textual information beyond that of the question stem that is related to the question, and might be useful for the respondent to produce a response. In the question example in Figure 2, the question context would include:

"The following graph and table show annual crude oil production in million tonnes of oil equivalent (mtoe) from 1960 to 2014. For a group project, you and your classmates have to present the overall historical patterns of annual oil production as a poster. Due to limited space, only one of the following representations can be included in the poster: a table, a graph, or a set of equations. Which representation should you use?"

We note that not all of the included items have question context text beyond that of the question text itself. For such items, it was not possible to include question context as part of the input.

1.3.2.2.3 Rubric Text

An important step in a measurement process is defining the outcome space for each item through a scoring guide, which is essentially a rubric. The scoring guide includes a detailed description of the reasons for which a response would be rated at a certain score and is used as a guide for human raters. In addition, the scoring guide often includes example responses for each rating level. An example of a scoring guide for the 'Crude Oil 4ab' item in Figure 1.1 is included shown in Figure 1.2. When the rubric text is included in the input text, we include both the level descriptions as well as the example response(s).

Figure 1.2

An Example of a Scoring Rubric Corresponding to the Item in Figure 1.1

Level	Description	Example Response
4	Student provides a fully correct positive and negative justification for ...	"The graph best illustrates the visual trend of oil production. The table or the equation won't provide an overall ..."
3	Student provides a fully correct positive justification for selected representation ...	"The table cannot show a trend in the oil production effectively. "
2	Student provides a partial/general positive justification for selected ...	"The graph helped me with more questions."
1	Student provides incorrect justification for selected representation.	"Because the graph gives us better projections."
0	Student makes no attempt to provide a justification for selected representation.	"I am not sure why."

1.3.2.2.4 Bundle Identifier

As described above in the data section, most of the questions belong to a bundle of questions - those that are linked based on a similar image, or context text. As a question bundle identifier, we concatenate a one-hot vector to the input vectors. Although items within the same bundle will often share the same context text, we include the one-hot bundle identifier

within our extra input text experiments so that we can infer whether the model makes use of semantics within the context text, or rather just a general indication of similar questions.

1.3.2.3 Classification Models

We compare a multinomial logistic regression model with a simple neural network classification model. We chose these classification methods representing a linear transformation of the feature space to a label (regression) and a non-linear transformation (neural network). A brief overview of each model is included.

Multinomial logistic regression (MLR) is a classification model that predicts probabilities of different outcomes for a categorical dependent variable. In order to generalize to a K-class setting, the model runs K-1 independent binary logistic regression models where one outcome is chosen as a “pivot” and other K-1 outcomes are separately regressed against the pivot outcome. We use the Limited-memory Broyden-Fletcher-Goldfarb-Shannon (LBFGS) algorithm for optimization (Fletcher, 2000), and incorporate L2 regularization.

Additionally, we use a simple feed forward neural network on a categorical cross entropy loss function with 2 hidden layers of size 100, using rectified linear unit (ReLU) activation functions for both hidden layers. We include a dropout of 0.4 and utilize Adam optimization. We train the model for 16 epochs and use a batch size of 36.

1.3.2.4 Evaluation and Model Comparison

In this section, we enumerate our experiments and the evaluation methods chosen for comparison.

1.3.2.4.1 Experiments

As input to our model, we experiment with eight different combinations of content to vectorize and concatenate to the response vectors before training our classification models:

- 1) response
- 2) question + response
- 3) question context + response
- 4) scoring rubric + response
- 5) bundle one-hot + response
- 6) question + scoring rubric + response
- 7) bundle one-hot + question + scoring rubric + response
- 8) bundle one-hot + question + question context + scoring rubric + response

For each of the 8 combinations listed above, we create three different vector representations with the aforementioned methods: 1) Sentence-BERT (SBERT), 2) Word2Vec (W2V), and 3) Bag-of-words (BOW), resulting in 24 distinct input types. We fit a classification model for each of the input types, for both of our classification models. Thus, we compare 48 separate versions of an ASAG model with three types of evaluation.

1.3.2.4.2 Leave-one-question-out Evaluation

In order to assess the generalizability of the ASAG model to out-of-training-sample questions for each of the 48 experiments, we average the results of N=31 (where N is the number of questions) independent models. For each of the N models, we train the classifier on data from N-1 questions, and test on data exclusive to the left-out-of-training question. In the case of the leave-one-question-out results, it is important to note that although the model has not seen

data specific to the left-out question, it has seen questions that are part of the same question bundle and are therefore related.

For comparison, we also include results from a standard N -fold cross validation. We chose $N=31$ (number of questions) so that the test set sizes were comparable to those in the leave-one-question-out experiments, although not exact since each question has a unique number of responses. Our data is randomly partitioned into N splits, and for each $1/N$ sized partition, we use the one partition as the test set, and the remaining $N-1$ partitions as the training set. This is repeated N times such that each partition is used as a test set once, and reported results are the average of the N experiments. This allows us to evaluate the model in such a way that is not specific to one particular training data set.

1.3.2.4.3 Evaluation Metrics

We report our results in multilabel accuracy because it is both widely used and easy to interpret. Multilabel accuracy represents the degree to which our model classifications agree with the ground truth labels (for this project, human ratings). It is calculated simply as the number of correct predictions divided by the total number of examples.

1.3.2.4.4 Empirical Monte Carlo Confidence Intervals

We conceptualize the results of each leave-one-question-out experiment as data points in a distribution - for each experiment we have a distribution of 31 data points. Thus, we are able to simulate Monte Carlo sampling distributions from each observed distribution in order to construct a sampling distribution of sample means. We specifically focus on the neural network models to create the Monte Carlo Confidence Intervals in order to have a more robust comparison between models. For simplicity, we do not calculate the condence intervals for the logistic regression experiments because it appears almost universal across results that the neural network models perform better.

For each leave-one-question-out experiment, we calculate a cumulative distribution function from the empirical probability density function by ordering the unique observations in the data sample and calculating the cumulative probability for each as the number of observations less than or equal to a given observation divided by the total number of observations. We then sample from this distribution by drawing a random number within the range of our observed distribution, and getting the minimum CDF density where it is greater than the random draw. We sample $M=1000$ distributions of size $N=31$ (number of questions). For each distribution, we calculate the mean value. We thus have a distribution of sample means that is size 1000 for each neural network experimental setup.

Empirical 95% confidence intervals are calculated with the lower bound as the 2.5th percentile of the 1000 means, and the upper bound as the 97.5th percentile of the 1000 means for each neural network experimental model. So, we know from direct observation that 95% of our sampled means fall within the CI reported. In order to justify the conclusion that one model is superior to others, the lower bound of the empirical confidence interval should be higher than the upper bound of the CI for the model to which we are comparing.

1.3.3 Results

Results of our experiments are detailed in Table 1.1, reported in multi-label accuracy. In the left-most part of the table, an “X” is present for a given row if the information type, indicated by the column header, is included in the model input. For example, results in the first row

represent an input of only the response text and results in the second row represent an input of both the Bundle ID and the response. Additionally, the top half of the table results are those from the multinomial logistic regression classifier, and the bottom half of the table results are those for the neural network classifier (as indicated by the leftmost column). For each of our evaluation methods, random holdout, question holdout, and bundle holdout, we present results for the three textual representation methods: SBERT, W2V, and BOW. Finally, as previously mentioned, for the neural network experiments in the bottom half of the table, we present 95% empirical confidence intervals from Monte Carlo sampling.

Table 1.1
Experiment Results: Multilabel Accuracy

	Resp Text	Bundle ID	Quest Text	Context Text	Rubric Text	Random Holdout			Question Holdout			Average (ACC)
						SBERT	W2V	BOW	SBERT	W2V	BOW	
Majority Class						0.34715			0.37044			0.34093
LogReg	x					0.58034	0.49765	0.54665	0.35870	0.35338	0.36517	0.34093
LogReg	x	x				0.59603	0.53154	0.55602	0.37225	0.37595	0.36097	0.40698
LogReg	x		x			0.63062	0.55748	0.58702	0.40378	0.41290	0.32506	0.42678
LogReg	x			x		0.62972	0.56036	0.58486	0.40352	0.41815	0.33053	0.42886
LogReg	x				x	0.61657	0.55982	0.58846	0.38488	0.38379	0.42198	0.42049
LogReg	x		x		x	0.61818	0.56432	0.59152	0.40967	0.38968	0.37571	0.42940
LogReg	x	x	x		x	0.62484	0.56144	0.59261	0.41534	0.40174	0.36048	0.42441
LogReg	x	x	x	x	x	0.61406	0.55963	0.59009	0.41494	0.39999	0.36104	0.42380
NN	x					0.60249	0.56450	0.60973	0.35736	0.34029	0.36930	0.40922
NN	x	x				0.60540	0.59802	0.61963	0.39011	0.37083	0.35526	0.42006
NN	x		x			0.65800	0.61009	0.65532	0.40633	0.37576	0.34075	0.44411
NN	x			x		0.66070	0.61388	0.66198	0.39309	0.38079	0.33176	0.44500
NN	x				x	0.64017	0.61226	0.62234	0.36721	0.35595	0.40597	0.43395
NN	x		x		x	0.62503	0.61009	0.62198	0.41137	0.39257	0.33095	0.43394
NN	x	x	x		x	0.63206	0.59981	0.62938	0.40885	0.36204	0.39455	0.44276
NN	x	x	x	x	x	0.60557	0.59405	0.61478	0.42270	0.39130	0.35288	0.43002
Average (ACC)						0.62124	0.57468	0.60452	0.39500	0.38157	0.36140	0.33223

Table 1.2
Experimental Results: 95% Monte Carlo Confidence Intervals

	Resp Text	Bundle ID	Quest Text	Context Text	Rubric Text	Monte Carlo CIs		
	x					SBERT	W2V	BOW
NN						(0.3199, 0.4046)	(0.3033, 0.3892)	(0.3323, 0.4157)
NN	x	x				(0.3498, 0.4390)	(0.3078, 0.4774)	(0.3185, 0.4011)
NN	x		x			(0.3371, 0.4905)	(0.2986, 0.4541)	(0.2986, 0.4541)
NN	x			x		(0.3136, 0.4757)	(0.3060, 0.4637)	(0.3060, 0.4638)
NN	x				x	(0.2903, 0.4537)	(0.2798, 0.4375)	(0.2797, 0.4375)
NN	x		x		x	(0.3396, 0.4883)	(0.3194, 0.4786)	(0.3193, 0.4786)
NN	x	x	x		x	(0.3378, 0.4929)	(0.2802, 0.4451)	(0.2802, 0.4450)
NN	x	x	x	x	x	(0.3475, 0.5017)	(0.3079, 0.4774)	(0.3079, 0.4774)

In terms of the general performance of our classification models, we consider the random holdout evaluation method. Overall, it looks as if the SBERT representations performed best when averaging across the classification methods and input combinations, followed by the BOW representations (accuracy of 0.621 for SBERT compared to 0.575 and 0.605 for W2V and BOW, respectively). Additionally, the neural network generally achieves higher accuracy than the logistic regression in general. Both SBERT and BOW seem to perform notably well when the input includes the question text, or the question content.

To assess the generalizability to grading answers to questions unseen in the training set, we focus on the question holdout results. Across the board, we see much lower accuracy for the question holdout experiments than that of the random holdout. This is in line with what one might expect because for the question holdout, the model has not yet seen responses for the particular question in the test set whereas with the random holdout experiments, each question should be represented in the training set (although there is a very low probability that a particular question might not be represented in the training set, due to random chance).

Similar to the random holdout experiments, we see the same overall pattern for the question holdout experiments: SBERT seems generally superior, followed by BOW and W2V, respectively. One notable difference for the question holdout experiments compared to those of the random holdout is that we see increased performance when we include multiple extra sources of information. We might explain this result as, when the model is lacking previous information about the test question from training, extra input information might provide guidance for the model.

We might take inferences from the row averages in the rightmost columns of the table that across all experiments and text representation types, the response and question text, as well as the response and context text achieve the highest evaluation scores. Additionally, the column averages seem to show that the SBERT representations perform best.

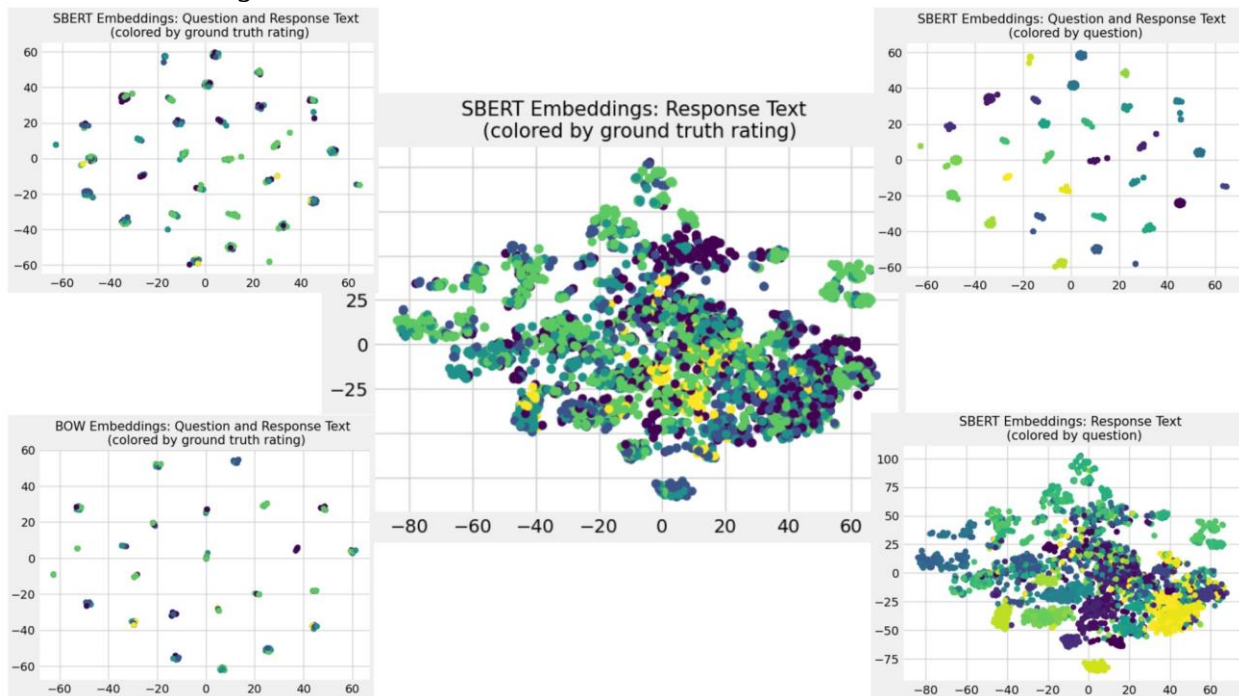
However, the Monte Carlo Confidence Intervals (CIs) in Table 1.2 confirm that no strong conclusions can be made as far as which vectorization method or extra question-related information helps the autograder's performance in terms of accuracy. We see that all, with no exceptions, the empirical confidence intervals overlap.

We include results from a majority class classifier on the top row for a baseline comparison. We emphasize that, across all random holdout experiments, the classification models outperform the majority class classifier significantly. However, this is not the case for the question holdout experiments. For the question holdout experiments, many of the SBERT experiments outperformed the majority class classifier, but when looking at the Monte Carlo confidence intervals, the lower bounds of our condence intervals are generally still below the majority class classification accuracy.

Further interpretations can be made from Figure 1.3, which includes dimension-reduced, t-distributed Stochastic Neighbor Embedding (TSNE) vector representations of student responses. Depending on what text is included as input, we see very different dimension-reduced visualizations of the data. The center image includes SBERT embeddings of only the response text and the colors represent the ground truth ratings. With only response text, it is apparent from the image that the responses do not cluster based on their rating level. However, when comparing the center image to the bottom right image where we color the data points based on the question, we see that even without question text included, the student responses tend to group by question. But, the groupings are not very distinct and overlap throughout the two dimensional space.

Figure 1.3

2D Visuals of Input Vector Representations. Plots Vary by Representation Type, Input Content and Color Labeling.



In contrast, when we include the question text with the response text, the two dimensional vector space looks very different. Referencing the top left, bottom left, and top right images the embeddings are distinctly clustered by question (top right image) but not necessarily by the rating (top left and bottom left image). However, from the bottom right image, we can visualize question specific clusters, but they are not as distinct as the representations that include the question text. From these images, although we are not able to conclude anything about which

vector space is more helpful for an automated grading model, we can see that the embedding space changes dramatically with the inclusion of extra text.

Overall, we cannot conclude that in terms of question holdout, the model can generalize to out-of-sample questions with no clear improvement over the majority class.

1.3.4 Discussion

Although our results are not promising for the generalizability of autograding models to unseen questions, we emphasize the importance of finding more generalizable models to decrease time spent on the laborious task of creating ground-truth human ratings. Our intention is that this work will influence researchers to consider further innovative methods to increase the generalizability of ASAG models. Further, because we did find that including certain question-related text may improve model performance, it may be of use to the ASAG research community to continue to explore how extra sources of information about a question may be incorporated into an ASAG system.

As is evident in our literature review, there has been increased adoption of state-of-the-art textual representation methods such as SBERT, and large language models such as BERT and XLNet, within the field of NLP in Education. Our results support that such models may achieve superior performance for certain tasks.

1.3.4.1 Next Steps

To build on this work further, we could consider other methods, beyond that of concatenation to the input text, to include the extra question information in our model. We could further pre-train a transformer-based model such as BERT or XLNet with the extra textual information by either tuning the existing weights or altering the existing architecture with an extra encoder layer of weights trained on our text alone. Moreover, we may focus more closely on how the classification model itself might be altered such that it might better generalize to out-of-training-sample questions, instead of only focusing on the input content.

We believe that beyond model performance, the practical utility of an ASAG system must be considered in order for educators to continue to adopt new technologies that employ advanced methods in artificial intelligence. Recent years have seen vast improvements in the field of machine learning and language processing. Embracing such technologies for applications in education may be pivotal to provide the assistance that both educators and learners need. However, we do not suggest that machine learning systems such as ASAG should be used to replace human judgements in education, especially in high stakes testing scenarios.

1.4 Chapter 1, Study 2: Utilizing a Scoring Rubric to Increase Model Performance

Many assessments are created via collaboration between teachers, researchers, and test developers where scoring rubrics for each item are carefully designed. Rubrics can provide raters with both exemplar student answers as well as competency descriptions corresponding to each grading category. Rubrics are created to increase both reliability and validity of scores (Waltman et al., 1998). When a rubric clearly and rigidly defines scoring criteria, inter and intra rating consistency can be increased (Wainer, 1993). There are, however, grading scenarios in which rubrics are not created such as contract grading where evaluation is solely based on completion of a task (Danielewicz & Elbow, 2009), or intuitive grading where instructors rely on their experience to grade (Ecclestone, 2001).

In order to exploit the advantages of using scoring rubrics, we propose to incorporate item-specific scoring rubrics into an ASAG model. We do so by further training a model called BERT (Bidirectional Encoder Representations of Transformers) using the scoring rubric text and structure. BERT is a type of language model called a transformer, which has been pre-trained on the English language and can be later trained for downstream tasks such as classification (Devlin et al., 2018). BERT, as-is, encapsulates latent relationships in natural language from the text on which it has been pre-trained. However, we want it to capture such relationships in the scoring rubric text specifically. Before we train it on our classification task of grading, we continue the model’s pre-training with the scoring rubrics. We hypothesize that this will improve the ASAG accuracy beyond that of similar models not trained on rubrics. Additionally, we look at model performance at the question level to examine if certain types of questions and their corresponding rubrics exhibit varied results.

The original BERT model is a neural network consisting of 12 transformer encoder layers with pre-trained weights. To provide the model with the rubric text, we add one more (13th) transformer encoder layer to the model to train with the masked language model (MLM) task, using the rubric text as input. Thus, the model will also incorporate relationships in the text specific to the rubric as it classifies student responses. Additionally, because the ordinality of the rubric is important - namely that its text would not mean much to a human grader if they could not interpret that a level 3 score is ranked “higher” than a level 2 score - we incorporate a sampling method called Node2Vec (Grover & Leskovec, 2016) for creating a vector representation of the rubric text. If we conceptualize the rubric as a graph network to capture the ordinality in the scoring levels, Node2Vec will learn a latent representation of the structure of the rubric.

The novelty of this research is the modification of BERT to incorporate a rubric for each item. The purpose is to increase an automated grading system’s performance for practical use in terms of producing scores that match that of human graders. Using BERT for downstream classification tasks is common for applied machine learning research, as well as for ASAG-specific tasks. Modification of BERT to incorporate a rubric has not yet been studied.

We hypothesize that pre-training BERT on rubric text, termed “further pre-training” will improve the model’s ability to match human ratings of the student responses. However, we do anticipate that this effect may differ depending on the type of questions and their corresponding rubrics.

1.4.1 Background

In this section, we provide a brief background of the BERT model and the Node2Vec sampling method.

1.4.1.1 BERT

BERT (Devlin et al., 2018) was the first language model to successfully learn relationships within sequences of words by pre-training with a bi-directional prediction task, masked language modeling (MLM). It does so by randomly masking 15% of tokens, or words, throughout the text during training and uses a self-attention mechanism to predict the masked texts. Additionally, BERT learns representations between sequences of sentences by its pre-training on a next sentence prediction (NSP) task, where the model is given two sentences and predicts whether one sentence should come directly after the other. The standard BERT model was pre-trained on large amounts of natural language text from Wikipedia and BooksCorpus, and can be fine-tuned for different tasks, such as classification, by adding only one additional layer to

the existing neural network (Devlin et al., 2018). We use a compressed version of the original BERT model called BERT-base, consisting of 12 encoder layers and a total of 110 million parameters. With limited computational resources, the base version is faster to train yet still retains much of the impressive performance on standard benchmark tasks.

1.4.1.2 Node2Vec

Node2Vec is an approach for learning latent representations of vertices in a network such that graph structures can be represented as a continuous vector space (Grover & Leskovec, 2016). By modeling random walks over the graph structure, Node2Vec creates input sequences conceptually similar to that for a sequence of words, however the sequences capture the structure of the graph. The algorithm works as follows: firstly, a random walk generator takes in a graph structure, and uniformly samples a node as the root of the walk. Then, the walk uniformly samples from the direct neighbors of the last visited node, and this process continues until a maximum length has been reached. This sequence of nodes would represent one input sequence, and as many sequences can be generated as specified. In addition to the number of walks and the length of walks, two other parameters exist which specify how likely it is for the walk to wander far from the starting node (breadth of the search), and how likely it is to return to the start node (depth of the search).

1.4.2 Methods

In this section, we describe the data used for this project, outline the methods used to alter the architecture of the BERT-base model, explain the Node2Vec sampling approach for creating input sequences to further pre-train the extra layer of the model, and detail the process of training the classifier.

1.4.2.1 The Dataset

We used two data sets for this project, consisting of short responses to science questions. The first dataset we used was sourced from a 2019 field test of an assessment created at the Berkeley Evaluation and Assessment Research (BEAR) center. Further details of this dataset can be found in section 1.3.1.

We also show results with an open-source data set called the Automatic Student Assessment Prize (ASAP) Short Answer Scoring (SAS) data (Hamner et al., 2012). The data was used in a 2012 Kaggle competition sponsored by the Hewlett Foundation, and consists of almost 13,000 short answer responses to 10 science and English questions. We used only the 5 science questions since the domain is the same as the BEAR data set. The questions were scored from 0 (incorrect) to 3 (fully correct), and each question includes a scoring rubric. We refer to this data set as ASAP. An example of a question from the ASAP data set is:

"Starting with mRNA leaving the nucleus, list and describe four major steps involved in protein synthesis."

The rubric consists of a list of key elements of an exemplary response such as:

"mRNA exits the nucleus via nuclear pore"

and specifies how many key elements need to be included for the answer to be rated at a given level.

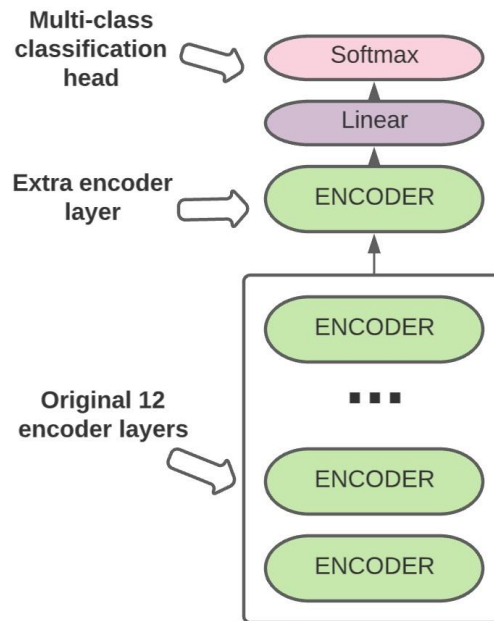
Standard pre-processing of both the data sets included removing null responses, adding a period to the end of sentences, and concatenating a numeric question identifier to the beginning of student responses. We split the data sets into training, validation and testing sets with a 70/15/15 percent split.

1.4.2.2. Architecture Modification

To further pre-train the model on the scoring rubric text, we add a 13th encoder transformer layer to the original BERT-base model. The weights of the 13th layer are randomly initialized, and the weights of the other 12 encoder layers are frozen during the additional pre-training. Thus, the original 12 layers maintain the weights that capture latent relationships in general English language from the model's original pre-training, and the new weights of the 13th layer start as essentially a blank slate. We then train the 13th layer with the MLM task using our scoring rubric text as input. Standard neural network parameters were tuned to minimize the MLM loss, in addition to considering the best configuration for our computational limitations. Optimal parameters include a batch size of 8, a learning rate of $4e-4$, and 12 training epochs. Figure 1.4 includes an illustration of the architecture of the modified BERT model.

Figure 1.4

The Modified BERT Architecture



1.4.2.3 Sampling Methods

To produce input sequences that capture the ordinal structure of the scoring rubric for pre-training the BERT model, we use the Node2Vec sampling method. We create a graph representation of the structure of the scoring rubric for each question in the data set. An example of a scoring rubric and its corresponding graph structure is included in Figure 1.5. In order to conceptualize the rubric as a graph, we make each level of the scoring rubric a node in the graph, and undirected edges connect each level to both the level below and above it. The description of

each level as well as the example responses are included as features of the level nodes. We use the python package, ‘networkx’ (Hagberg et al., 2008) to create the graph structure.

Figure 1.5

An Example of a Scoring Rubric and its Corresponding Graph Structure

Level	Description	Example Response
4	Student provides a fully correct positive and negative justification for ...	"The graph best illustrates the visual trend of oil production. The table or the equation won't provide an overall ..."
3	Student provides a fully correct positive justification for selected representation ...	"The table cannot show a trend in the oil production effectively. "
2	Student provides a partial/general positive justification for selected ...	"The graph helped me with more questions."
1	Student provides incorrect justification for selected representation.	"Because the graph gives us better projections."
0	Student makes no attempt to provide a justification for selected representation.	"I am not sure why."



Parameters to be tuned for Node2Vec include the walk length, number of walks, and the p and q parameters that indicate how far the random walk strays from the original node, and how likely it is to return to the previous node. The best set of parameters were chosen based on the performance with our downstream classification task and consisted of a walk length of 10, number of walks being 50, a p of 0.5 and a q of 2.

Included as features of each node of our scoring rubric graph is both the description of the grading level as well as exemplary student responses. When the random walk lands on a node, the method randomly samples one piece of text from the node (rubric level). For example, if the algorithm starts on the level 2 node, we grab either the level description text, or one of the exemplary responses for the input sequence. If, during the random walk, we return to the same node, the algorithm will similarly randomly choose one of the pieces of text. The full input sequence for each walk is created by concatenating the text from each "step" of the walk in the order they are sampled. The sampling process is repeated for each question included in the data, as each question has a unique corresponding scoring rubric. Resulting input sequences for each question are combined to create all the input text. All of the input sequences are used to pre-train the weights of the 13th encoder layer of our model.

1.4.2.4 Fine-tuning Experiments

To test our hypothesis that providing the BERT model with information about the scoring rubric will increase the performance of the classifier for ASAG, we performed four different

experiments, with four separately trained models. We choose the particular experiments to help us clarify the implication of our results such that we can distinguish between whether the text itself or the latent graph structure is helping the ASAG model.

Each model is fine-tuned for our multi-class classification task - i.e. we add a classification head to the model after its encoder layers (regardless if it has 12 or 13 encoder layers) and train the classification head with the labeled data. We performed a standard parameter search and achieved the best validation set results with a batch size of 8, 4 training epochs, and a learning rate of $5e-4$. We include two standard evaluation metrics for both datasets (BEAR and ASAP) in our results table: Cohen’s Kappa (CK) and weighted F1 score (F1). The F1 score includes both precision and recall and accounts for class imbalance, and CK reports the agreeability between two scores beyond random chance - a more robust measure than accuracy. Additionally, we provide a majority class classifier for the F1 score results as a baseline comparison. The CK results have an ‘n/a’ for the majority class classifier because a majority class model is the same as classification by random chance for multi-class classification, which would always result in a CK metric of 0.

A 5x2 cross validation (CV) paired t-test was used to evaluate the statistical significance of the difference in models. The 5x2 CV paired t-test is based on five iterations of twofold cross-validation, and is presented in (Dietterich, 1998) as the recommended approximate statistical test for whether one machine learning model outperforms another because of its more acceptable type I error, and stronger statistical power than other methods such as McNemar’s test, or a paired t-test based on 10-fold CV. We used the CK metric for the 5x2 CV paired t-tests (with 5 degrees of freedom). For each of the ‘Rubric’ models, there is a corresponding t-test comparing the CK metric for the BERT-base model with the given model. The null hypothesis for each t-test states that the CK metric for the specified model is no different than the CK metric for the BERT-base model.

1.4.3 Results

We present our results below, in Table 1.2, for the four models we wish to compare. The first model, ‘BERT-base’, is a standard BERT-base classification model (no extra 13th layer). This model serves as a baseline with which to compare our modified architecture (‘Rubric’) models. The second model, ‘Node2Vec Rubric’, incorporates the Node2Vec sampling method in the further pre-training of the 13th layer. This is the model we hypothesize will perform best because it captures both the text and structure of the scoring rubric in the added 13th layer. The third model, ‘Top-level Rubric’, also has an added 13th encoder layer but the extra layer is trained with only the top level of scoring rubric text (i.e., explanations and examples of the most correct responses) and the text is randomly sampled instead of using Node2Vec Sampling. In this experiment, we still sample 10 pieces of text for each input sequence, and 50 sequences total for each question for consistency in the amount of training data. Finally, the fourth model, ‘Random Rubric’, we add the 13th encoder layer and train it with all of the scoring rubric text, randomly sampled (no Node2Vec sampling). Thus we can compare a regular BERT-base model to three other models that have been altered in structure with a new 13th layer to examine: if just the rubric text itself with no structure is helpful (Random Rubric), if only including exemplar level descriptions and text is helpful (Top-level Rubric), or if providing the model a signal about the structure of all the rubric text, along with the text itself is helpful (Node2Vec Rubric) in terms of the performance of our ASAG model.

Table 1.3*Test Set Weighted F1 Score and Cohen's Kappa*

	CR4CR F1	CR4CR CK	ASAP F1	ASAP CK
BERT-base	0.6742	0.6831	0.7755	0.8677
Graph Rubric	0.6946	0.7361	0.8001	0.8808
Top Rubric	0.6932	0.7170	0.7879	0.8739
Random Rubric	0.6815	0.6961	0.7972	0.8825
Majority Class	0.2082	n/a	0.2737	n/a

Results of the 5x2 CV paired t-test are shown in Table 1.4. For a paired one tailed t-test, the critical values for an alpha, or probability of a type I error, of 0.05, 0.10 are 2.015 and 1.476 respectively. In Table 1.4, statistical significance at the 0.05 level is represented with an appended *.

Table 1.4*5x2 Cross Validation Paired One-tailed t-test*

BERT-base versus:	CR4CR t(5)	CR4CR p-val	ASAP t(5)	ASAP p-val
Graph Rubric	1.814	0.0647	3.098*	0.0135
Top Rubric	1.880	0.0594	3.191*	0.0121
Random Rubric	1.778	0.0678	4.110*	0.0046

From Table 1.3, we see that all of the models, including BERT-base, on both the BEAR and ASAP data, perform much better than the majority class classifier. Additionally, across both data sets and considering both metrics, the 'BERT-base' model performs slightly worse than all the others. For the BEAR data, the 'Node2Vec Rubric' model performs best and the difference between this model and the 'BERT-base' model is considerable, especially in terms of the CK metric. The 'Top-level Rubric' model still seems to be an improvement to the 'BERTbase' model, and the 'Random Rubric' model shows only slight improvements from the 'BERT-base' model. With the ASAP data, we see similar results when comparing the 'BERT-base' model and the 'Node2Vec Rubric' model - we get at least two percentage points improvement in terms of both F1 and CK metrics. Additionally similar, we get improved performance, but to a lesser extent, with the 'Top-level Rubric' model. However, one considerable difference is that we see about the same performance improvement with the 'Random Rubric' model as the 'Node2Vec Rubric' model, when comparing against 'BERT-base'.

In Table 1.4, we further validated our results with the 5x2 CV t-test for statistical significance for whether each of the three models with an added 13th layer model performs superior to the 'BERT-base' model in terms of the CK metric. With the BEAR data, results are

statistically significant at the 0.10 type I error level, but not quite at the 0.05 level as shown by the p-values for each of the three ‘Rubric’ models. However, with the open-source ASAP data, all three models show statistically significant improvement over ‘BERT-base’ at the 0.05 type I error level.

Additionally, since the ASAP data showed more promising results for all three ‘Rubric’ models in terms of statistical significance, we broke down the results for this data set by question in order to see if our method is more useful when autograding a particular type of question or with a certain rubric type. We saw the biggest performance increase between the ‘BERT-base’ model and the ‘Node2Vec Rubric’ model with one specific question that elicited a very straightforward rubric. The question context presents an experimental procedure, and asks students to describe what other appropriate pieces of information are needed to replicate the experiment. The rubric is simply broken down into the number of pieces of information that the student provides; to achieve the highest score, a student must provide three additional pieces of information, and to achieve the second highest score, two additional pieces of information. In contrast, the least performance increase (actually, a slight performance decrease) was observed for a question that had a much more open-ended nature and abstract rubric. The question context presents an experimental design and corresponding results, and asks students to describe two ways to improve the experimental design and/or the validity of the results. Rubric guidance is vague, including statements such as: "The response draws a valid conclusion supported by the student’s data but fails to describe, or incorrectly describes, how the student could have improved the experimental design and/or the validity of the results".

1.4.4 Discussion

Most evident with the ASAP data set results, adding an extra 13th layer to the model, further pre-trained with some version of the rubric text improves the classification performance of the ASAG model notably (from the t-test, we see a statistically significant improvement in performance, at a significance of 0.05, when comparing all three models to the BERT-base model with no 13th layer). Less evident, but also noteworthy, is the performance increase for all three ‘Rubric’ models with the BEAR data (from the CV t-test, we see a statistically significant improvement in performance, at a significance of 0.10, when comparing all three ‘Rubric’ models to the BERT-base model with no 13th layer). Thus, we conclude that incorporating rubric text to BERT by further pre-training an extra encoder layer adds value to the ASAG model.

However, using Node2Vec sampling to capture the structure of the rubric text prior to further pre-training BERT does not add value. We do not see consistently lower p-values in the 5x2 CV t-test results for the ‘Node2Vec Rubric’ model than the other ‘Rubric’ models (‘Top-level Rubric’ and ‘Random Rubric’), nor do we observe a noticeable increase in performance for the results in Table 1 when comparing the ‘Node2Vec Rubric’ model with the other ‘Rubric’ models. So, although it appears that allowing the BERT model to learn about the text in a rubric is beneficial for the model’s performance, the structure of the rubric seems not as important as we hypothesized. It may be the case that the model benefits simply from ingesting the subject-specific jargon in the rubric, versus the more general English text on which the other 12 encoder layers have been pre-trained. Further, there are methods outside of Node2Vec that may better capture the rubric structure and further research may attempt to include other methods to represent structured text for language models.

Finally, after investigating the results broken down by question, we conclude that our method of further pre-training a BERT model using rubric text may be useful only for specific

types of questions that elicit a very straightforward rubric structure. Questions with rubrics that call for more abstract interpretation or subjective judgment may not be a good application for our proposed method. Further investigation on different short answer question types and rubric structures is necessary to further generalize this conclusion.

The outlook of automatic grading continues to improve with advances in machine learning and natural language processing (Haller et al., 2022). Incorporating human-created, domain related text such as a scoring rubric may be one way to ensure that ASAG models reflect human judgment, even more than with supervised classification (i.e. training the model to match human ratings), while still improving model performance. However, if it is the case that our methods are only applicable to questions of a simplistic nature, future research may need to focus on how we can automatically grade questions that elicit more complex responses. For example, the knowledge integration perspective on science inquiry supports that students can maintain multiple ideas, sometimes conflicting, about a particular topic (Linn et al., 2014). Automatically identifying a student's various, and potentially contradicting, ideas within a response may be an important step in creating quality autograders for more complex science questions (Gerard et al., 2016).

Chapter 2

Peering inside the “black box” of an ASAG Model

2.1 Introduction

The use of educational technology with automatic short answer grading (ASAG) models may help alleviate a time burden of grading while encouraging educators to frequently incorporate OE items in their curriculum. However, current state-of-the-art ASAG models consist of “black box” neural networks (NN) that provide little to no explanation for which characteristics of an input were used to arrive at the produced output. Specifically, large language models (LLMs) (Vaswani et al. 2017) are NNs frequently used for ASAG. These models learn a mapping, often with billions of parameters, from a student’s response to a given score, and a well-trained model should grade responses in a way that mimics that of the training set. Whereas educators typically use a scoring rubric to provide an explanation to stakeholders about which features of a given response substantiate the assigned score, NN ASAG models have no ability to provide justification for their predictions. The inexplicable nature of these models can be frustrating to both teachers and students when trying to make sense of, or learn from an automated grade. Teachers are unable to monitor student understandings at a fine-grained level, and students may not productively learn without knowing why they received a certain grade (Schlippe et al., 2022).

In this chapter, we address the lack of explainability in typical ASAG models. We examine the use of a Neural Additive Model (NAM) for ASAG, which combines the explainability of additive models with the predictive power of a neural network. Further, we train a deep reinforcement learning (RL) agent to revise a student’s response by inserting key phrases or deleting portions of the response to achieve a high grade from an ASAG model in the least number of revisions. The idea is that the agent’s revisions may provide some explanation as to what was missing or unnecessarily included in a student’s response. If the ASAG model’s grading performance is acceptably high, adding correct phrases that a student did not already include, and deleting portions that should not be include should also increase the ASAG model’s score. We use the RL agent to expose shortcomings in the ASAG model by finding revised responses that achieve a high grade from the automatic grader, but may not be considered high quality responses by a human rater.

2.2 Related Work

In this section we describe previous work relating to Explainable AI and more particularly Explainable ASAG, applications of RL for education, and applications of Neural Additive Models.

2.2.1 Explainable AI and ASAG

A recent review article outlines the increase in demand for explainable AI (XAI), especially for those who are affected by AI driven decisions (Xu et al., 2019). They describe the rising popularity of neural network models for their state-of-the-art performance, despite that their inference processes are not known or interpretable. Many researchers aim to increase the explainability of AI systems. Initiatives such as DARPA’s “Explainable AI (XAI) program” (Gunning and Aha, 2019) and the European Union’s “General Data Protection Regulation”

which demands citizen's rights to an explanation for decisions made by AI (Goodman and Flaxman, 2017) are contributing to an increased demand for XAI.

Researchers are now beginning to think about XAI in terms of ASAG models. Schlippe et al. (2022) surveyed over 70 educators about their preferences for explainability in ASAG. They learned that educators would most prefer to see matches between student answers and exemplary answers to justify scores over other explainability methods like highlighting the importance of certain words or confidence percentages for the ASAG model's scores. Poulton et al. (2021) proposed an explainability framework using Shapley Additive Explanation (SHAP) values to assess popular LLMs for ASAG such as BERT, RoBERTa, DistilBERT, and ALBERT. Zeng et al. (2022) investigated whether autograding models align with human graders in terms of the important words they use when assigning a grade. They conduct a randomized controlled trial to see if highlighting words deemed important by an autograder can assist with human grading. Singh et al. (2023) introduced Summarize and Score (SASC) as a method to explain 'text modules', which map text to a scalar continuous value, within LLMs. The SASC method provides a natural language explanation of the module's selectivity as well as a measure of the reliability of the explanation. Finally, Tornqvist et al. (2023) created ExASAG, an explainable framework for ASAG that uses integrated gradients to generate natural language explanations for each prediction, and Gombert et al. (2023) compared a variety of transformer and feature-based models to recognize ideas in students' writing and inspect what features played a role for which prediction outcome.

2.2.2 Reinforcement Learning for Education

Although research regarding the use of AI has recently surged in education, leveraging one such type of machine learning, RL, for applications in education is not yet broadly explored despite its flexible implementation of reward functions, allowing algorithms to be optimized toward specific use cases (Memarian & Doleck, 2024). As such, the utilization of RL within the field of education holds potential to bring about a shift in the way students learn and teachers evaluate students (Fahad Mon et al., 2023).

A survey article by Singla et al. (2021) summarized that the main contributions in RL for education have been personalizing curriculum, providing hints, adaptive experimentation, modeling students, and content generation. Further, Fahad Mon et al. (2023) examined existing literature in the field and provided best practices for incorporating RL into education settings. They emphasize that because there are various configurations and design choices to be made when designing and RL system, the consequences of such decisions must be weighed. Notable works include Fazazi et al. (2021) which presented a multi-agent approach to deep Q-learning for designing an adaptive e-Learning system which is customized to a particular learner's characteristics. MacLellan and Gupta (2021) used Proximal Policy Optimization (PPO) to learn expert models for educational tasks and compare their RL approach to existing computational approaches. Doroudi et al. (2019) reviewed a variety of RL based attempts at instructional sequencing, and concluded that RL has been most successful when it is constrained by theories from cognitive and psychological sciences. Park et al. (2019) introduced an RL "social robot" for personalized and adaptive education, and Cai et al. (2021) presented MathBot, a bandit algorithm-based chatbot that explains math concepts, provides practice questions, and offers personalized feedback to students. RL has been used to identify critical pedagogical decisions within an ITS setting (Ju et al., 2020) and further make such decisions for individual assistance with hints and keywords. Schmucker et al. (2023) showed that with a multi-armed bandit and

offline policy evaluation techniques, student success is optimized in terms of correctly answering a given question and overall session performance. Finally, Abdelshiheed et al., (2023) compared adaptive learning interventions from an RL agent which considered a student's cognitive load to a static control group and found that the RL agent was superior in its effect on future learning.

2.2.3 Applications of Neural Additive Models

Although Neural Additive Models (NAMs) were introduced only a few years ago (Agarwal et al., 2021), there has been interest in their use as the machine learning community has increased efforts to promote understandable models. Some researchers have proposed altered or improved versions of NAMs for specific applications. Mariotti et al. (2023) explored the tension between interpretability and performance of NAM models and introduced a constrainable NAM (CNAM) which includes specifications for model regularization. The CNAM model is able to consider the performance-interpretability tradeoff during training according to various definitions of performance and interpretability, and is demonstrated to work well for both regression and classification tasks. Luber et al. (2023) introduced Structural NAMs (SNAMs) which incorporate neural splines to a typical NAM model. SNAMs offer improved intelligibility over NAMs by enabling direct interpretation of estimated parameters, as well as quantification of parameter uncertainty and allow for post-hoc analysis of the underlying model's behavior. Bouchiat et al. (2023) employed a Bayesian approach and developed a Laplace approximated NAM (LA-NAM) to enhance interpretability. LA-NAMs provide an enhanced estimation of uncertainty, yield tractable estimates of the marginal likelihood, and can rank feature pairs as candidates for second-order interactions. Jo and Kim (2022) introduced NAMs for multivariate time series modeling called NAM nowcasting (NAM-NC), and Xu et al., (2022) empowered NAMs with feature selection and improved generalizability with their sparse NAM (SNAM) model that employs group sparsity regularization. Others have used NAMs for applications in finance (Chen and Ye, 2022), inventory management (Siems et al., 2023), mortality prediction (Moslehi et al., 2022), and survival analysis (Utkin and Konstantinov, 2022).

2.3 The Data Set

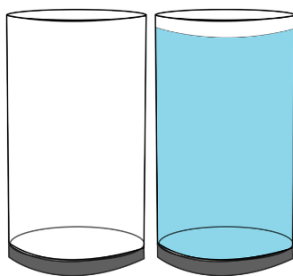
In this chapter, we use the same data set to investigate each of our research questions. The data was collected during a previous research project at the Web-based Inquiry Science Environment (WISE) research center at the University of California, Berkeley consisting of OE science items designed for a formative assessment of middle school students (Gerard et al., 2016; Riordan et al., 2020). Students accessed the items via an online classroom system, and responses were scored with a Knowledge Integration (KI) rubric (Linn, 2000).

For this chapter, we use one item from a unit designed for 8th grade students about the physics of sound waves that engages students to refine their ideas about concepts such as wavelengths, frequency and pitch. For this item, students must distinguish how the pitch of sound made by tapping a full glass of water compares to the pitch made by tapping an empty glass. They are asked to explain why they think the pitch of the sound waves may be the same or different for the two glasses (Riordan et al., 2020). See Figure 2.1 for a visual of the item from the WISE platform.

Figure 2.1

KI Soundwave Item

Arlene has two glass cups. She leaves one empty and fills the other one with water. She then uses a chopstick to gently strike each glass. What will she hear?



How does the pitch of the sound made by the tapped full glass compare to the pitch of the tapped empty glass?
(For reference: A whistle makes a high-pitched sound and a bear makes a low-pitched growl.)

- ☐ There is no change; the pitch stays the SAME.
- ☐ The pitch of the tapped full glass is LOWER than the pitch of the tapped empty glass.
- ☐ The pitch of the tapped full glass is HIGHER than the pitch of the tapped empty glass.

SAVE

Explain why you think the pitches of the sound waves generated by striking the two glasses will be the same or different.

The data includes 1313 OE student responses that were carefully rated from 1 to 5 by subject matter experts and educators. A detailed KI rubric was created by researchers and used for scoring. The rubric includes a description of the rating level, examples of correct/incorrect mechanisms and conclusions, and examples of student responses that would fall into each category. See Table 2.1 for a sample of the KI scoring rubric.

Table 2.1
KI Soundwave Scoring Rubric Sample

KI Score	Description	Category of responses	Examples
1	I don't know or off topic: Response is off topic or "I don't know"	Response is off topic or "I don't know" • I don't know • Irrelevant ...	• "Because there is water in the cup" • "Idk" ...
2	Irrelevant or incorrect: Response is on topic but irrelevant or incorrect (contains no valid ideas)	<u>Irrelevant or incorrect mechanism</u> • sound bounces • amount of air matters (or blocks) • sound wave has less space to move • the material inside the glass (water or air) makes the sound echo	• "One of them has water, which makes the sound bounce, the other has air, which doesn't." • "There is more room in the glass to make a sound"

		• sound is heavier in water so it sinks ... <u>Incorrect conclusion</u> • sound is quieter (confusing or conflating frequency/pitch and volume/amplitude) 5 ... <u>Incorrect mechanism + incorrect conclusion</u>	then the water full one.”
3	Partially correct: Response contains a valid idea (either assertion or rationale) but without a link made between the assertion and supporting rationale (why)	<u>Correct mechanism only</u> • making note of water as something as opposed to air (which is “nothing”) • more weight, more/higher density, more/higher mass or more stuff (more particles) affects pitch ... <u>Correct conclusion only</u> • sound (or pitch) will be lower (or deeper) • sound/pitch is different (sufficient for a 3) ... <u>Correct mechanism + incorrect conclusion / incorrect mechanism + correct conclusion</u>	• “One is thicker than the other.” • “The sound travels through the water slower so it comes out lower while sound travels through the empty space quicker so it come out higher.” ...
4	Emerging understanding (1 valid link): Response contains 1 valid link (correct assertion supported by a correct rationale/why)	<u>One set of correct mechanism + correct conclusion (about pitch)</u> Two components are linked, can be either talking about the full or the empty glass • Lower pitch is linked to lower frequency in full glass / Higher pitch is linked to higher frequency in empty glass • Higher/lower density of material is linked to higher/lower pitch ...	• “The pitch is lower because the water density changes the sound wave since it has to travel through the water. The empty glass sound wave will be a higher pitch because it doesn't have to travel through the water, keeping its length and amplitude the same.” ...
5	Full understanding (2 or more valid links): Response contains 2 or more valid links (correct assertion supported by a correct rationale/why)	<u>Two sets of correct mechanism + correct conclusion (about pitch)</u> Three components are linked, can be either talking about the full or the empty glass • Full glass has more mass, so it is harder to vibrate, creating a lower pitched sound ...	• “The full glass has mass inside of it making it more dense. This would cause it to have a lower pitch/frequency than an empty glass.” ...

2.4 Chapter 2, Study 1: A Neural Additive Models for Explainable ASAG

Many of the best performing ASAG models include some variation of a deep neural network (NN; Haller et al., 2022). NNs are impressive predictors for high dimensional inputs like text embeddings, but predictive power tends to come at the cost of explainability. In an effort to create a powerful yet intelligible ASAG model, we experiment with a type of model that combines the performance of a NN with the explainability of an additive model called a Neural Additive Model (NAM; Agarwal et al., 2021). NAMs allow us to visually examine the contribution of each feature to the final predicted score for each response, similar to testing the significance of a regression coefficient. For this type of model, we must engineer features of a

response as inputs to the model instead of allowing the model to create its own text features like typical Large Language Model (LLM) classifiers do. The research questions we seek to answer include, can NAMs provide intelligible automatic grading such that stakeholders can understand which features of a response are important for its prediction, and is the predictive performance of NAMs better than that of legacy explainable models like a logistic regression and commensurate with that of an LLM classifier? This research is unique as NAM models have not yet been explored in the context of explainable automatic grading.

We use a Knowledge Integration (KI) perspective (Linn, 2000) and rubric to guide our feature engineering to create model inputs that reflect whether or not a student included certain ideas in their response. KI is a framework for strengthening science understanding that emphasizes incorporating new ideas and sorting out alternative perspectives with evidence (Linn, 2000). We hypothesize that the inclusion (or exclusion) of predefined KI ideas as features will be sufficient for the NAM to have good predictive power, as this is precisely what would guide a human scorer using the KI rubric shown in Table 2.1. We compare the performance of the NAM with another explainable model, a logistic regression using the same features as those used for the NAM, and to a non-explainable model, DeBERTa (Decoding-enhanced BERT with disentangled attention) an improved version of the popular BERT model (He et al., 2020). The DeBERTa model does not utilize the engineered features that are used by the NAM or logistic model, but creates its own features from the text.

2.4.1 Background

In this section, we provide a brief overview of NAMs, the DeBERTa model, and logistic regression.

2.4.1.1 Neural Additive Models

Neural Additive Models (NAMs) impose a restriction to the structure of neural networks (NNs) in order to make the model interpretable. NAMs belong to a family of models called Generalized Additive Models (GAMs) which have the form:

$$g(E[y]) = \beta + f_1(x_1) + f_2(x_2) + \dots + f_k(x_k)$$

Where $x = (x_1, x_2, \dots, x_k)$ represents the input with K features and each f_i is a univariate shape function with $E[f_i] = 0$ (Agarwal et al., 2021). NAMs learn a linear combination of jointly-trained NNs where each NN attends to a single input feature. They can approximate complex, high-dimensional functions and their predictions are easily interpretable with regard to the engineered features. NAMs provide advantages over standard GAMs such as their superior scalability with the use of GPU/TPU hardware developed for NNs, differentiability, and visual interpretability.

2.4.1.2 Logistic Regression

Multinomial logistic regression (MLR) is a classification model that predicts probabilities of different outcomes for a categorical dependent variable. In order to generalize to a K-class setting, the model runs K-1 independent binary logistic regression models where one outcome is chosen as a “pivot” and other K-1 outcomes are separately regressed against the pivot outcome. We use the Limited-memory Broyden-Fletcher-Goldfarb-Shannon (LBFGS) algorithm for optimization (Fletcher, 2000), and incorporate L2 regularization.

2.4.1.3 DeBERTa

DeBERTa (Decoding-enhanced BERT with disentangled attention) is a Large Language Model (LLM) that improves upon the popular BERT (Devlin, et al., 2018) and RoBERTa (Liu et al., 2019) models using a different version of the standard attention mechanism (Vaswani et al., 2017) and a novel position encoding method. He et al. (2020) introduces the ‘disentangled attention’ mechanism where each word is represented with two vectors - one to encode the word’s context, and another to encode the word’s position. The attention weights are computed via disentangled matrices on both their contents and relative positions. Further, an enhanced masked decoder incorporates absolute positions to predict masked tokens prior to the softmax layer during training. Finally, DeBERTa uses a unique adversarial training method to fine-tune the model’s generalizability performance on downstream tasks (He et al., 2020).

We use an improved yet smaller version of the original DeBERTa model called DeBERTaV3-base (He et al., 2021) which replaces the masked language modeling (MLM) training task with a more sample efficient method called replaced token detection (RTD). The V3 model also uses a novel gradient-disentangled embedding sharing method which improves training efficiency (He et al., 2021). DeBERTaV3-base consists of 12 layers with a hidden size of 768 and 86 million parameters. Its vocabulary contains 128,000 tokens.

2.4.2 Methods

In this section, we describe the methods used to train each model, feature engineering for the NAM model, and the model evaluation procedures and metrics.

To fit a NAM model, we utilize the python library, "nam" (Kayid et al., 2020), created by the researchers who introduced NAMs for easy implementation of the model. We altered the package code to accommodate our multi-class classification setting and allow for our particular evaluation procedure and corresponding metrics. Further described in the "Evaluation" section below, we use a cross validation evaluation procedure for model comparison, so for each model class (NAM, DeBERTa, Logistic Regression) we train a total of 10 models on a different sample of the data. We train each of the 10 NAM models for 120 epochs with a batch size of 64 and a learning rate of 0.002. An epoch is one complete pass of the training data through the model, batch size represents the number of samples used in a single forward and backward pass through the network, and the learning rate governs the pace at which the model updates its parameter estimates. Further, we use a regularization technique called dropout with a value of 0.15 to help avoid over fitting. Hyperparameters were chosen using a standard grid search where different combinations of hyperparameter values are tested, and those that result in the best performance are chosen.

Each of the DeBERTa models were trained for six epochs using a batch size of eight, and a learning rate of 0.0007. The Logistic Regression model was fit using the scikit-learn "LogisticRegression" library with the maximum number of training iterations to convergence set at 200.

2.4.2.1 Feature Engineering

To create features from student response texts for the NAM and Logistic Regression models, we utilize KI phrases directly from the rubric that represent both correct, and incorrect mechanisms and conclusions. We chose to use KI ideas to create features because these are what a human identifies to assign a grade to a student response within the KI framework, according to the scoring rubric. Because students often express their ideas using a variety of different words (i.e., the idea that *"sound moves faster in air"* could be stated as *"sound travels faster through*

air"), we scan the responses for n-grams that are similar to the chosen KI phrases. An n-gram is a series of n consecutive words within a string of text. For example, 2-grams of the phrase *"the dog ran"* would be *"the dog"* and *"dog ran"*. We use n-grams of size n=1 through 5 for each response as students can also express the same ideas in a different number of words.

To calculate the similarity between a given KI phrase and an n-gram from a student response, we first embed both the n-gram and the phrase using sentence-BERT. Sentence-BERT utilizes siamese and triplet network structures to create semantically meaningful sentence embeddings (Reimers and Gurevych, 2019). We then calculate the cosine similarity of the KI phrase and the n-gram. Cosine similarity is a metric used to measure the similarity of text using the cosine of the angle between the vector embeddings of the two texts being compared. Once the KI phrase has been compared with all respective n-grams in a response, we use the largest cosine similarity found as the feature for that phrase. Thus the feature representation for each phrase provides a measure of whether or not some form of the phrase is included in the student response.

We use a total of 34 phrases and corresponding 28 key words from the KI Sound Wave Rubric (shown in Figure 2) for a total of 62 features. The phrases include:

"I don't know", "idk", "sound bounces", "water blocks sound", "water mutes ringing", "sound moves more in air", "sound moves less in water", "sound echoes", "sound sinks in water", "sound moves fast in air", "frequency is height of wave", "pitch higher in full glass", "the density is different", "water is more dense", "water vibrates less", "affects vibration", "pitch is lower in water", "pitch is different", "higher frequency in air", "frequency is different", "sound moves faster in water", "water has more mass", "mass is different", "sound is denser in water", "sound is slower in water", "amplitude is number of waves", "pitch lower in empty glass", "air is less dense", "empty glass vibrates more", "vibration is different", "pitch is higher in air", "lower frequency in water", "sound moves slower in air", "empty glass less mass"

And the 28 words were chosen directly from these phrases.

2.4.2.2 Evaluation

We include two standard evaluation metrics in our results table: Quadratic Weighted Cohen's Kappa (QWK) and an Accuracy score (ACC). The accuracy score is easily interpreted as the proportion of correct predictions out of all predictions, and QWK reports the agreeability between two scores beyond random chance - a more robust measure than accuracy.

A 5x2 cross validation (CV) paired t-test was used to evaluate the statistical significance of the difference in models. The 5x2 CV paired t-test is based on five iterations of twofold cross-validation, and is presented in (Dietterich, 1998) as the recommended approximate statistical test for whether one machine learning model outperforms another because of it's more acceptable type I error, and stronger statistical power than other methods such as McNemar's test, or a paired t-test based on 10-fold CV. We used the QWK metric for the 5x2 CV paired t-tests (with 5 degrees of freedom). We perform a t-test comparing the QWK metric for the NAM model with the Logistic Regression model and the DeBERTa model. The null hypothesis for each t-test states that the QWK metric for the NAM model is no different than the QWK metric for the Logistic Regression and DeBERTa model.

Further, we report the magnitude of the difference in performance of the NAM model to the Logistic Regression and DeBERTa models using the QWK metric from the 10 cross validation model fits using a Cohen's D effect size. Cohen's D is essentially a difference of

means, scaled by a standard deviation value (Kelley and Preacher, 2012). A low value of Cohen's D which represents a small magnitude of difference is typically below 0.20, and a large value would be around 0.80 or higher. We use a pooled standard deviation to calculate Cohen's D, which represents a weighted combination of the standard deviations from each model.

2.4.3 Results

A comparison of each ASAG model's performance results are presented in tables 2.2 and 2.3. In Table 2.2, we present results of the 5x2 cross validation paired one-tailed t-test using the QWK metric. We compare the NAM model against both the Logistic Regression model and the DeBERTa model. The NAM model performs significantly better than the Logistic Regression model at a significance level of 0.05 ($t=2.1732$, $p=0.0409$) but the NAM model does not perform significantly better than the DeBERTa model at a significance level of 0.05 ($t=0.8410$, $p=0.2194$). Particularly, as we can see from the cross validation average metrics in Table 2.3, the DeBERTa model, on average, performs better than the NAM model. Additionally, in Table 2.2 we present the Cohen's D effect sizes for the difference in performance from the NAM model to the two other models. The magnitude of difference between the NAM model and both models is quite large, with effect sizes greater than 1.0 for each comparison.

Table 2.2

5x2 Cross Validation Paired one-tailed t-test (5 df) and Cohen's D Effect Sizes

NAM versus:	t(5)	p-value	Cohen's D
DeBERTa	0.8410	0.2194	1.111
Logistic Regression	2.1732	0.0409	1.003

Table 2.3

5x2 Cross Validation Averages

Model	QWK	ACC
DeBERTa	0.7475	0.6471
Logistic Regression	0.6929	0.6202
NAM	0.7174	0.6382

A visual of feature importance from the NAM model is presented in Figure 2.2. We show the top 40 most important features (out of 62) in order of increasing importance. We see that the feature importance for the words "*density*" and "*higher*" is much larger than that of most other words and phrases. The phrase "*the density is different*" also has a comparatively high importance. Further, in Figure 2.3, we present NAM Shape functions for the top 8 important words and phrases, ordered by increasing importance. As each feature of the NAM model is handled independently by a learned shape function, we can see how the model makes its predictions, or in our case ratings, by graphing the shape functions for each feature. The scores are averaged and centered so that the shape functions can be directly compared on the same scale. In our case, as our NAM model is a multi-class classifier, we visualize two shape functions for each feature in Figure 2.3, one representing the lowest rating category and the other

representing the highest rating category. Our ASAG model rates responses into five categories, but including five shape functions on one graph presents a cluttered image that is difficult to interpret. Including the shape function for the highest rating and the lowest rating is more visually pleasing, and provides the same level of intelligibility of the model. However, we included all five shape functions for a sample of features in Figure 2.4 for an idea of what including all of them looks like. On the y-axis, a negative score represents a decreased probability of a certain class, and a positive score represents an increased probability. The x-axis shows the maximum n-gram cosine similarity found in the response of the given phrase. Noticeably, the shape functions are jagged-like with sharp jumps. Agarwal et al. (2020) emphasizes the unique capability of NAMs to model highly “jumpy” 1-dimensional functions due to the use of an exponential-centered (ExU) non-linear function, whereas standard neural network models are bias towards smoothness with the use of Rectified Linear Unit (ReLU) functions. We could hypothesize that the true shape functions for the features in this project are more smooth, and in future work we could employ regularization techniques such as weight decay to provide smoother shape functions. See Agarwal et al. (2020) for a more detailed description of the ExU and ReLU functions as they relate to the shape functions of NAMs.

On the same plot, we see the data density in the form of pink shaded bars. The darker the shade of pink, the more data there is in that region. In some areas with a low data density, the model may not have had enough data to adequately learn the shape function.

Figure 2.2
NAM Mean Feature Importance

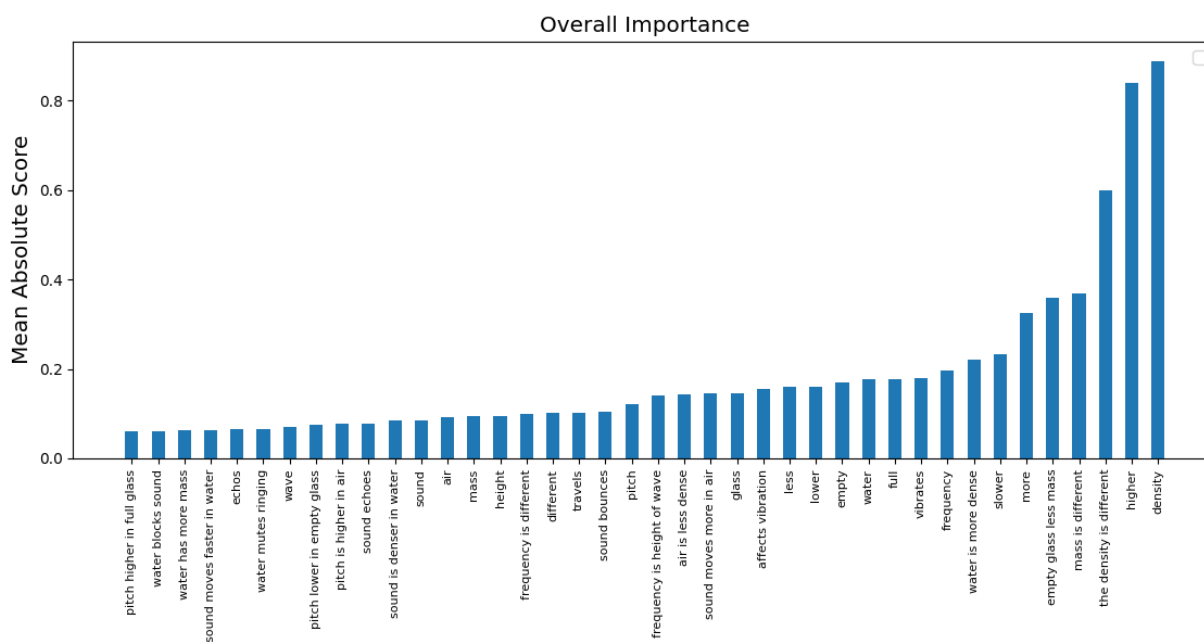


Figure 2.3
NAM Shape Functions (with highest and lowest rating categories)

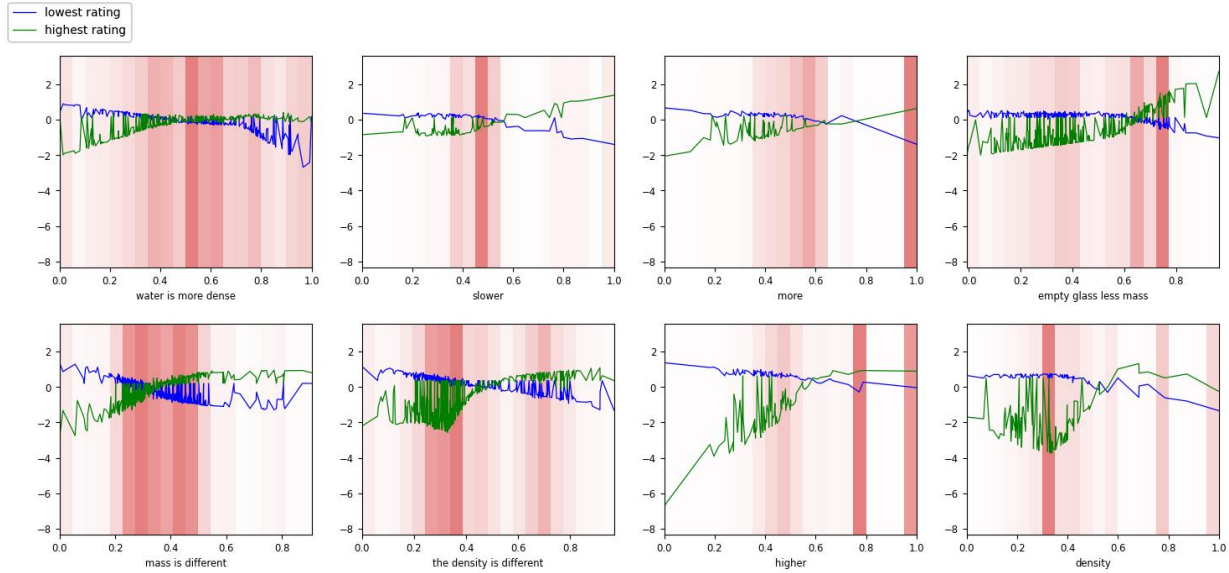
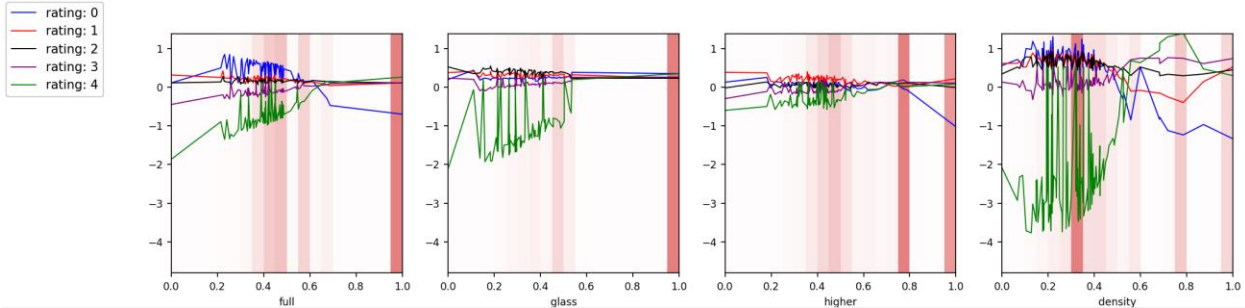


Figure 2.4
NAM Shape Functions (with all rating categories)



2.4.4 Discussion

The cross-validation averages in Table 2.3 give us an idea for how well each model performs for ASAG. We are not surprised to see that the NAM model provides more predictive power than the Logistic Regression, but not as much as the DeBERTa model. The DeBERTa model uses the entire student response as input text and learns many hundreds of features for its predictions, whereas with the NAM model, we are limited to the particular features we choose and it would be an unreasonable task to engineer the number of features as the DeBERTa model learns. Additionally, the results of the 5x2 cross validation paired one-tailed t-test in Table 2.2 give evidence that the NAM model outperforms the Logistic Regression model with the same features. Further, the Cohen's D effect size values confirm that the magnitude of difference in performance of the models is quite large. Not only is the difference statistically significant, but it seems to be practically meaningful.

Figure 2.2 allows us to visually interpret the NAM ASAG model through overall feature importance, and Figure 2.3 gives us more detail about how each feature contributes with NAM shape functions. With a framework like Knowledge Integration, grading can consist of identifying which ideas - correct or incorrect - are included in a student's response. Using a NAM model with features constructed from these KI ideas for ASAG can help stakeholders identify which ideas are most indicative of the student's grade. However, one downfall of using a NAM

model with the KI framework is that KI places importance on how ideas are related, but the NAM cannot provide an interpretation for the connectedness of students' ideas. With the NAM shape function, we get an idea of how the directionality of the model's prediction changes for different values of different rating classes. For example, for the *"higher"* feature, the probability for the highest rating class (shown by the green line) goes down significantly with decreasing cosine similarities. So, including the word *"higher"* in a response seems to correspond to higher scores. And, with the density shading in the NAM shape function plots, we can infer the density of responses that include the given word or phrase. For example, the words *"more"*, and *"higher"* both have dense regions near high cosine similarities, so we can infer that many responses include these words or similar words. This can give an educator insight into which ideas students are more or less likely to write about.

Limitations of using a NAM model for ASAG include that the performance of the model in terms of its match to human ratings is less than that of a LLM ASAG model. Also, engineering features for the NAM model is more costly than preparing student responses for a LLM ASAG model, as the model will create features from the text itself. Further, we only show results for one question, and a science question more specifically. We cannot conclude that our results would generalize to other short answer items, or other subject areas.

Our results suggest that the NAM model may be a sufficient and useful alternative to legacy explainable models, like a logistic regression. In our experiments, the NAM model performed better than the logistic regression with the same features, and the effect size of the difference was quite large. Further, the NAM model provides easily interpretable visualizations of the model's prediction functions which can give an educator insights about their students' understanding of KI ideas. In future work, we hope to experiment with using NAMs for ASAG with different question types from different domains, and perform qualitative interviews with educators to see if the NAM visualizations are understandable and helpful.

2.5 Chapter 2, Study 2: Training an RL Agent to Revise a Short Response

In this section, we explore how an Automatic Short Answer Grading (ASAG) model can be coupled with deep Reinforcement Learning (RL) to train an intelligent agent to revise a student's short response. We use batch (offline) learning to simulate an environment, such as an online tutoring system, where students are prompted to answer OE questions. A pre-trained ASAG model then provides a rating of the response, and an RL agent makes a series of revisions to the text until the response receives a high rating from the ASAG model. The agent is able to add key phrases to the response or delete a section of the response for its revisions.

The key idea of this work is to train an RL agent to choose the best sequence of revisions for a given OE response, in order to arrive at a more correct response in the least number of revisions. Although the space of student responses to a given question is infinite, we hypothesize that the agent can learn which of a finite set of revisions will be most useful to improve a particular student's answer. The main contribution of this work is to explore a real-world application of deep RL for an education technology. There has been minimal work so far studying the use of deep RL in educational contexts.

For this section, we focus specifically on the implementation of the deep RL algorithm and its ability to learn to revise a response. We expect that if the agent is able to learn how to consistently revise a student's response, the revision may serve as an explanation of why the student received the original rating from the ASAG model. More specifically, a student or educator might extrapolate from the agent's revisions which key ideas a student did not include

in their response (phrases that the agent chose to add), or which parts of their response were not fully correct (portions of the response that the agent chose to delete). It is noteworthy that, a current flaw in the RL agent’s ability to revise is that it is not able to directly connect or synthesize ideas through its revisions. It provides correct information through inserting key phrases, but its corrected responses are not exemplary according to frameworks like Knowledge Integration (KI; Linn et al., 2014) which emphasizes distinguishing among and integrating ideas that are either correct and incorrect.

Further, in Section 2.6, we use the RL agent to Audit an ASAG model by creating revision sequences that receive a high score from an ASAG model but would not be considered high scoring responses from a human grader. In Chapter 3, we will evaluate the agent’s ability to revise a response qualitatively by having nine subject matter experts rate the quality of a sample of revisions, and quantitatively by the average number of revisions the agent makes on a hold-out data set of responses not seen by the agent during training. Also in Chapter 3, we will expand on this work by using the agent’s revisions to provide formative feedback to students to encourage revision of their own short responses.

2.5.1 Background

In this section, we provide a brief overview of deep RL and the specific variants we use in this project, as well as the ASAG model.

2.5.1.1 Deep Reinforcement Learning

RL employs learning to control a dynamical system by providing feedback to an agent via a reward system, in order to choose an action to take in a given state. The action consists of what the agent can do in the state, and the state represents the current condition of the agent. When the agent takes an action, it receives a reward (either positive or negative) as feedback from the environment. Thus, the agent learns a policy to maximize its total expected sum of rewards. The system can be defined as a fully or partially-observed Markov decision process (MDP) (Sutton and Barto, 2018). Deep RL utilizes deep neural networks as the policy in conjunction with RL. The incorporation of deep networks has improved performance for applications of RL in many domains (Levine et al., 2020). With deep RL, the agent can make decisions from unstructured data without any manual engineering of the state space and the algorithms can handle very large amounts of data for learning an optimal policy.

2.5.1.1.1 Offline (batch) Learning

Traditionally, RL algorithms implement online learning by iteratively collecting experiences, while actively interacting with the environment. The collected experiences are used by the model to improve the policy (Sutton and Barto, 2018). For some applications, however, using online data collection can be resource heavy or dangerous or impractical for settings that necessitate a large dataset (Levine et al., 2020). In contrast, offline RL algorithms utilize previously collected data. Data can be stored in a replay buffer and is not altered during training, allowing for the incorporation of very large or pre-existing datasets.

For this project, our policy is trained using offline learning with a previously collected dataset of student responses with corresponding human ratings. We utilize the existing student responses to simulate an equivalent online-learning setting where a real-life student would be prompted to answer a short response item, and an agent would provide its revision to the student in real time while the agent is still learning how to choose an optimal revision sequence. Training

a policy with online data collection is often not feasible for applications of RL in education, as a trial-and-error learning approach by the agent could be harmful to students because the agent will make many “bad” or unuseful revisions as it learns. Instead of allowing students to interact with the agent while it is learning, we use the previously collected student responses to train the agent, and then deploy the agent for real use once it has learned to consistently choose an optimal sequence of revisions for student responses.

2.5.1.1.2 Proximal Policy Optimization

Proximal Policy Optimization (PPO) algorithms (Schulman et al., 2017) are a family of policy gradient methods for reinforcement learning distinguished by alternating between sample data through interaction with an environment (or simulation of interaction with an environment via offline sampling) and optimizing a surrogate objective function with stochastic gradient ascent. Previous policy gradient methods perform one gradient update per data sample, but PPO allows for multiple epochs of minibatch updates. PPO methods are simpler to implement and have better empirical sample complexity than a popular family of policy gradient algorithms called trust region policy optimization algorithms (TRPO). Further details of PPO algorithms can be found in Schulman et al. (2017).

2.5.1.2 The ASAG Model

The ASAG model that we use to provide ratings for student responses as well as feedback to the agent through a reward function is a BERT-base multi-class classification model. For more information about the BERT-base model, please see section 1.4.1.1 of this dissertation.

The model is fine-tuned as a supervised classifier using human ratings of the Soundwave KI Item from Figure 2.1 as ground-truth training ratings, just as in previous ASAG works such as (Condor, 2020). We trained our ASAG model with a standard train-test holdout split using 70% of the data for training, 15% for hyperparameter tuning and validation, and 15% to test the model’s ability to generalize to a set of data not used during training. The final model held-out test set achieved a Quadratic Weighted Kappa of 0.7919 and a ROC AUC value of 0.9185

2.5.2 The RL Formulation

In this section, we describe the offline RL formulation consisting of the state, action space, policy and reward for the agent. The agent’s training objective is to iteratively revise a response to achieve a high grade from the ASAG model in the least number of revisions.

2.5.2.1 The Algorithm

An overview of the proposed algorithm is as follows. For each episode, we will first initialize by randomly sampling one of the previously collected student short answers. The student answer text comprises the state representation. Based on this state input, the agent will choose an action to take via the current policy, i.e., either adding one of the phrases or deleting a portion of the response. Next, we send the newly revised student response to the ASAG model and receive a probability distribution of rating categories as output. Then we will calculate our reward as a function of the difference in class probabilities from the previous response, to the newly revised response, such that an increased probability of the highest rating category for the revised response would correspond to a higher reward. If the response has surpassed a predefined rating threshold, the episode will terminate. Otherwise, the process is repeated with the revised student response (the original response plus the added phrase, or minus the deleted

portion) as the new state, and the agent will choose another action until the ASAG rating threshold is achieved, or it reaches the maximum number of revisions which we predefined as eight. The algorithm is summarized with pseudocode in Table 2.4 and further details of the RL formulation follow.

Table 2.4

Pseudocode for the RL Algorithm

```

Repeat (for each episode):
    Randomly sample student response
    Get response rating from ASAG model
    Repeat (for each time step):
        Choose action and alter response
        Get rating of altered response
        Calculate reward
    Until max steps or rating threshold reached
  
```

2.5.2.2 The State

The state consists of the response that the agent is currently revising. For a given student's response, if the agent has already made a revision, the state would consist of the revised response (i.e., the original response plus the added text or without the deleted text).

2.5.2.3 The Action Space

The agent can choose one action per revision. Actions consists of the phrases that the agent may add and where it chooses to place the phrase within the response - either in the front (before the first word) or at the end (after the last word) of the current response. The agent may also choose to delete any 1/5th sized section of the existing response. Thus, the action space is size 25 (2 locations * 10 phrases) + 5 delete options. We only allow the agent to add phrases to two locations and limit the flexibility of its' delete actions to keep the action space at a reasonable size. An RL agent with a large action space will take much longer to learn, or may potentially be unable to learn its task. The maximum number of revisions (additions or deletions) the agent can make in one episode (to one original response) is eight. The 10 phrases that the agent can add were chosen from the KI Soundwave Scoring Rubric in Table 2.1. The phrases are as follows:

- 1) *"sound moves faster in air"*
- 2) *"sound moves slower in water"*
- 3) *"water is more dense"*
- 4) *"water has more mass"*
- 5) *"higher frequency in air"*
- 6) *"pitch is lower in water"*

- 7) *“pitch higher in empty glass”*
- 8) *“air is less dense”*
- 9) *“less vibration in water”*
- 10) *“the pitch is different”*

2.5.2.4 The Policy

We use a BERT-base model as the policy for the RL formulation. We chose to use a BERT model as the policy because of the model’s impressive ability to learn representations of text. The input to the BERT policy is the state, and output is a probability distribution of the action space. For more details about the BERT-base model, see section 1.4.1.1 of this dissertation.

2.5.2.5 The Reward

The reward is a function of the difference in the ASAG model’s classification probability distribution of ratings between the old response and the revised response, with a penalty for each revision the agent makes. We include the penalty because we want the agent to learn how to revise the student’s response in the least number of revision steps. The reward is essentially a scaled difference in the expected value of the old rating to the expected value of the new rating, minus 1 for the penalty. We scale the difference in expected values by multiplying by 3 because the overall difference tended to be small and we needed to send a stronger signal to the agent. The reward formula is as such:

$$3 * \left(\sum_{x=0}^4 x_{new} * p(x_{new}) - \sum_{x=0}^4 x_{old} * p(x_{old}) \right) - 1$$

Where x_{new} represents the newly revised response score from the ASAG model and x_{old} represents the previous response score from the ASAG model.

2.6 Chapter 2, Study 3: Auditing an ASAG Model with deep RL

To test the goodness of ASAG models, researchers typically follow the standard evaluation methods of machine learning (ML) researchers: using a hold-out or cross validation approach where a portion of the data is used for training and another portion used to test the model (for cross-validation, this process is repeated on various train-test splits of the data and results are aggregated). Results are presented with metrics like accuracy and Quadratic Weighted Kappa which signify a degree of agreement to the human ratings used to create the held-out test data. If an ASAG model shows concordance to human ratings, researchers generally say that the model performs well. However, we argue that even when training data are carefully constructed via meticulous grading from subject matter experts (SMEs), metrics which signify high agreement to an educator’s ratings do not provide sufficient evidence that an ASAG model’s scoring is robust.

The purpose of this work is to audit an ASAG model by finding unique responses that are granted high scores from a state-of-the-art ASAG model but would not receive a high score from an educator, according to a scoring rubric. We train a reinforcement learning (RL) agent via Proximal Policy Optimization (PPO) to revise a student’s response with the objective of obtaining a high score from an ASAG model in the least number of revisions. A revision consists of adding a phrase or deleting a portion of the student’s response. Because the RL agent has no knowledge of the scoring rubric and learns by trial-and-error over many thousands of revisions,

we hypothesize that some of the high-scoring responses it learns to create would not qualify as a high scoring response according to a scoring rubric created by SMEs. Analyzing such responses may allow us to peer into the "black box" of the NN by providing evidence of flaws in the ASAG model. Although a human could find ways to exploit an ASAG model similarly through trial and error probing, the advantage of using RL is that the agent can create many thousands of responses quickly through its revisions, and does so specifically to achieve a high rating.

We use a BERT-base multi-class classification model as our ASAG model which is a specific type of LLM (Devlin et al. 2018) that has been used extensively in recent ASAG research (Bonthu et al., 2021; Camus and Filighera 2020). The novelty of this research is not in training and evaluating the ASAG model, but in finding ways to probe a typically used ASAG model using RL. The novelty of this research is not in training and evaluating the ASAG model, but in finding ways to probe a typically used ASAG model using RL.

We analyze results from three different experimental settings, distinguished by which phrases the agent can add to a response. In all three experiments, the agent is able to delete a section of the student's response in the same way. To define the phrases that the agent is able to add to a response, we reference the scoring rubric that was created by subject matter experts for the original human ratings. In the first experiment, the agent is able to insert any of 10 key phrases which each represent an idea determined to be important for achieving a high score as specified in the rubric. In the second experiment, the agent can add any of 10 unhelpful phrases that should not increase the probability of achieving a high rating, but that include relevant terminology. In the third experiment, the agent can add any of 20 phrases including the 10 key phrases from the first experiment and the 10 unhelpful phrases from the second experiment.

In the first experiment, we set the agent up for success by giving it revision choices that are likely to increase a response's rating but we hypothesize it may sometimes use the phrases in unexpected ways. In the second experiment, we set the agent up for failure by only allowing it to make additions to responses with unhelpful phrases. If the ASAG model was flawless, the agent would not be able to increase a score by adding an unhelpful phrase but we hypothesize that it will nonetheless find ways to choose successful revision sequences that achieve a high score. In the third experiment, we hope that the agent will favor adding key phrases from experiment one over unhelpful phrases from experiment two, but we give the agent flexibility to choose either to investigate whether it may still choose unhelpful phrases in certain revision sequences.

2.6.1 The Experiments

We investigate how the RL agent finds ways to exploit the ASAG model with three different action spaces. We will refer to each as a separate experimental set up. In all three experiments, the agent can delete any 1/5th response length section as 5 of its actions. What differs between the experiments is the set of phrases the agent can choose to insert either before the first word of the response or after the last word of the response. In the first experiment, the agent may insert any of 10 key phrases into a response. These phrases were taken directly from the SME-created rubric as important mechanisms or conclusions that a student could include in their response to achieve a high rating. In the second experiment, we include 10 phrases that include words related to the subject matter, but that do not contain information that should lead to a higher score. We will refer to these as unhelpful phrases. In the third experiment, we allow the agent to insert any of 20 phrases including both the important phrases and unhelpful phrases. Below are the phrases for the first two experimental setups.

Experiment 1 - Helpful Phrases: *'sound moves faster in air', 'sound moves slower in*

water', 'water is more dense', 'water has more mass', 'higher frequency in air', 'pitch is lower in water', 'pitch higher in empty glass', 'air is less dense', 'less vibration in water', 'the pitch is different'

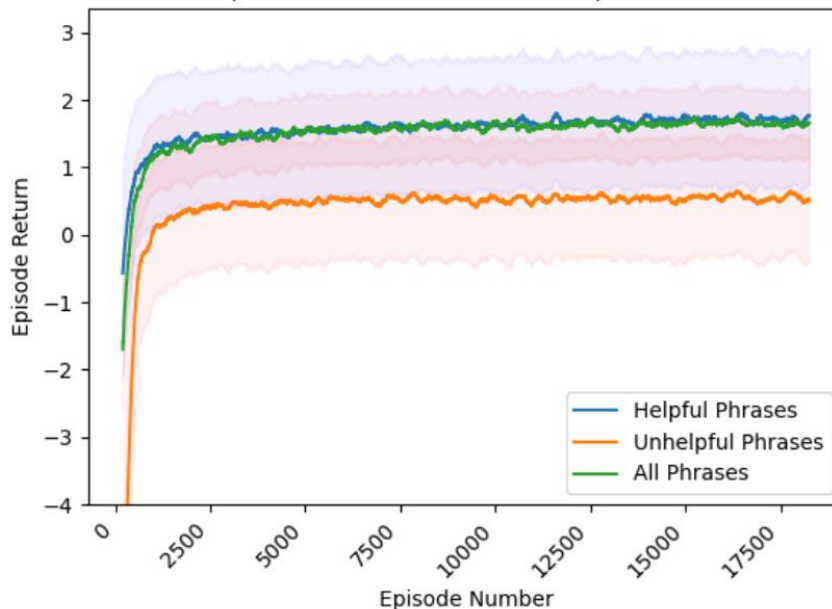
Experiment 2 - Unhelpful Phrases: *'I am not sure', 'tapping the glass', 'sound bounces in the glass', 'sound sinks in water', 'sound will echo in glass', 'the pitch is the same', 'water blocks the sound', 'frequency is height of wave', 'amplitude is number of waves', 'sound is more dense'*

2.6.2 Results

For each experiment, we train the agent 10 separate times for 75,000 time steps, differing only by the seed used for random initialization of the policy network. Each training run yields a slightly different number of episodes depending on how many revisions it takes the agent to arrive at the threshold high rating, so for consistency we analyze the first 18,250 episodes of each run. Visually, we can compare the mean episode return over time, smoothed with a rolling average of $n=200$, of the 10 runs across each experiment in Figure 2.5. We include 95% confidence bands for each smoothed mean as well. It is apparent that the training runs for Experiment 1 (Helpful Phrases) and Experiment 3 (All Phrases) consistently yield higher average episode returns than those for Experiment 2 (Unhelpful Phrases). We are not surprised by this result as less revisions correspond to a higher return, and we hypothesized that the agent should be able to achieve the high threshold rating with a lesser number of revisions using helpful phrases than with the unhelpful phrases. We did not necessarily expect the mean episode returns for Experiment 3 to be so close to that of Experiment 1, but it seems that the agent heavily favors the helpful phrases over the unhelpful phrases in Experiment 3.

Figure 2.5

A Comparison of Smoothed Mean Episode Returns

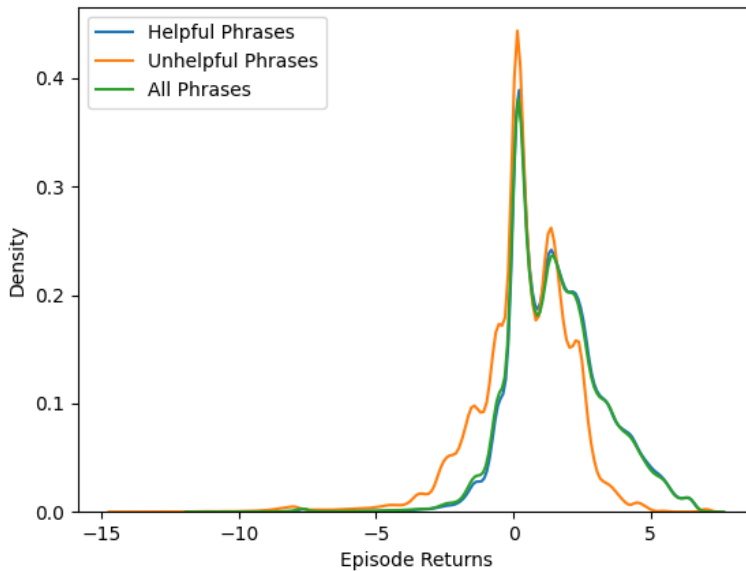


In Figure 2.6, we get a slightly different perspective of the differences in returns between the three experiments by comparing kernel density estimated distribution plots of all the episode returns over the 10 training runs, for each experiment. The density of the Experiment 2 returns

(Unhelpful Phrases) is shifted to the left from that of Experiments 1 and 3, which is, again, not surprising as Figure 2.5 showed that the mean returns for experiment 2 were consistently lower than those of experiments 1 and 3, although the confidence intervals overlapped significantly. In Figure 2.6, we also see significant overlap of the distributions, and the distribution for Experiment 1 (Helpful Phrases) is almost indistinguishable from that of Experiment 3 (All Phrases).

Figure 2.6

Kernel Density Estimated Distribution Plots for each Experiment



In the sections that follow, we provide results of an exploratory data analysis of the returns and actions sequences chosen for the top 5% of episode returns, for each of the three experiments. Essentially, we are investigating how the agent is able to construct a revision sequence that achieves the highest score from the ASAG model in a low number of revisions, for each experiment. Since the purpose of this study was to find ways that the RL agent can exploit the ASAG model, we specifically look for revised responses within the top 5% of returns that would not be considered a high scoring response from a SME, given the rubric. Further, since we did not have an exact hypothesis for how the agent would do so, our analysis is exploratory and not confirmatory. When showing examples of student responses and corresponding revisions in subsequent sections, we show added phrases in brackets, and deleted pieces of a response with a strike-through.

2.6.2.1 Experiment 1: Helpful Phrases

We analyze 9,125 total high scoring revision sequences that correspond to the top 5% of returns over all 10 training runs in Experiment 1. The mean number of actions necessary to achieve a high ASAG rating is 3.4 with the minimum being two and the maximum being five. Interestingly, the agent rarely chooses to delete a section of a response in the top scoring sequences. A delete action is present in only 21 sequences out of 9,125 (0.23%). The most compelling finding from the experiment 1 revision sequences is that the agent frequently chooses to add the same phrase multiple times within a revision sequence. In fact, the number of

sequences with at least one repeated phrase addition is 2,353 which is almost 26% of all 9,125 revision sequences we analyze. Further, in 161 revision sequences, the agent chooses to add the same phrase at least three times in a row (1.76% of the time). See two examples below of high scoring revision sequences with repeated phrase additions.

Example 1: "[water has more mass] in a pool its lower and slower a glass wouldnt be different [water has more mass] [water has more mass] [water has more mass]" (after 4 revisions)

Example 2: "because it has more water to slow it down [higher frequency in air] [water has more mass] [higher frequency in air]" (after 3 revisions)

2.6.2.2 Experiment 2: Unhelpful Phrases

We analyze 9,146 total high scoring revision sequences that correspond to the top 5% of returns over all 10 training runs in Experiment 2. The mean number of actions necessary to achieve a high ASAG rating is 3.12 with the minimum being two and the maximum being seven. Although the top 5% of scores for experiment two is lower than that of Experiment 1 (as is evident in Figure 1), we emphasize that the revised responses in the top 5% of experiment two are still able to achieve the high threshold rating from the ASAG model, using the supposedly unhelpful phrases.

The most interesting finding from Experiment 2 is that 84.3% of actions chosen, across all revision sequences, correspond to adding the phrase, "*sound is more dense*". Further, the phrase, "*frequency is height of wave*" is chosen 14.8% of the time, and the rest of the phrases are chosen less than 1% of the time. Similar to experiment 1, the agent rarely chooses to delete a section of a response in the top 5% of revision sequences. We only see a delete action chosen in 58 (0.63%) revisions. Even more pronounced than that of Experiment 1 is the repeating of the same action choice multiple times within a revision sequence. For experiment two, this happens in 7,916 total revision sequences (86.6%) and the majority of these cases consist of adding the "*sound is more dense*" phrase repeatedly. Two examples of these findings are shown below.

Example 1: "the waves travel farther making a higher pitch [sound is more dense] [sound is more dense]" (after 2 revisions)

Example 2: "[sound is more dense] the molecules vibrate faster through the liquid in the glass making the pitch sound higher [frequency is height of wave] [frequency is height of wave]" (after 2 revisions)

2.6.2.3 Experiment 3: All Phrases

We analyze 9,125 total high scoring revision sequences that correspond to the top 5% of returns over all 10 training runs in Experiment 3. The mean number of actions necessary to achieve a high ASAG rating is 3.5 with the minimum being two and the maximum being five. We find that 90% of all actions taken in the top 5% consist of adding 'Helpful Phrases' from Experiment 1, and only 10% consist of adding 'Unhelpful Phrases' from Experiment 2. As was the case in experiments 1 and 2, the agent rarely chooses to delete a portion of the response for its revision: 29 times or 0.09% of actions. Of the actions that the agent chooses from the 'Unhelpful Phrases', results are consistent to that of Experiment 2 in that by far, the most frequently chosen phrase to add was: "*sound is more dense*". This phrase was chosen about 95% of the time an unhelpful phrase was chosen. Further, consistent with experiments 1 and 2, we see the agent repeats the same phrase addition at least once within a sequence in 2172 (24%) of the 9,125 revisions analyzed.

Example 1: "[pitch is lower in water] i dont know [water has more mass] [water has more mass] [sound is more dense]" (After 4 revisions)

Example 2: "[pitch is lower in water] the water density will make it harder to hear [sound is more dense] [higher frequency in air]" (after 3 revisions)

Because the objective of this work is to audit an ASAG model, the results focus on cases where the RL agent finds undesirable ways to achieve a high rating like repeating the same phrase or adding unhelpful phrases. However, it is apparent that in many cases the agent makes revisions in line with what an educator would agree are helpful - i.e., adding phrases deemed as helpful by the question rubric.

2.6.3 Discussion

In all three experiments, it appears that the agent is able to increase ASAG response scores by adding the same phrase multiple times. This should certainly not be the case as repeating a phrase does not indicate an increase in understanding. For students attempting to "game the system", or find ways to achieve a high score without increasing their understanding of the subject matter, adding the same phrase repeatedly could suffice. We note that this may be a major flaw of typical ASAG models. Further, we note that in all experiments, the agent very rarely chose to delete a portion of a student's response. We hypothesize that this may indicate that shortening a response in any manner is not useful to increase the score of the response from the ASAG model. This may explain part of the reason the agent repeatedly adds the same phrase to achieve a high score - the length of the response itself may be an important factor that the ASAG model has learned to use when assigning a score. This is not ideal as the scoring rubric certainly did not indicate that a longer response should indicate a higher score.

There were a couple interesting takeaways from Experiment 2. The idea that the agent is able to improve a response's automatic grade with any of the unhelpful phrases is evidence that there are issues to be resolved with the ASAG model. We noticed that the agent disproportionately favored the phrase "sound is more dense" which is a typical misunderstanding of students, as specified in the scoring rubric. The ASAG model seems to be tricked by this phrase, as it includes relevant terminology like "more dense". If a student were to state that water is more dense, it would be reasonable to think that this would increase the rating from an ASAG model or a human rater. We arrive at similar conclusions when hypothesizing about the second highly added phrase from Experiment 2: "frequency is height of wave" is a common misunderstanding that was included in the scoring rubric. Terms like "frequency" and noting the height of a sound wave are important ideas to address when answering the question at hand, but the idea that a frequency consists of the height of a wave is simply incorrect.

Finally, we note that in the third experiment, although the agent highly favors the helpful phrases, it does choose the unhelpful phrases about 10% of the time. This provides evidence that not only was the agent choosing to include the misunderstanding, "sound is more dense", when it was only able to choose unhelpful phrases, but that it chooses this incorrect phrase over helpful ones in many cases.

There are many limitations with our attempt to audit an ASAG model. Firstly, our analysis is dependent on the action spaces we chose for the agent in each experiment. There may be a better or different assortment of unhelpful phrases that would have been useful to include in the action space for experiment two. Or, we could have included individual key words for the agent to add to a response to allow the agent a bit more flexibility in its choices. Further, our

analysis was specific to a ASAG BERT model and although this is a particularly popular option for recent ASAG research, many researchers are using different variants of LLMs like RoBERTa or Llama. Additionally, we analyze results from one specific question related to science understanding. An ASAG model may perform differently for other types of questions. Finally, as our analysis was exploratory, it was only reasonable to find general patterns in the agent’s revisions as opposed to examining each high scoring revision sequence. We may have missed an assortment of findings if we were able to look into each case, especially for Experiment 2.

We have found a few interesting flaws in a frequently used, state-of-the-art, ASAG model. For this work to be practically useful, we hope that these findings may be helpful for improving similar LLM ASAG models. In future work, we propose to use these undesirable examples to retrain the ASAG model. For instances where the agent added incorrect phrases, or inserted the same phrase multiple times, we can rate those responses as lower than their original, un-revised scores, and add the altered responses and ratings to the original data set to retrain the ASAG model. We think of this method as data-augmentation such that our training set is more robust. Thus, we would hypothesize that the ASAG model retrained on these examples may not reward responses with repeated phrases or non-useful phrases.

Further, our results provide evidence that state-of-the-art ASAG models may not be robust enough to use in high stakes scenarios. We have shown that there are still issues to be sorted out with such models and that by using unexplainable, black-box models, we are left to wonder which (potentially undesirable) attributes of a response were used to estimate a rating.

Chapter 3: AI-Assisted feedback for student revisions of short responses.

3.1 Abstract

In this chapter, we explore how an automatic short answer grading (ASAG) model can be coupled with deep reinforcement learning (RL) to create a system of AI-assisted feedback (AIF) that encourages engagement in a revision process for open-ended (OE) responses. We train an RL agent to revise a student's response by either adding a predefined key phrase or deleting an existing portion of the response such that the revised response achieves a high rating from an ASAG model in the least number of revisions, as described in more detail in Chapter 2. We examine the practical use of the revision agent by conducting a randomized controlled trial wherein undergraduate students respond to an OE item through an online tutoring platform. Before being encouraged to revise their original response, students are asked to critique the agent's attempt to improve their response in order to promote active engagement in the feedback process. We compare results from the RL agent's feedback with a similar intervention using a ChatGPT revision agent, and a control group that does not include a revision example. Student critiques are examined to measure students' perceptions of the feedback, and the efficacy of the feedback for short answer improvement is estimated by differencing the ASAG model's rating of their original response with that of their revised response. Further, we investigate if a student's prior knowledge plays a role in the comparison of interventions, and evaluate the ChatGPT feedback for accuracy.

3.2 Introduction

Writing out explanations can help students reflect on what they know, and how they know it (Fellows, 1994; Klein, 2015; Ray et al., 2016) as previous “writing-to-learn” literature shows that the engagement of cognitive processes like retrieval, reorganization, and elaboration through writing is associated with higher post-test scores and more generally that curricula which include more essay-like items can lead to better downstream performance (Arnold et al., 2017). Further, engaging in a revision process can prompt students to reconsider evidence they have to support their ideas and refine their original explanations (Rivard, 1994; Ryoo & Linn, 2014), bridging the gap between knowledge (familiarity and awareness) and understanding (comprehension and discernment) of course material (Mynlieff et al. 2014). By altering written content, students are more likely to remember concepts than by simply generating ideas (Langler & Applebee, 1987).

However, encouraging meaningful written revisions can be challenging. When students do have the opportunity to alter a written response, they often fix mechanical errors like spelling mistakes, make single word substitutions as opposed to word, phrase, sentence level additions (Crawford et al., 2008), or fail to change any part of their original response (Tansomboon et al., 2017). Shortcomings in student revisions may be due to a lack of trust in revision feedback (Zhu et al., 2017) or an insufficient understanding of the subject matter (Zhu et al., 2020) and related terminology (Jonsson & Panadero, 2018). Nevertheless, recent research has shown increased success in encouraging written revisions by prompting students to first engage with and reflect upon peer ideas (Matuk & Linn, 2023).

More generally, scholarship has shown that providing timely and useful feedback to students can lead to positive learning outcomes although this effect varies across different forms of feedback and learning types (Hattie, 2008; Hattie & Zierer, 2019; Shute, 2008; Wisniewski et al., 2020).. Hattie and Timperley (2007) define feedback as “information provided by an agent regarding aspects of one’s performance or understanding” where the term “agent” may broadly refer to a human or thing that takes an active role in producing a certain effect. When teachers act as the agent, providing immediate and personalized feedback to students is often infeasible, considering limited classroom resources and individual student differences (Gerard & Linn, 2022; Pardo et al., 2019; Tansomboon et al., 2017). However, the usefulness of feedback for learning gains has been shown for not only face-to-face guidance (Black & Williams, 1998; Jonsson, 2013) but also for computerized guidance (Cavalcanti et al., 2021; Gerard et al., 2015). When comparing the effect size of learning gains between human tutoring and computerized tutoring in online learning environments, the effectiveness of Automatic Feedback (AF) is almost equivalent to that from human tutors (Cavalcanti et al., 2021; VanLehn, 2011).

As outlined in chapters 1 and 2 of this dissertation, ASAG models can provide instant scores for OE responses but do not provide an explanation for why a student received a given score, and do not provide guidance for improvement. In this chapter, we move beyond assessment and correctness and center on building a system of AI-assisted Feedback (AIF) to promote revision of short OE responses, rooted in a constructivist theory of learning, with a focus on how students engage with, and learn from feedback - i.e. a “learner-centered focus” (Winstone et al., 2022). Broadly, constructivism states that knowledge is constructed by the learner, and builds on a foundation of prior knowledge and experience (Brunner, 1990; 1991). Constructivism emphasizes that the learner takes an active role in the process of constructing knowledge, modifying and reflecting upon their understandings in the face of new experiences (Bada & Olusegun, 2015). Moreover, student engagement in the context of learning has been

defined to include three to four dimensions including cognitive, behavioural, emotional, and social, where cognitive engagement broadly refers to students' self-regulatory approaches, behavioural engagement centers on students' effort and attention, emotional engagement focuses on students' feelings and perceptions, and social engagement incorporates interactions with peers (Finn & Zimmer, 2012; Fredricks et al., 2004, 2016). Our feedback was designed primarily to elicit deep cognitive engagement, but also promote multidimensional engagement, described in a later section of this paper.

We explore how an automatic grading model can be coupled with deep reinforcement learning (RL) to automatically improve a student's short OE response. After the RL agent described in Chapter 2 has learned to consistently revise a student's response in a simulated environment, as deemed sufficient by quantitative measures and qualitative evaluation from subject matter experts (SMEs), we deploy the agent in a real-world setting where students interact with an online tutoring system. To create guidance from the RL agent's revisions, we tell students that an AI bot is learning to improve their responses, and present their original response altered with the first of the agent's revisions. We promote active engagement, and elicit perceptions of the feedback by asking students to critique the AI bot's revision before prompting them to engage in their own revision based on what they learned from the bot.

We conduct a randomized controlled trial to collect data from undergraduate students in Florida schools and compare results from the RL agent's feedback with a similar intervention using ChatGPT, a popular language model chatbot (OpenAI, 2023), and a control group. Students in the ChatGPT group will receive a revision example, similar to that of the RL group, except the ChatGPT agent can add any natural language to the response whereas the RL agent can only choose among 10 predefined helpful phrases to construct the revision example. We compare the constrained RL agent to an unconstrained generative chatbot like ChatGPT to investigate the differences in using a generalized, generative AI language model versus a more task-specific trained RL agent that has no ability to produce unique, natural language responses. The RL agent has the disadvantage that it cannot generate natural language, and the ChatGPT agent has the disadvantage that it can add potentially incorrect phrases in its revisions. Students in the control group will not be provided with AIF, but will be shown a sentence that provides them with generic, but useful information regarding the question. The main disadvantage of the control group feedback is that it is not specific to a given student's response but instead contributes a static hint that may or may not help a certain student. Students will answer basic demographic questions and are given a short pre-test consisting of four multiple choice questions to assess prior knowledge. Our research questions for this study are as follows:

- 1) Is the RL AIF superior to AIF from ChatGPT and non AI feedback in encouraging students to improve their original response?
- 2) Does the feedback intervention effect vary for students with different prior knowledge?
- 3) How do students perceive and act on the feedback they received, and do perceptions differ between intervention groups?
- 4) Can ChatGPT generate correct and useful revisions, and correspondingly what issues arise with using feedback from an unconstrained, generative AI?

The user interface for data collection was built using Open Adaptive Tutor (OATutor), an open source tool for researchers to develop custom educational technologies and conduct educational research, developed at the Computational Approaches to Human Learning (CAHL) research lab at the University of California, Berkeley (Pardos et al., 2023; Anastasopoulos et al.,

2023). To train the ASAG model and the RL agent before deployment, we use a data set collected during a previous research project at the Berkeley Evaluation and Assessment Research (BEAR) research lab at UC Berkeley consisting of OE mathematics problem solving items designed for a formative assessment of high school students (Arneson et al., 2019). The items were created under the BEAR Assessment System (BAS) which outlines principles for assessment development (Wilson, 2023), further described in section 3.5.2. The OE item and pretest items used in our data collection come from this data set.

3.3 Related Work

In this section, we describe previous work related to actionable automated feedback, AI-assisted feedback, prior knowledge as a feedback moderator, and uses of ChatGPT for education.

3.3.1 Actionable Automated Feedback

Although the benefits of receiving feedback for learning gains are not consistent across all scenarios, researchers tend to share the view that feedback designs which promote students to actively engage as constructive contributors in the feedback process are more effective (Winstone et al., 2017). Yet, there is still a lack of literature to provide evidence of which pedagogical approaches best facilitate student engagement with feedback (Mao & Lee, 2023; Zhang & Hyland, 2022). Notable works that focus on how students can actively participate with feedback include Jonsson (2013) who provided a literature review specifically to investigate why some students do not use the feedback they receive. A few themes are evident in the existing literature such as a lack of useful and individualized feedback, that students sometimes lack strategies for productive use, and that students may not understand the particular academic terminology used in feedback. Winstone et al. (2017) conducted a systematic review of research pertaining to the issue of how learners actively engage with and implement feedback. They identified factors proposed to influence the likelihood of feedback being used, and outline components of interventions that support learners' engagement with feedback.

Iraj et al., (2020) examined students' engagement with feedback, and distinguished between an early notion of feedback which emphasized providing relevant information, and a new paradigm which focuses on actionability, and change in student behavior as a consequence of receiving feedback. Carless (2019) explored student uptake of feedback through a five year longitudinal study. He analyzed feedback loops of both short-term (i.e. those that elicit immediate actions) and long-term (i.e. those that encourage more complex, longer-term adjustments) nature, and proposes a model of long-term student engagement with feedback.

Adams et al. (2020) focused on an individual's self-efficacy as a predictor of effective engagement with feedback, and found that academic self-efficacy operates as mediator of the association between perceptions of feedback and academic outcomes. Zhang and Hyland (2022) explored how a group of students engaged with pedagogically different types of feedback including that from a peer, teacher, and automated system. They conclude that there is no single "best" approach to written feedback, and that integration of different types offers benefits like encouraging students to devote more time and effort into their writing and revising, reducing the wait for timely feedback, and requiring students to analyze, evaluate and monitor their writing revisions.

3.3.2 AI-assisted, and Automated Feedback

AF provides increased opportunity for quick feedback without the direct presence of an educator, although empirical support is inconsistent regarding the effects of feedback timing across different types of learning and variations of feedback (Fyfe et al., 2016; Mertens et al., 2020; Shute, 2008; Swart et al., 2019). Theoretical models of feedback like connectionism (Clariana, 2000) tend to favor immediate over delayed feedback for strengthening mental associations between a question stem and the correct response (Clariana, 1999), however, delayed feedback may only be helpful in very special situations. For example, the interference-perseveration hypothesis speculates that when a learner is presented immediate feedback showing the correct response after an error, the error response interferes with the correction of the response due to the immediacy of the feedback (Mory, 2004).

Recent advances in computation and machine learning have contributed to an increase in interest from education technology researchers to find improved methods for Artificial Intelligence Assisted Feedback (AIF). Deeva et al. (2021) provided an extensive review of 109 recently developed educational AF systems and emphasized that many of these tools were designed to tackle distinctly different educational tasks across various domains. They created a common reference framework to classify AF systems based on the educational setting for which the system is used, the type of feedback delivered, and approaches to design and evaluation. Lee et al. (2023) emphasized a responsible and humanistic design to the use of AI in Education for evaluating, and providing feedback for, student essays in a large online course as a surrogate for one-on-one instructor interactions. They explained that AI is not likely to take over the role of instructors for teaching and learning because the humanistic role is profoundly different from any current AI equivalent. Moons et al. (2022) introduced a semi-automated feedback approach where human feedback is reused for future AF on similar student responses and mistakes. They deployed their feedback system in an open-sourced learning management system (LMS) called Moodle, with 36 teachers, to explore its efficacy as well as how teachers used and perceived the system.

Works relating more specifically to AF for revision include Cang et al. (2023) which utilized a pre-existing software to provide automated feedback for students' short essays to promote engagement in revision. They categorized types of student revisions and found that most students do not use automated feedback for their revisions. Similarly, Lee et al. (2021) examined different types of AF for student engagement in revisions of science explanations and found that about half of students participated in revision. Further, they found that scores for students' post-AF revised explanations were significantly higher than their original responses. Matuk and Linn (2023) explored the extent to which students revised their explanations after automatically demonstrating similar or diverging student ideas. Students were prompted to reflect upon student ideas, and found that most students engaged in substantial revisions, and increased their subject-matter understanding measured by post-test scores regardless of receiving either similar or divergent peer ideas as AF.

3.3.3 Prior Knowledge and Feedback

Although many factors may moderate the effect of different types of feedback on learning gains (Wisniewski et al., 2020), one moderator frequently examined is prior knowledge, as learning from feedback, from the constructivist's point of view, can be viewed as an interaction between information stored in long-term memory (prior knowledge) and new information from the feedback (Mory, 2004; Narciss & Huth, 2004). Learning something new is predicated on what one already knows, and prior knowledge has an important influence on

learning by reducing cognitive load (van Riesen et al., 2019), and by interacting with self-regulation (Yang et al., 2018). Studies have generally found that the effectiveness of feedback is more pronounced for students with lower prior knowledge (Fyfe et al., 2012; Nussbaumer et al., 2014; Razzaq et al., 2020), with some providing causal evidence for the moderating role of prior knowledge (Fyfe et al., 2016), and showing that prior knowledge can predict the effectiveness of learning from feedback (Fyfe et al., 2012). However, it may be the case that under certain conditions, a moderate to high level of prior knowledge hinders learning from feedback (Fyfe et al., 2012).

3.3.4 ChatGPT for Education

Since the release of ChatGPT in November of 2022, there has been much attention given to its potential applications in Education, as is evident from the substantial body of literature that exists only one year later relating to ChatGPT and Education. Lo (2023) conducted a rapid literature review to understand ChatGPT's capabilities for education and issues raised by researchers. Their findings suggest that ChatGPT has the potential to serve as an assistant for instructors and virtual tutor for students, although issues around the bot generating fake information and bypassing plagiarism detectors must be addressed. Adiguzel et al. (2023) provided guidance on how educators can successfully incorporate AI technologies like ChatGPT in their teaching practices but gave caution about the ethical and practical challenges. They commented on the potential for personalized learning, and feedback applications, as well as the opportunity for improved learning outcomes, productivity and student engagement. Kasneci et al., (2023) noted the opportunity for remote learners to use ChatGPT for real-time, personalized feedback, and for teachers to create personalized learning experiences through AI from ChatGPT. They cautioned that LLMs can amplify existing biases in society, and that relying too heavily on ChatGPT without continuous human review and validation of its responses may negatively impact learning. Grassini (2023) described the rapidly changing landscape of educational practices due to advances in Artificial Intelligence, and the mixed emotions of educators around technologies like ChatGPT. They suggested mitigation strategies in response to concerns over the use of ChatGPT for education, for example to decrease opportunities for cheating, educators may need to use more sophisticated plagiarism detection tools and reassess assessment methods to ensure fairness.

Thomas et al. (2023) generated short responses and corresponding ratings using ChatGPT as a method to create training data for an ASAG model. They compare autograding results using the generated training set with that of real student responses and human ratings, and find that the ASAG models trained on human responses outperform those trained with ChatGPT generated data. Aleedy et al. (2022) created a taxonomy of chatbot design approaches for educational settings. They describe challenges relating to building chatbots like ChatGPT in languages other than English, and more specifically the linguistic complexities of languages like Arabic with gender-specific verbs, adjectives and pronouns. Bevilacqua et al. (2023) evaluated differences in how LLM-based scoring models assess human generated content versus ChatGPT generated content. Interestingly, they find that LLM scoring models classify ChatGPT generated content at 10-15% higher accuracies than human authored text. They hypothesized that this difference may be due to an overlap in pre-training data for ChatGPT and the LLM. Researchers have also examined the uses of ChatGPT to create conversational tutoring systems (Ahmed et al., 2023; Nye et al., 2023; Schmucker et al., 2023), automatically grade various types of questions (Chang & Ginter, 2024; Kooli & Yusuf, 2024; Latif & Zhai, 2024; Lee et al., 2024), generate educational

content (Koraishi, 2023; Lee et al., 2023), and provide automatic feedback (Dai et al., 2023; Jacobsen & Weber, 2023; Jukiewicz, 2024). Most relevant to our study, Steiss et al., (2024) examined the use of ChatGPT to provide automatic formative feedback for student writing. They found that well trained humans provided higher quality feedback than ChatGPT, but emphasized the tradeoffs between the two as using ChatGPT saves a considerable amount of time.

3.4 Feedback Design

AF without careful design is often not sufficient for positive learning outcomes (Hattie & Timperley, 2007; Winstone et al., 2017), so a challenge for this project was to design a feedback mechanism using the RL agent's improved response that would encourage students to participate in a meaningful revision process. Many existing AF mechanisms simply provide information without considering how students will use the information to produce actionable changes (Boud and Molloy, 2013; Malecka et al., 2020), and students are often dissatisfied with the feedback they receive when it includes encouraging comments without actionable advice (Nicol, 2010). The "feedback gap" is defined as the difference between the potential and actual use of feedback (Dawson et al., 2018) and there has been a recent increase in researchers seeking ways to narrow this gap by exploring how students receive and interpret feedback (Carless, 2019; Jonsson, 2013; Price et al., 2010; Winstone et al., 2017).

We developed our feedback design with the goal of presenting actionable, suggestive, task-adaptive advice to support and afford analysis of all four dimensions of engagement. Actionable and suggestive advice indicates that we provide a hint that describes how to change or improve, and encourages an action as opposed to corrective feedback which presents a simple "right" or "wrong" response (Deeva et al., 2021). Additionally, we aimed to provide action-based, task-adaptive feedback constrained with material specifically curated from SMEs, where action-based signifies immediate feedback based on a student's action as opposed to a student requesting feedback, and task-adaptive corresponds to feedback that is custom to student's answer or action (Deeva et al., 2021).

To design actionable feedback with an opportunity for deep cognitive engagement, we borrow from the ICAP theory of cognitive engagement (Chi et al., 2018) which establishes a hierarchy of cognitive engagement modes that correspond to observable task-driven behaviors. Activities that fall under the "interactive" mode of cognitive engagement elicit the deepest category of knowledge change, and consist of collaborative interactions where participants contribute ideas beyond what was originally presented and engage with others' contributions. The interactive mode is presented as a cooperation between humans, but we expand upon their definition to include an interaction between a human and an AI. Further, prior research has shown that modeling the revision process by providing an example of a similar, hypothetical classmate's response may help students understand what is expected in the revision process, whilst not giving them the exact changes needed for their own response (Gerard et al., 2022). Thus we present the RL agent's revised version of the students original response as an attempt from an "AI bot in training", inviting students to view the AI as a peer. We give students agency by asking for their help in improving the bot, so they must reflect upon the "bot's" changes to their own response in order to provide their critiques, as prior literature shows success in encouraging written revisions by prompting students to first engage with and reflect upon peer ideas (Matuk & Linn, 2023).

The RL agent was designed to contribute an idea beyond what was included in the original student's response thereby providing subtle corrective advice, and the subsequent

student reflection provides an opportunity for the student to expand upon their original conceptions, eliciting a collaborative and interactive activity within the ICAP theory and drawing upon not only cognitive engagement but also social engagement. By collecting student critiques, and further asking for a revision, we can analyze student's post-feedback behaviors and perceptions, providing a lens with which to view the emotional and behavioral engagement of students.

We consider the role of the educator or SME as an important party in the feedback process, even if feedback is fully automated, such that an effective AIF system should be “open”, or customizable to an educator's preference (Shum et al., 2023). This is given from the design of the RL agent's action space which consists of adding phrases deemed important by the SME-created scoring rubric, or in practice, by an SME or educator themselves. In our design, we include 10 phrases but an educator could certainly choose more or less, or different phrases on which the agent would be trained. The same feedback mechanism could be applied to various OE items from different subject matters by design, although we do not have evidence to substantiate the effectiveness of our mechanism generalized to other items or subjects.

We present feedback to students as follows. In the RL and GPT groups, students write their response to the OE hiking item, and upon pressing submit, they immediately receive the following message with the machine-revised response:

“We're training an Artificial Intelligence (AI) bot to change your response so that it may be more correct. Look at how the AI bot revised your response below, then proceed to the next question.”

[***MACHINE REVISED RESPONSE HERE***]

Then, we ask for students' help with training the AI bot by providing a critique of its revision. We prompt students as follows:

“We need your help! The bot is still in training so it doesn't write in perfect English, and sometimes messes up. Write a couple sentences to critique the bot's changes: what did the bot add or remove, and why does this make your response better or worse?”

Students in the control group receive immediate feedback as well, but in a different form. They are not seeing a revised version of their own response, so their feedback does not come from an “AI bot” like that of the RL and GPT groups. We present a static hint and similarly ask students to elaborate on whether they find the information useful to improve their response as shown:

“Consider how the slope of each line can give us a hint as to whether Angel or Robin reached their respective destinations faster. If you could re-do your response to the previous question, would you use this information? Why or why not?”

3.5 Methods

In this section, we describe ChatGPT, the data set and student model, as well as the ChatGPT prompt. Details about how the RL agent was trained can be found in Chapter 2, section 2.5 of this dissertation, and details about the BERT ASAG model can be found in Chapter 2, section 2.5.1.2.

3.5.1 ChatGPT

ChatGPT is an AI-based, online chatbot, launched in November 2022 by OpenAI (OpenAI, 2022). It has the capacity to generate cohesive, human-like responses to a user's prompt and since its release, researchers have sought to explore the model's capabilities and limitations for various applications in healthcare, finance, education and many others.

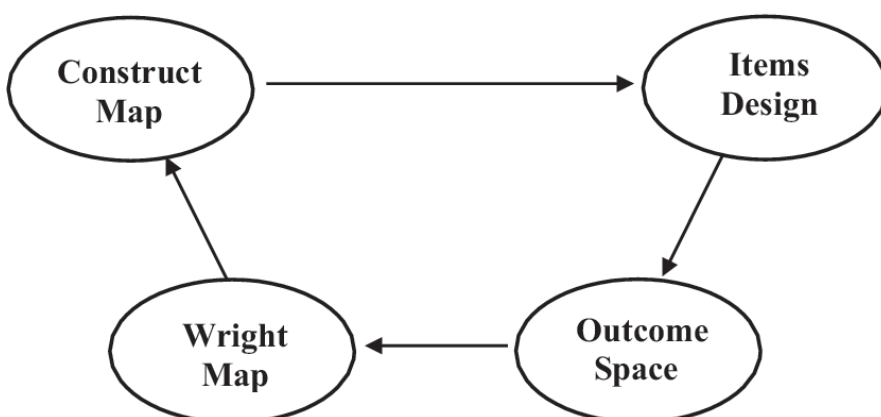
ChatGPT was built from OpenAI’s Generative Pre-trained Transformer 3 (GPT3; Brown et al., 2020), an autoregressive language model with a transformer architecture (Vaswani et al., 2017) and 175 billion parameters - 10 times more than any previous LLM. GPT3 has achieved impressive performance on various NLP tasks including translation, question-answering and reading comprehension. ChatGPT was trained using Reinforcement Learning from Human Feedback (RLHF; Christiano et al., 2017) using similar methods to that used to train InstructGPT in Ouyang et al. (2022). RLHF promotes the inclusion of human preferences in the model’s output by first having human labelers compare and rank multiple versions of output from a single prompt. A reward model is then trained using the human preference rankings, and further used to optimize the LLM using proximal policy optimization (PPO; Schulman et al., 2017). We use ChatGPT version 3.5 for this project.

3.5.2 The Data Set and Student Model

We use data collected at the Berkeley Evaluation and Assessment Research (BEAR) center, using the BEAR Assessment System (BAS; Wilson & Sloane, 2000). BAS outlines four “building blocks” that can be considered an application of a construct modeling approach to measurement (Wilson, 2023). Construct Modeling is an example of an evidence-centered design for measurement (Mislevy et al., 2003) such that it serves as a guide to collect evidence to support the inferences we wish to draw - i.e. reasoning from evidence. The four building blocks, shown in Figure 3.1, constitute steps in an iterative assessment development process including (1) the construct map, which embodies our theoretical model of student’s cognition such as an understanding of an academic topic, (2) the items design, which includes the way in which we observe a real-world manifestation of the theoretical construct, (3) the outcome space, which consists of the way responses to items from (2) are mapped to the construct map, and (4) the Wright Map, which involves the statistical model that transforms the item scores into measures that enable us to compare results between different respondents and occasions in relation to the construct from (1).

Figure 3.1

The Four Building Blocks in the BEAR Assessment System (BAS)



The item we use comes from an assessment of Critical Reasoning for College Readiness (CR4CR) that included items from multiple related constructs under a meta-construct called Problem Solving using Mathematics (PSM). The item we use was designed to measure a sub-construct of PSM called Interpreting Mathematical Results (IMR) that examines a student’s ability to critically evaluate the appropriateness of results of mathematical work by connecting

results to real-world scenarios. We present the construct map for IMR in Table 3.1. A construct map provides the “form” of a construct, and outlines a roadmap of change along the construct. In our cases, the construct map outlines a model student’s proficiency in IMR, from the lowest conceptualized level of proficiency to the highest level. This roadmap consists of qualitatively described milestones called levels that range from the lowest conceptualized level of proficiency to the highest level, which allow us to easily interpret a student’s degree of expertise. As a student learns and increases their skills relating to IMR, their location along the construct map changes. See Table 3.1 for more details about the IMR construct map.

Table 3.1
The IMR Construct Map

Level	Construct Level Description
4 - Step beyond	Extends a correct and complete interpretation which involves a claim, evidence, and a warrant, by coming up with an alternative interpretation or critiquing assumption(s) built into the interpretation.
3 - Interpretation	Makes a correct and complete interpretation, integrating (a) a claim (what the result means in the problem space), (b) evidence from the result (e.g., a graph, a table, or a mathematical operation), and (c) a warrant that logically connects the claim and evidence.
2 - Observation	Makes a correct but incomplete interpretation, (a) using TWO of the three elements: claim, evidence, and warrant. OR (b) three elements are present but they are vague / only partially correct.
1 - Partial observation	Makes a correct statement only about ONE of the three elements: claim, evidence, and warrant.
0 - No observation	Makes an incorrect claim/ interpretation.

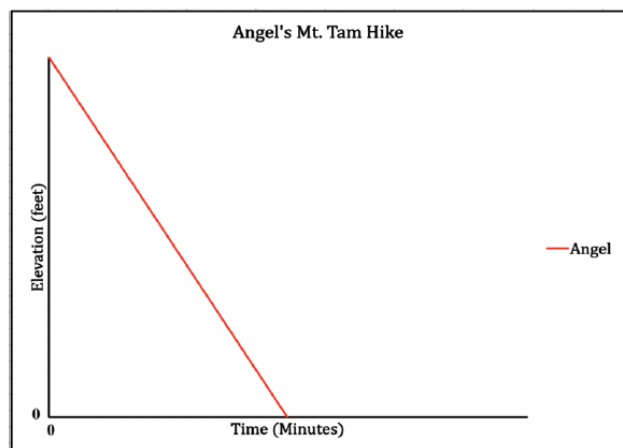
Note. The IMR construct asks: “how critically does the student interpret the results of mathematical work (e.g., graphs), by connecting the results with relevant aspects of the real-world problem space?”

We use items from an item bundle, or a grouping of items that are related by a unifying prompt or context, for our data collection. In the survey, we include a short pretest consisting of four multiple choice items in order to assess prior knowledge, and an OE item that serves as the AIF and revision item. For the pretest items, students are presented with the prompt and image shown in Figure 3.2. The prompt describes a person’s hike, at a constant speed, from the top of a mountain to the bottom, and the graph gives a visual representation of the hike with the y-axis representing the elevation, and the x-axis showing the time in minutes. Students are asked to identify the “real-life” component of Angel’s hike represented by each symbol in the linear equation $y = mx + b$. The options presented for each four symbols include (a) Angel’s change in elevation in one minute, (b) Starting elevation in one minute, c) Number of minutes, (d) Angel’s elevation after a given number of minutes, and (e) Total distance of Angel’s hike.

Figure 3.2
The Context and Graph for the Pretest Questions

Angel is hiking down to the bottom of Mt. Tam, a popular hiking destination with an elevation of 2000 feet (ft). She hikes at a constant rate of speed on her way down. The graph below depicts Angel's hike (as a red line) from the top of Mt. Tam, 2000 ft. back down to the bottom.

Note: the intervals on the graph have been intentionally left blank.



Each of the symbols in the linear equation $y=mx+b$ represents a "real-life" component of Angel's 2000 ft hike to the bottom of Mt. Tam.

The OE item builds upon the pretest items with a similar context and prompt, but introduces a second person hiking upwards on the same mountain, from the bottom to the top also at a constant rate of speed. The graph shows the original line that represents the downward hike, and now also shows a second line that represents an upward hike. Students are first prompted to decide who made it to their destination faster with a multiple choice question, and are then asked to explain which features of the graph led them towards their selection. Figure 3.3 shows the OE item context and graph.

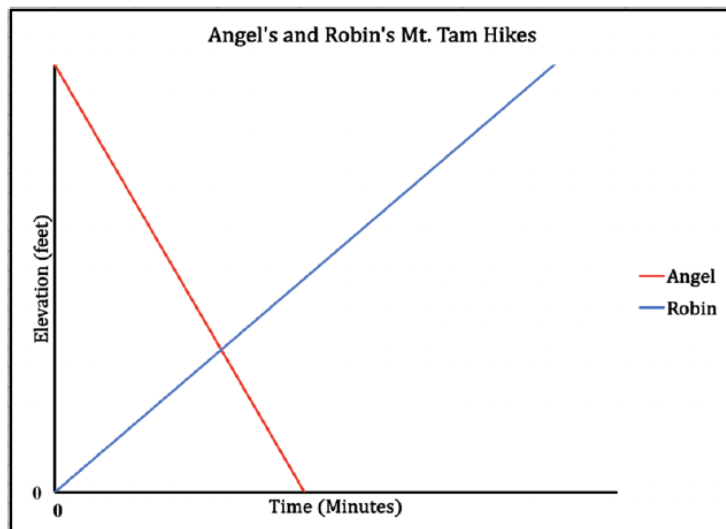
Figure 3.3

The Context and Graph for the OE Item

At the same time Angel begins her hike to the bottom of Mt. Tam, Robin begins her hike to the top, on the same road. Robin also hikes at a constant rate of speed.

The red line in the graph above shows Angel's downhill hike while the blue line shows Robin's uphill hike.

Note: the intervals on the graph have been intentionally left blank.



The scoring guide that maps students' responses to the construct map, representing their proficiency with the IMR construct for the OE item, is included in Table 3.2. The scoring guide is similar to a rubric, but specifically provides a manifestation of the outcome space from the BAS. In Table 3.2, we can see that the scoring guide provides a description of how the student response relates to each level of the IMR construct map, and examples of student responses or pieces of student responses that would fall under each level.

Table 3.2

Scoring Guide for the OE Item

Description	Example of Student Response
IMR4 [a] Selects (a) (Angel) = CLAIM [b] Provides BOTH (i) a comparative statement about the steepness of the line / slope / rate of change (as EVIDENCE), AND (ii) time / x-axis / speed (as WARRANT that links the claim and evidence).	[a] a. Angel, AND [b] The slope for Angel is more vertical compared to Robin (slope), which suggests that Angel completed her journey faster than Robin (time/speed) [b] Angel's graph is steeper (slope) meaning that every minute, she walked more distance, getting the destination faster (time/speed) [b] Angels graph from beginning to end (point A to B) has a faster rate of change (slope/rate of change), therefore making his line reach the bottom of the graph quicker (time/speed)
IMR3 [a] Selects (a) (Angel) = CLAIM	[a] a. Angel, AND [b] The slope of the graph was steeper (slope only)

[b] Provides EITHER (i) a comparative statement about the steepness of the line / slope / rate of change OR (ii) time / x-axis / speed (as EVIDENCE), BUT NOT BOTH. OR [b] Mentions BOTH the slope AND the time/x-axis, BUT very vaguely, for example using colloquial/everyday language	[b] Angel's line reaches 0 faster than Robin' reached 2000 (time only) [b] Angel's graph shows she took less time than Robin (time only) [b] Angel's time at the end of her line is less than that of robins (time only) [b] Angel's line ends close to the origin while Robin's line is far from it. (x-axis only) [b] Angel declines much faster than Robin inclines (vague slope & vague speed) [b] Angel's decline is a shorter line, making it a smaller time period (vague slope + time; if states 'shorter line' only, it's IMR1)
IMR2 [a] Selects (a) (Angel) = CLAIM, AND [b] Mentions EITHER (i) steepness of the line / slope / rate of change OR (ii) time / x-axis / speed (as EVIDENCE), BUT NOT BOTH & WITHOUT a comparative statement	[a] a. Angel, AND [b] slope [b] x-axis [b] How long each line was in relation to the x-axis which shows time
IMR1 [a] Selects (a) (Angel) = CLAIM [b] Provides a context-relevant response showing engagement with the problem, BUT it is largely incorrect or vague (requiring multiple inferential leaps to make sense of the response.	[a] a. Angel, AND [b] Angel stops first [too vague; if referred to the x-axis, it would be IMR2] [b] The red line for Angle is shorter [too vague] [b] It is consistently going up [too vague; what is going up?] [b] Because Angle is going down as opposed to going up [using prior / everyday knowledge]
IMR0 [a] Selects (a) (Angel) = CLAIM [b] Provides a context-IRRELEVANT (off-topic) response showing NO engagement or attempt, or no response. OR [a] Selects (b) (Robin), or (c) (It took the same amount of time) = INCORRECT CLAIM [b] It doesn't matter what the student writes, since they made an incorrect claim in [a].	[a] a. Angel [b] ??? [a] a. Angel [b] I don't know. [a] b. Robin [b] Cuz she is faster. [a] c. Same [b] They traveled the same distance.

As mentioned, the RL agent is trained as described in Chapter 2, with the only change being that we alter the 10 phrases included in the action space, or those that the agent can add to a response in order to accommodate this specific OE item. The 10 phrases were chosen from the scoring rubric and are shown below.

“a steeper line”, “the rate of change”, “line reaches zero faster”, “more feet per minute”, “Angel took less time”, “the line is shorter”, “time on x axis”, “a

greater slope”, “slope of the line”, “Angel was faster”, “Angels graph is steeper.”

3.5.3 ChatGPT Prompt Design

Our aim was to construct a prompt that would elicit responses that meet two criteria. One being that ChatGPT’s changes to student responses would be subtle - five words or less - just like the RL agent’s, and the second criteria being that the revised student’s response would still be recognizable to the student as their own. We set these criteria because we want the ChatGPT intervention to be as close to the RL intervention as possible for a more equal comparison, and we want students to believe that the AI agent changed their own response, not gave them a different and more correct response. Ekin et al. (2023) emphasized the importance of prompt engineering for maximizing the effectiveness of the language model in terms of bridging the gap between the user’s intent, and the model’s understanding. They outlined techniques for effective prompt engineering including using clear and specific instructions and using explicit constraints, and performing iterative testing and refining. We follow their advice in constructing our revision prompt. One challenge with creating the prompt was that the question context is mainly interpretable via the graph (shown in Figure 3.3), but ChatGPT seems to understand the question well with a textual description of the graph. The prompt to ChatGPT is shown below:

I am presented with a graph of two lines with elevation on the y-axis and time on the x-axis. One line is longer, blue, has a positive slope of 1 and represents Robin's uphill hike. The other line is shorter, red, has a slope of -3 and represents Angel's downhill hike. My response to who made it to the destination faster and why is:

"[STUDENT'S RESPONSE HERE]"

Please change no more than five words of my response to make it more correct. Respond only with your updated response.

An example of a student's response and the corresponding ChatGPT revised response is shown below.

Student’s response: *"Angel's line ends first, indicating that he arrived earlier."*

ChatGPT revised response: *"Angel's line ended first, indicating that he arrived earlier than Robin."*

3.6 Evaluation of the RL Agent and Machine Revisions

We evaluate the quality of the RL agent to ensure it is suitable to be used with real-life students for our experiment, and evaluate machine-revisions from both the RL agent and the ChatGPT agent. First, we quantitatively evaluate the agent in terms of its ability to learn the given task: make iterative revisions to a student’s response to achieve a high score from the ASAG model in the least number of revisions. We do so by viewing a plot of training rewards over time and by reviewing summary statistics of the number of revisions needed to achieve a high ASAG rating. We also qualitatively evaluate the agent by surveying mathematics SMEs to rate the quality of the agent’s revisions. We include ChatGPT revisions in the SME survey, and compare survey results between the RL revisions and ChatGPT revisions.

3.6.1 Quantitative Evaluation

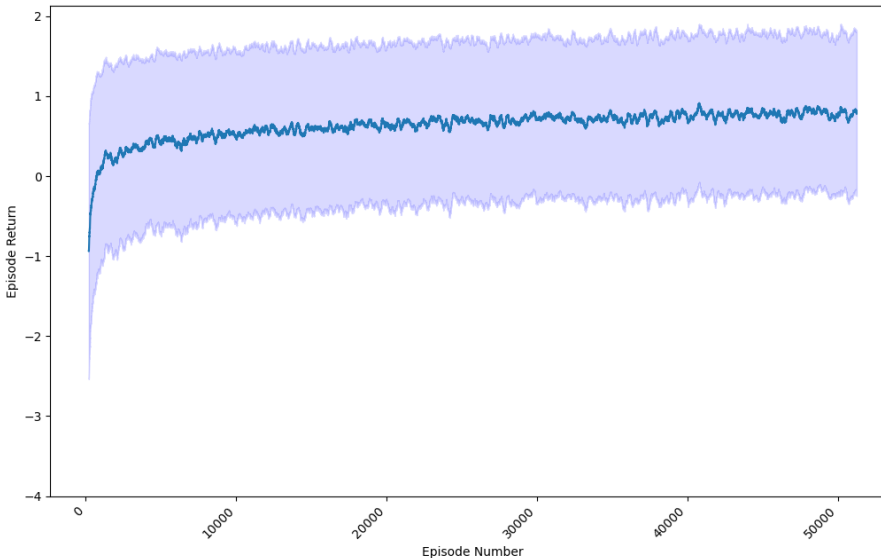
We consider that the revision agent has learned to sufficiently increase a student's response score - i.e. location along the construct map - from the ASAG model if the training episode returns converge to a value that corresponds to an average of 4 revisions or less. The ASAG model gives ratings from 0 to 3 where 0 is the lowest level of the construct and 3 is the

highest, and we deem an expected score of 2.7 as the cutoff value for the agent’s revision sequence to terminate. So, on average, the agent must be able to achieve an expected response score of 2.7 in 4 revisions or less. Noteworthy, we restrain the agent to only make 10 total revisions per episode.

3.6.1.1 Quantitative Evaluation Results

First, we visualize the agent’s mean episode returns in Figure 3.4 with a smoothed rolling average of size 200 over 51,250 episodes for 10 training runs, differing only in the random seed used to initialize the policy network. We include 95% confidence bands around the mean of the 10 trends as well. A higher episode return generally corresponds to less revisions per episode, but is calculated as the average reward per episode. The reward calculation is presented in section 2.5.2.5 of Chapter 2. From Figure 3.4, we see that the agent learns very quickly within the first 5000 episodes, and then average returns do not increase much over the remaining training runs.

Figure 3.4
Smoothed ($n=200$) Mean Episode Returns



We combine the last 1000 episodes from each of the 10 training runs, since the returns appear to have converged and calculate summary statistics on the number of actions taken, shown in Table 3.3. We see that the mean number of revisions taken to achieve the cutoff expected rating of 2.7 is 2.296 so we consider our revision agent sufficiently trained. Further, we note that even at the 75th percentile of revision sequences in the last 1000 training episodes the agent is only making 3 revisions.

Table 3.3
Summary Statistics for Number of Revisions

n	mean	std	min	25%	50%	75%	max
10,000	2.296	1.372	1.000	1.000	2.000	3.000	8.000

Note. The last 1000 training episodes for each of the 10 randomly initialized training runs are used to calculate the summary statistics.

When we utilize the revision agent in practice, we only show students a revised response with one revision - specifically the agent's first revision. So, we also examine the change in expected score, which maps to levels of the construct map, for the last 1000 revisions for the 10 training runs for only the first revision. We score the original student responses with the ASAG model, and the agent's first revised response, and subtract the expected score of the original response from the expected score of the revised response. Results are shown in Table 3.4. We see that on average, the revision increases about a half of a construct level.

Table 3.4

Summary Statistics for Construct Level Increase

n	mean	std	min	25%	50%	75%	max
10,000	0.557	0.0562	-1.835	0.035	0.469	0.092	2.60

3.6.2 Qualitative Evaluation

Because we intend for the algorithm to be student-facing, we must consider how an educator would view the agent's revisions. We recruited seven educators and SMEs to each evaluate a sample of the agent's revised responses, as well as the ChatGPT revised responses. We define a SME in this context to be a person who is qualified to teach, or has taught, high school level mathematics or higher. All reviewers will be presented with the OE item to get an idea for the question context and subject matter. The reviewers will be given an overview of the way the feedback is presented to students including the prompt to critique the "AI Bot's" revisions, and the prompt asking students to revise their original response. Each SME is then shown eight examples of real student responses, and the corresponding machine revisions - four from the RL agent and four from ChatGPT. They are not explicitly told whether each machine revision is from the RL agent or from ChatGPT, although they are aware that they will see examples of both. SMEs are asked to indicate their level of agreement to three statements on a seven-point Likert scale (Likert, 1932) for each response revision example. A Likert scale is an ordinal scale that allows respondents to indicate the degree to which they agree or disagree with statements. The 7 point Likert scale options include "*Strongly Agree*", "*Agree*", "*Somewhat Agree*", "*Neither Agree Nor Disagree*", "*Somewhat Disagree*", "*Disagree*", and "*Strongly Disagree*". A downfall of the use of an ordinal scale is that the difference between levels is not necessarily equidistant or measurable. The three statements are:

- 1) "*The machine-revised response is recognizable as the original student's response.*"
- 2) "*The machine-revised response includes a concept that was not present in the original student response.*"
- 3) "*The machine-revised response provides a clue about how the original response can be improved.*"

Because the feedback to students was designed such that students are asked to critique a revised version of their own response, one of our criteria for a successful revision was that the revision should not alter the original response to the extent that it is unrecognizable to the student as their own. With this idea in mind, the RL agent's revision options were designed such that the

agent can only include three to five word phrase additions or delete 20% of the response. Relatedly, the prompt to ChatGPT specifies to only change up to five words of the response. Statement 1 was designed to evaluate our success in this constraint. Further, one intended advantage of task-adaptive AI-assisted Feedback is that the RL agent, or ChatGPT can choose its feedback based on what is included or not in the student's response, whereas in the control group, all students receive the exact same piece of feedback. Statement 2 allows us to evaluate whether the customized feedback is uniquely helpful for the given response. Finally, as our main objective for feedback is to encourage students to engage in revision of their original response, Statement 3 prompts SMEs to evaluate whether or not the feedback would be helpful for the student to improve their own response.

3.6.2.1 Qualitative Evaluation Results

To analyze responses from the SME surveys, we code the Likert scale response options as numbers from 1 to 7 where a 1 corresponds to “Strongly Disagree”, and 7 corresponds to “Strongly Agree”. Thus, larger numbers represent higher agreement. We caution that, because Likert Scale response options are considered ordinal and not interval level data, calculating the mean or standard deviation of the corresponding interval-level coded values has been highly criticized, although commonplace (Wilson et al., 2022). For the purposes of this study, we present means and standard deviations in Table 3.5 to provide a quick idea of the SMEs level of agreement with the three statements, and for an easy comparison of how SMEs tended to view the RL revisions versus the ChatGPT ones. Further, we provide a bar chart for each statement in Figures 3.5, 3.6 and 3.7 to show the frequency of SME ratings in each Likert response category, broken down by the RL revisions and the ChatGPT revisions.

In Table 3.5, we see that the summary values for Statement 1 tend to be high for both the RL revisions and ChatGPT revisions. Correspondingly, Figure 3.5 allows us to see that there are much higher frequencies of ratings in the “Somewhat Agree”, “Agree”, and “Strongly Agree” categories than those in the “Disagree” spectrum, although for both RL and ChatGPT, there are exceptions. One example from a ChatGPT revision that received a “Strongly Disagree” rating for Statement 1 is shown below.

Original student response: *“it takes angel much less time to make it to his destination”*
Machine-revised response: *“Angel reached destination significantly faster due to less time.”*

Further, an example from an RL revision that received a “Somewhat Disagree” rating is shown:

Original student response: *“The line angle and the time”*
Machine-revised response: *“Angel walked faster, the line angle and the time”*

We see that ChatGPT does not always follow the prompt instructions to change no more than five words of the response in its revision, and although the RL agent is constrained to adding short three to five word phrases, when the student's response is already short, the addition of a phrase may substantially change the content of the response.

Further, the summary values in Table 3.5 for Statements 2 and 3 appear to be quite different for the RL revisions and the ChatGPT revisions. Respondents tended to disagree that the ChatGPT revisions included a concept that was not present in the original response, whereas respondents tended to agree to the same statement for the RL revisions. Similarly, respondents tended to disagree that the ChatGPT revisions provide a clue about how the original response can be improved, and tended to agree to the same statement for the RL revisions, This is confirmed by the frequencies shown in Figures 3.6 and 3.7.

Table 3.5
Coded Likert Scale Averages and Standard Deviations

<i>The machine-revised response ...</i>	<i>is recognizable as the original student's response.</i> (Statement 1)		<i>includes a concept that was not present in the original student's response.</i> (Statement 2)		<i>provides a clue about how the original response can be improved.</i> (Statement 3)	
	RL	GPT	RL	GPT	RL	GPT
mean	5.417	4.917	5.708	2.208	5.042	2.833
std	1.442	1.954	1.732	1.668	2.032	1.881

Figure 3.5
Frequency of Likert Agreement Levels for Statement 1

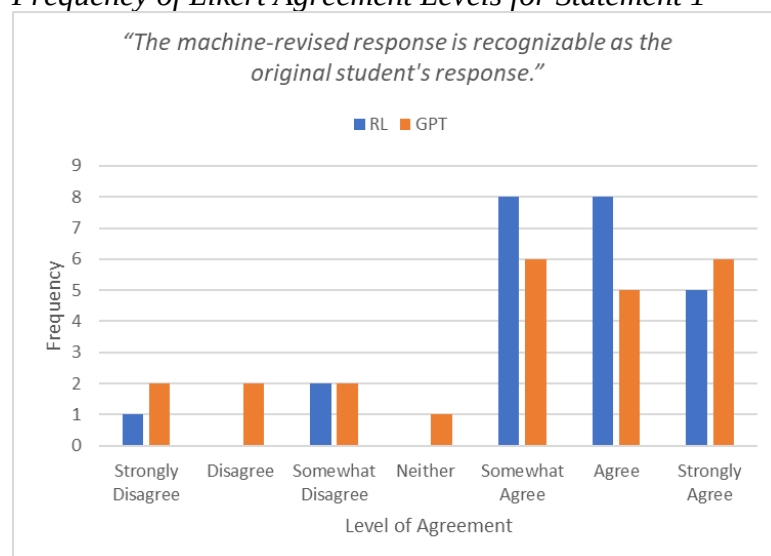


Figure 3.6
Frequency of Likert Agreement Levels for Statement 2

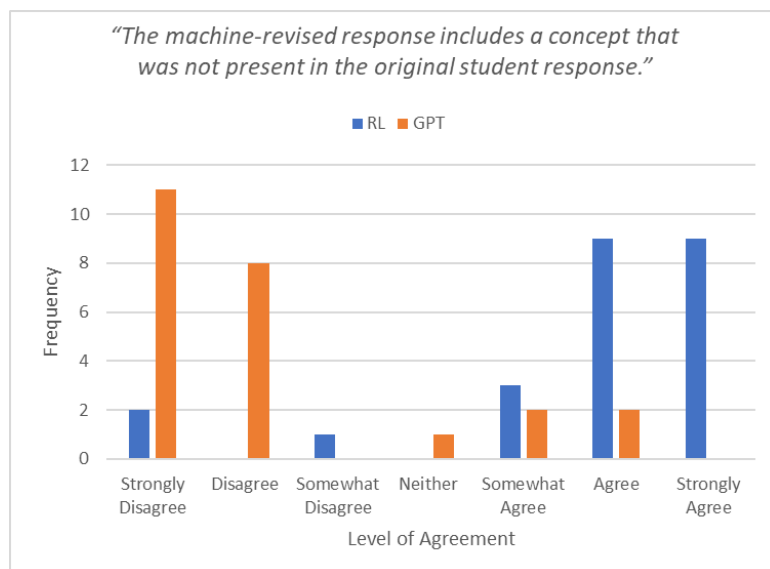
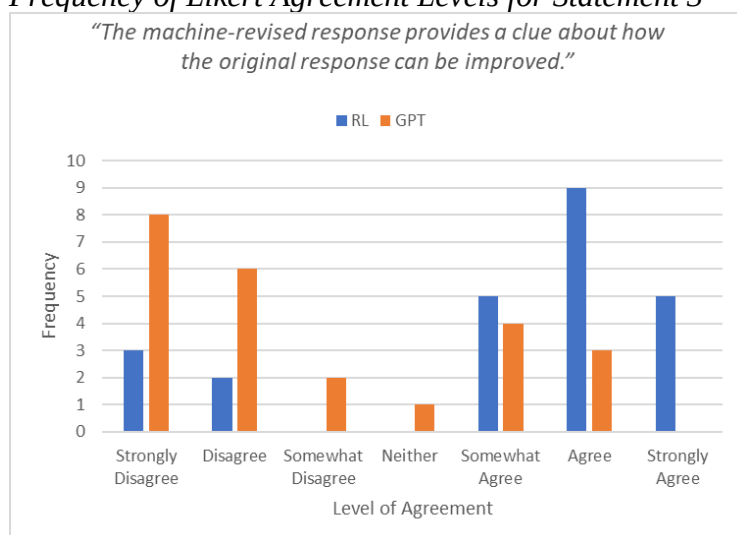


Figure 3.7
Frequency of Likert Agreement Levels for Statement 3



3.7 Data Collection: Processing and Results

In this section, we present the results of our data collection in terms of the three research questions stated in the introduction.

3.7.1 Data Processing

Before analyzing the collected data, we deleted records for participants who did not answer at least one demographic question, and records for participants who did not answer both the main OE item and the revision item. We also remove participants who specified that they are less than 18 years old via the demographic questionnaire as those participants would have needed parental approval to be included in the study.

We scored the pretest data according to the existing item bundle scoring rubric, and used the same ASAG model to score student's responses to the OE item and revision item. The ASAG model provides a probability distribution over the rating categories, so we use the expected value

as a one-number summary of the ASAG model's output. It is typical with classification models to choose the maximum probability category as the final label, but we choose to use the expected value for our final score as it changes with small changes in the probability distribution whereas the maximum probability category may stay the same. If a student's revised response shifts the ASAG model's rating probabilities just slightly towards a greater rating, we want to capture this change no matter how small. We will refer to the expected value of the ASAG model's predicted probability distribution for the original student response as the OE score and that of the revised student response as the revision score.

3.7.2 Summary Statistics

We analyze results from $n=329$ participants. Counts of participants in each treatment group, as well as demographic counts are provided in Table 3.6 below. We note that not all totals (n) add up to 329 because not all participants answered every demographic question. As mentioned, we eliminated participants only if they did not answer at least one of the demographic questions.

Table 3.6

Counts of Participants Partitioned by Treatment and Demographic Groups

group		age		gender		ethnicity		course	
ctrl	111	18-21	228	F	182	White	147	STA2023 (UCF)	191
rl	108	22-25	45	M	118	Hispanic	54	STA2023 (SCF)	42
gpt	110	25-30	21	Non binary	5	Asian	43	STA4102 (UCF)	39
		>30	10	Prefer not to say	4	Black/ AA	24	STA4173 (UCF)	23
						Multiple/ Other	32	STA1001 (UCF)	14
								MTH4420 (UCF)	4
n	329		304		309		300		313

3.7.3 Comparing Treatment Groups for Revision Efficacy

To answer Research Question 1 (Is the RL AIF superior to AIF from ChatGPT and non AI feedback in encouraging students to improve their original response?), we analyze the effectiveness of our intervention for student revisions by considering the OE scores as baseline measures, and the revision scores as follow-up measures. We use an Analysis of Covariance (ANCOVA) model to examine whether the revision score means, adjusted for OE scores, differ between the three treatment groups – RL intervention, GPT intervention, and control. Thus our dependent variable is the revision score, the independent variable is the treatment group, and the covariate is the OE score. We decided to *control* for the OE baseline measure with an ANCOVA, instead of regressing the *difference* in OE scores and revision scores on the treatment group variable. Scholarship has shown that using the differencing approach in a repeated measures

randomized trial leads to inflation of the variation of the treatment effect, resulting in a substantial loss of statistical power (Van Breukelen, 2006; Holland & Rubin, 1983; Rubin, 1974). Summary statistics of the baseline OE scores and revision scores for each group are shown in Table 3.7.

Table 3.7

Summary Statistics of OE and Revision Scores by Groups

	All		Control		RL		GPT	
	OE	Revision	OE	Revision	OE	Revision	OE	Revision
n	329		111		108		110	
mean	1.38	1.41	1.45	1.30	1.35	1.58	1.34	1.34
std	0.63	0.99	0.61	1.11	0.64	0.92	0.64	0.94

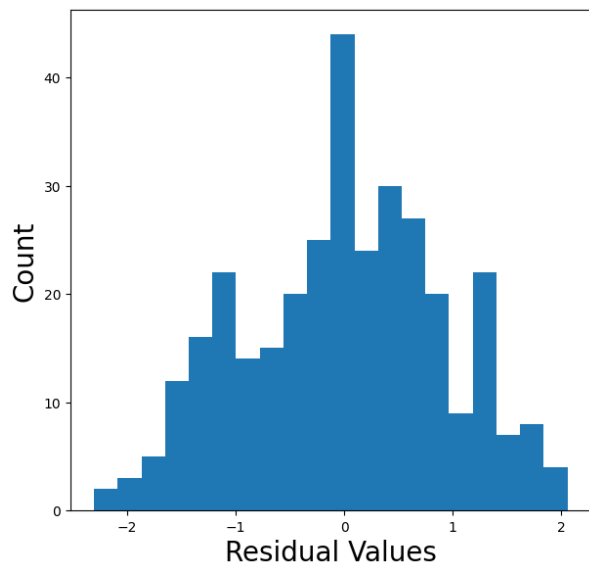
3.7.3.1 Assumptions of ANCOVA

Assumptions for the ANCOVA model include those of ANOVA: (1) approximately normally distributed residuals, (2) homogeneity of variance (homoscedasticity) of the dependent variable between groups and of the residual variance between groups, and (3) independent observations between and within groups. Additional assumptions for ANCOVA include (4) that the covariate should be linearly related to the dependent variable for each treatment group, and (5) the regression slopes for this relationship should be approximately the same (Porter & Raudenbush, 1987; Rutherford, 2011).

We assess (1) visually with a histogram shown in Figure 3.7 and perform a Shapiro-Wilks test (Shapiro & Wilk, 1965). The Shapiro-Wilks test tests the null hypothesis that the data was drawn from a normal distribution. Figure 3.7 shows that the distribution of residuals is not quite normal, and the results of the Shapiro-Wilks test (statistic = 0.988, $p = 0.007$) confirm this visual interpretation. We reject the null hypothesis at a significance level of 0.05 that the population distribution is normal, and conclude that the distribution of residuals is non-normal. However, Levy (1980) showed that non-normal conditional distributions with a non-normal covariate did not seriously affect the proportion of Type I errors made, and with reasonably large sample sizes for our groups, we are not particularly concerned about a loss of statistical power. To justify the violation of the normality assumption, we test whether the covariate is non-normal with the Shapiro-Wilks test and conclude that the covariate is also non-normal (statistic = 0.898, $p = < 0.001$).

Figure 3.7

Histogram of ANCOVA Residual Values



To assess homoscedasticity for assumption (2), we use box-plots for visual assessment and perform Levene's test. Levene's test is used to examine the null hypothesis of equality of variances for two or more groups (Levene, 1960). Firstly, we look at the variance of the dependent variable. The box plot in Figure 3.8 shows that the variances of our dependent variable are not the same. Most distinctly, the variance of the control group appears larger than that for the RL and GPT groups. Results of Levene's test (statistic = 4.344, $p = 0.014$) lead us to reject the null hypothesis with a significance level of 0.05 that the variance of the dependent variable across groups is equal, and conclude that the variances are not equal. Secondly, we assess the equality of the conditional variances across treatment groups by using the ANCOVA model residuals. The box plot in Figure 3.9 shows similar results to that of Figure 3.8: the variances of residuals do not appear to be equal across groups, and this is confirmed by the results of Levene's test (statistic = 9.328, $p = <0.001$). We reject the null hypothesis that the conditional variance is equal across treatment groups and conclude that the variances are not equal. However, Potthoff (1965) found that when group sample sizes are approximately equal and the variance of the covariate is the same across groups, ANCOVA is robust to the violation of the homoscedasticity assumption. So, we look at the variance of the covariate across groups with the box plot in Figure 3.10 and perform Levene's test. Figure 3.10 shows approximately equal variances of the covariate across groups, and the results of Levene's test (statistic = 0.546, $p = 0.580$) give evidence to suggest that variances are equal. We fail to reject the null hypothesis that the variances of the covariate are equal across treatment groups, and conclude that our ANCOVA model is robust to the violation of the homoscedasticity of variance assumption.

Figure 3.8

Box Plot of Expected Revision Score by Treatment Group

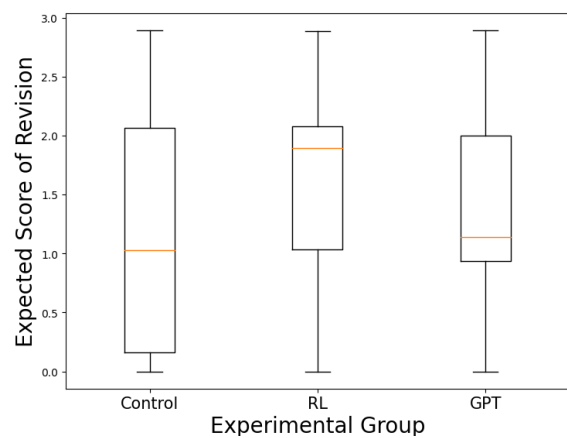


Figure 3.9
Box Plot of Residuals by Treatment Group

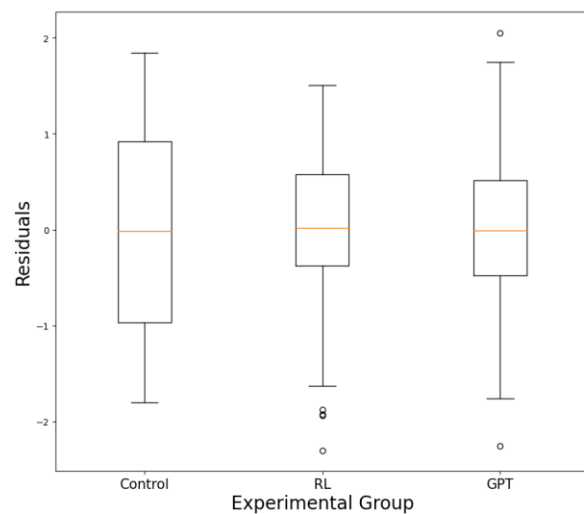
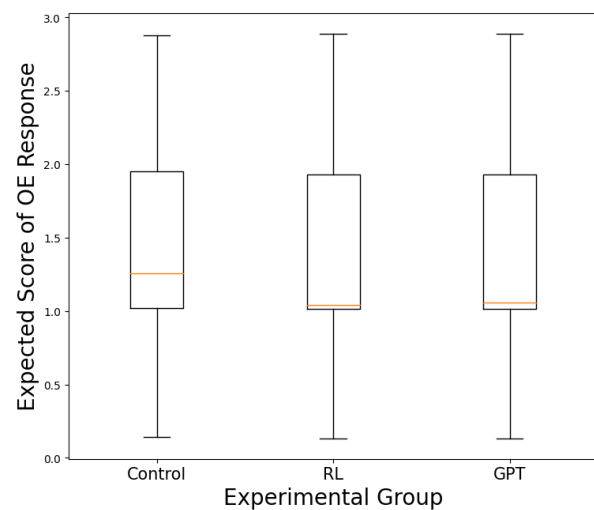


Figure 3.10
Box Plot of Expected OE Score by Treatment Group



The assumption of independence (3) is met by the design of the study, as each student only has one response per item and is only in one group, so observations between groups and within groups have no relation. To test the linearity assumption (4), we first calculate the Pearson correlation coefficient and perform a test of the null hypothesis that the population distributions are uncorrelated for each treatment group. We reject the null hypothesis at a significance level of 0.05 for all treatment groups, as results show favor for the alternative hypothesis (control: statistic = 0.271, $p < 0.001$, RL: statistic = 0.446, $p < 0.001$, GPT: statistic = 0.458, $p < 0.001$). Because Pearson's test assumes that the two samples of interest come from a normal distribution and it does not appear that ours do, we also calculate Spearman's rank-order correlation coefficient because it does not make strong assumptions about the underlying distributions of the samples. Results of Spearman's rank-order test confirm the results from Pearson's test (control: statistic = 0.261, $p = 0.006$, RL: statistic = 0.533, $p < 0.001$, GPT: statistic = 0.387, $p < 0.001$). Further, as Rutherford (2011) suggests the most obvious way to assess linearity is to plot the dependent variable against the covariate for each experimental group, scatterplots of the expected scores of OE responses versus the expected scores of revision responses with linear best fit lines are presented in Figures 3.11, 3.12, and 3.13 below for each of the treatment groups. We can see that although the relationships between OE and revision scores for each group do not appear to be perfectly linear, they also do not appear to have a clear polynomial fit. However, Huitema (2011) claims that the assumption of linearity is less important than the other ANCOVA assumptions because linearity provides an approximate fit to most behavioral data, and nonlinearity reduces the power of the ANCOVA F-test by only a small amount.

Figure 3.11

Scatterplot and Linear Fit for Expected OE versus Revision Score: RL Group

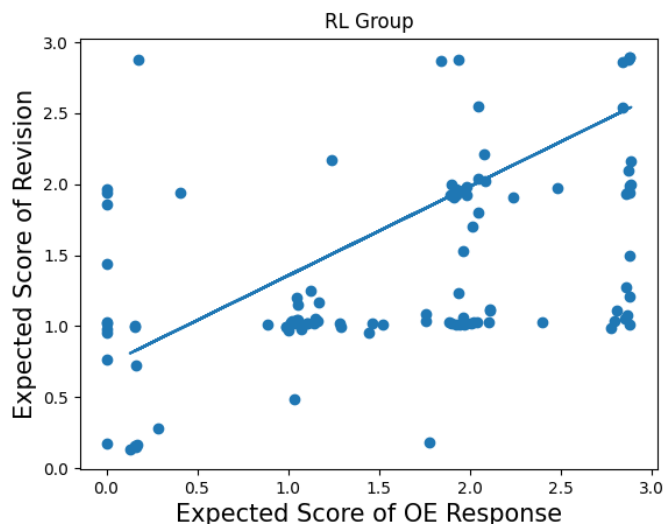


Figure 3.12

Scatterplot and Linear Fit for Expected OE versus Revision Score: Control Group

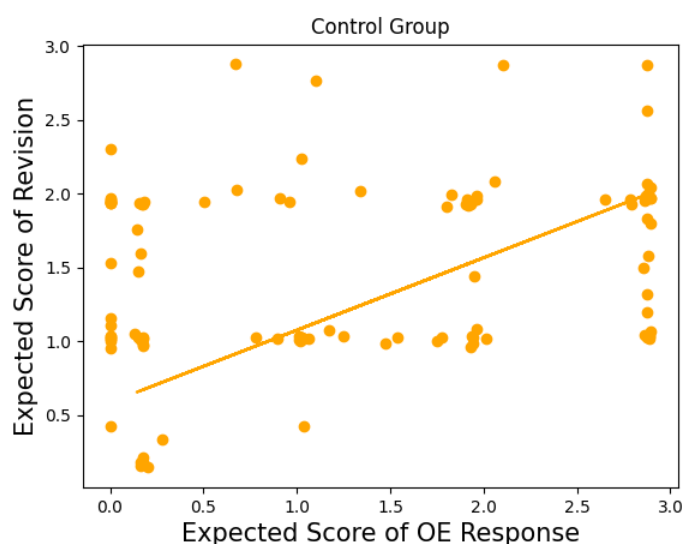
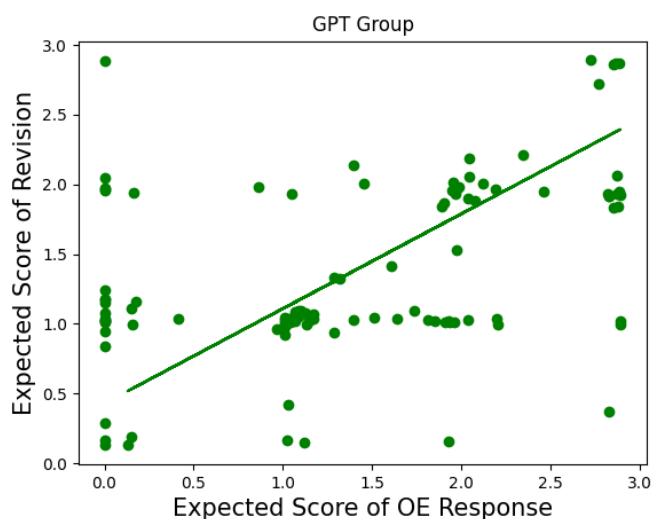


Figure 3.13

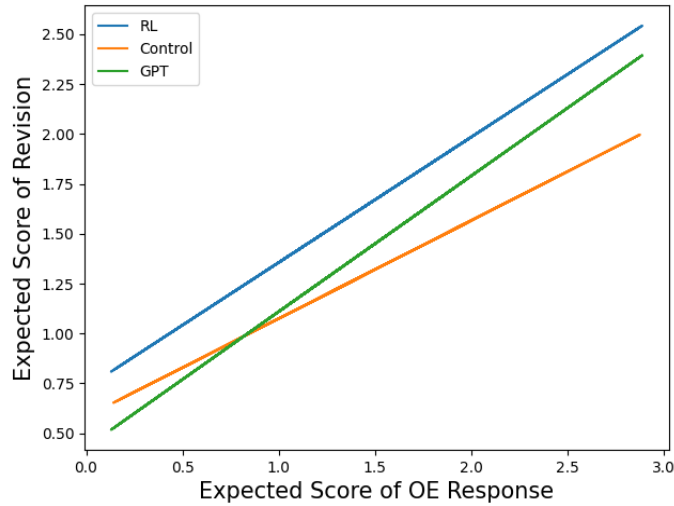
Scatterplot and Linear Fit for Expected OE versus Revision Score: GPT Group



Finally, to assess assumption (5) for equal regression slopes, we regress the dependent variable on the covariate for each treatment group and view each linear fit on the same plot. Figure 3.14 shows that the slopes of each line are quite similar, although not exactly the same. We are not concerned with a small violation of the equality of regression slopes assumption because scholarship has shown that with almost equal sample sizes for the treatment groups and a small difference in the linear relationship between the covariate and dependent variable across groups, the F-test for group differences still closely follows the F-distribution, and loss of statistical power is minimal (Levy 1980; Hamilton, 1977).

Figure 3.14

Linear Fit of Covariate to Dependent Variable by Treatment Group



3.7.3.2 ANCOVA Results

Table 3.8 shows results of the ANCOVA F- test. The large F-statistic and correspondingly small p-value (F-statistic = 3.869, $p = 0.022$) lead us to reject the null hypothesis at a significance level of 0.05 that the means of the revision score for each treatment group are equal when controlling for the OE score, or that the intercept-only model is sufficient. We have evidence that at least one of the treatment groups is significantly different from the others. The “np2” column represents the Partial Eta Squared effect size value that measures the proportion of total variation explained by a predictor after excluding the covariate. Generally, a partial eta squared value of 0.01 indicates a small effect and 0.06 indicates a medium effect. We see that the effect for treatment group is quite small. Further, as the assumptions for ANCOVA were not fully met, although most likely acceptable, we perform the Kruskal-Wallis H test on the adjusted means as an additional measure as it does not assume normally distributed residuals. Results of the Kruskal-Wallis test (statistic = 32.048, $p = <0.001$) provide a similar conclusion to that of the ANCOVA F-test: we reject the null hypothesis at a significance level of 0.05 that at the population medians of all the groups are all equal in favor of the alternative hypothesis that at least one group median is different.

Table 3.8
ANCOVA Results

Source	SS	DF	F	p	np2
group	6.602	2	3.869	0.022	0.023
OE	45.609	1	53.454	0.000	0.141
Residual	277.3	325			

To examine which particular pair(s) of groups are different, we run Tukey’s honestly significant difference (HSD) test on the adjusted group means as a post-hoc measure. Tukey’s HSD test performs pairwise comparison of means, where the null hypothesis is that the underlying distributions all have the same sample mean. Results of Tukey’s HSD test are shown

in Table 3.9. We see that the adjusted mean of the RL group is significantly greater than the adjusted means of the control and GPT groups, but the adjusted mean of the GPT group is not significantly greater than the adjusted mean of the control group.

Table 3.9

Results of Tukey's HSD Test

Comparison	Statistic	p-value	Lower CI	Upper CI
Control - RL	-0.282	0.000	-0.401	-0.163
Control - GPT	-0.046	0.629	-0.165	0.072
RL - GPT	0.236	0.000	0.116	0.355

3.7.3.3 Heterogeneity of Treatment Effect

We continue our analysis to investigate Research Question 2 by fitting an Ordinary Least Squares (OLS) regression model with an interaction term to test for heterogeneity of the intervention effect for revision score among students with varying prior knowledge while controlling for OE score. Our baseline ANCOVA model is an OLS model with the dependent variable being revision score, the independent variable the treatment group, and the OE score as a covariate. Although somewhat redundant, we provide the OLS output for the baseline model so that we can directly compare with the OLS output that includes the interaction of a second independent variable with the group effect.

Results of the baseline model fit in Table 3.10 confirm our previous conclusions that the expected OE item score is a significant covariate ($t = 7.311$, $p = 0.000$), the expected revision score for the RL group is significantly different from that of the control group when controlling for OE score ($t = 2.729$, $p = 0.007$), but we do not have evidence that the expected revision score for the GPT group is significantly different from that of the control group when controlling for OE score ($t = 0.892$, $p = 0.373$). On average, students in the control group with an average OE score will have a revision score of 0.440 as shown by the model intercept. Students in the RL group with an average OE score are expected to have a revision score 0.341 higher than those in the control group, and students in the GPT group with an average OE score are expected to have a revision score 0.111 higher than those in the control group. The model has an adjusted R-squared value of 0.146, where R-squared represents the proportion of the variance in the dependent variable explained by the independent variables and the adjusted R-squared implements a penalty for increasing the number of parameters. So, the baseline model does not seem to explain much of the variation in the dependent variable, although the model is significantly better than the intercept model with no independent variable(s) or covariate(s) included ($F = 19.76$, $p = < 0.001$).

Table 3.10

OLS Fit Results for the Baseline Model

	coef	Std err	t	P> t	[0.025	0.975]
Intercept	0.440	0.146	3.001	0.003	0.151	0.728

Group- GPT	0.111	0.125	0.892	0.373	-0.134	0.356
Group- RL	0.342	0.125	2.729	0.007	0.095	0.588
OE	0.590	0.081	7.311	0.000	0.432	0.749
R-squared	0.154	F-stat:	19.76	AIC:	885.4	
Adj R-squared:	0.146	F-stat p-val:	0.000	BIC:	900.6	

Next, we experiment with adding the interaction of a continuous independent variable, prior knowledge, with the intervention group to our baseline OLS fit. As our proxy for prior knowledge, we score each of the four MC pretest items according to the existing scoring rubric and sum the scores for a total pretest score. Summary statistics of the four pretest scores and the total combined score (prior knowledge) are included in Table 3.11 below:

Table 3.11
Summary Statistics of Pretest Scores

	Control					RL					GPT				
Pre test item	1	2	3	4	total	1	2	3	4	total	1	2	3	4	total
n	111					108					110				
mean	2.57	2.37	2.56	2.74	10.2	2.44	2.32	2.56	2.51	9.62	2.25	1.99	2.31	2.47	9.02
std	0.68	0.87	0.73	0.56	2.17	0.88	0.91	0.99	0.72	2.77	1.04	1.04	1.01	0.71	3.24

Results of the second OLS model fit with the prior knowledge and group interaction, Model 2, are shown in Table 3.12. We see that the main effect for prior knowledge (pre_total) is significant at a level of 0.05 ($t = 2.987$, $p = 0.003$) and that at least one of the interaction terms, Group-RL*pre_total, is significant at a level of 0.05 ($t = -2.087$, $p = 0.038$). So, the expected difference in revision score between treatment groups is not the same for students with different prior knowledge, when controlling for OE score. As the coefficients for both Group-GPT*pre_total and Group-RL*pre_total are negative, the difference in the expected revision score between the GPT group versus the control group, and the RL group versus the control group shrinks as prior knowledge increases, when holding OE score constant. We conclude that the treatment group effect is not the same for students with various prior knowledge scores at a mean OE score.

Table 3.12
OLS Fit Results for Model 2 with Prior Knowledge and Group Interaction

	coef	Std err	t	P> t	[0.025	0.975]
--	------	---------	---	------	--------	--------

Intercept	-0.660	0.419	-1.573	0.117	-1.485	0.165
Group-GPT	0.518	0.486	1.066	0.287	-0.438	1.474
Group-RL	1.420	0.520	2.733	0.007	0.398	2.443
pre_total	0.119	0.040	2.987	0.003	0.041	0.197
Group-GPT* pre_total	-0.030	0.048	-0.631	0.529	-0.124	0.064
Group-RL* pre_total	-0.105	0.051	-2.087	0.038	-0.205	-0.006
OE	0.509	0.082	6.238	0.000	0.349	0.670
R-squared	0.202	F-stat:	13.60	AIC:	872.3	
Adj R-squared:	0.187	F-stat p-val:	0.000	BIC:	898.8	

3.7.4 Examining Student Perceptions

In order to answer Research Question 3 (How do students perceive and act on the feedback they received, and do perceptions differ between intervention groups?), we first look at whether students engage in revision at all, and then investigate whether students are simply copying and pasting the machine revisions for the RL and ChatGPT groups. Further, we manually review the student critiques of their feedback in order to identify interesting patterns in student perceptions of the feedback.

3.7.4.1 Engaging in Revision

In this section, we present counts of cases for each treatment group where students did answer the OE item, but did not answer the revision item. We are not examining the quality of revision in this section, but simply the actions of students to engage in any type of revision after seeing their feedback from either the RL agent, ChatGPT or the control group static hint. It is worth noting that these cases were not included in the ANCOVA analysis, as we only included cases where a student answered both the OE and the revision item. 19 students from the control group, 10 students from the RL group, and 20 students in the GPT group responded to the OE item but not the revision item.

3.7.4.2 Similarity Analysis of Machine and Student Revisions

A potential issue with our feedback design for the GPT and RL groups is that by including an example of revision (the machine revision), students can simply copy the revision and use the exact text for their revision response. Students who are trying to game the system, i.e. achieve a high grade without understanding the material, may default to this method which nullifies our attempt to measure a student's revision capacity.

To explore the extent to which students are copying the machine revisions to use as their own, we embed the student revisions and machine revisions using Sentence BERT (Reimers & Gurevych, 2019), and calculate the cosine similarity of each machine-revision to the student

revision. SentenceBERT is described in more detail in Chapter 1, section 1.3.2.1.1 of this dissertation, and cosine similarity is a metric used to measure the similarity of text using the cosine of the angle between the vector embeddings of the two texts being compared. We find that only four students (4%) in the RL group and eight students (7%) in the GPT group provided revisions that are an exact match to the machine-revisions. Cosine similarity ranges from 0 to 1 where 1 signifies an exact match and 0 signifies no similarity. After manual review of a sample of the responses and similarities, we find that cosine similarities above 0.98 generally indicate an exact copy-paste as well with modifications being as minor as a punctuation or capitalization. There were 10 cases (9%) in the RL group and 10 cases (9%) GPT group with cosine similarities greater than 0.98.

3.7.4.3 Analysis of Student Critiques

Before being asked to revise their original response, students in the RL and GPT groups are asked to critique the “AI bot’s” attempt to improve their response. Students who are in the control group are similarly asked to critique and reflect upon the information given, although not from an AI bot. The student critiques are examined to understand students’ perceptions and interpretation of the feedback. We manually reviewed and coded the feedback critiques for all three groups. First, we assigned each student’s feedback a rating of positive, negative or neutral based on their comments. A positive rating was assigned if the student mentions that the bot’s response is better, added clarification, detail, etc., or that they would use the hint for revision. A few examples of positive coded critiques are:

“It clarified the slope to be the rate of change which made my response stronger”, “it added a better explanation”, “If I did not know the slope was related to the question before, this would have been a good hint to achieve the right answer.”

A neutral rating was assigned if the student did not express any preference about the hint or bot’s revision. Some students simply commented on what the bot added or what the hint said, instead of forming an opinion about it or commenting on whether they would use the information for revision. A few examples of neutral coded critiques are:

“Slope angle just means speed, Angel's slope still ended sooner”, “the bot took out my “mathematical” verbage”, “Changed less to fewer”, “I had the idea there, but it gave me a different suggestion”

A negative rating was assigned to a student’s critique if they expressed a negative sentiment about the revision, mentioned that the hint would not help them revise, or made other negative sounding comments. A few examples of negative coded critiques are:

“I would not because in my opinion the slope of the lines and the overall visual design of the graph is simple enough for me to understand.”, “It changed it to “Angel's line is more perpendicular.”, “That made it incorrect because steeper does not mean more perpendicular.”, “The bot didn't use punctuation which made it harder to read.”

Overall, many students (48%) expressed positive sounding sentiments about their feedback, many (35%) did not express positive or negative sentiments about the feedback, and some (17%) responded negatively about the feedback. The distribution of sentiments is mostly consistent across treatment groups: for control, RL, and GPT groups respectively, 49%, 47%, 49% gave positive critiques, 33%, 37%, 36% gave neutral critiques and 19%, 16%, 15% expressed negative feelings about the feedback.

We identified more specific patterns in our review of student critiques as well. Many students in the control group (33%) commented that they would not use the information provided

because either they already used it in their original response, or they did not believe it would be helpful, but more students in the control group (40%) said they would use the information provided. Further, in some of the RL critiques (15%) and some of the GPT critiques (15%), students commented primarily on the mechanistic changes that the bot made relating to grammar, capitalization, punctuation or spelling instead of focusing on the content added or removed. For example, one student wrote: *“It added a comma between line and the. Then it changed greater to faster. It is more or less the same answer.”*, and another wrote: *“The bot didn't use punctuation which made it harder to read.”* Additionally, in a portion of the RL critiques (13%) and a similar proportion of the GPT critiques (13%), students expressed that the agent didn't add or remove anything from their responses or that the machine-revised response was essentially the same as their original one.

More specific than an overall negative or positive sentiment, for the RL and GPT groups we identified whether the student indicated that the machine-revision was either improved/better or more incorrect/worse than their original response. In both the RL and GPT groups, many (40%) students indicated that the revision was better in some way. For example, one student in the GPT group wrote: *“It added context to the sentence. Made my answer better”*, and one student in the RL group said: *“the bot added something about the slope of the line. this made my response better as it was more mathematical”*. In the RL group, a small proportion of students (7%) indicated that the revision was worse, and in a slightly larger proportion of the GPT group (12%). One student in the RL group said: *“It removed angels name and it made the answer worse”*, and another student in the GPT group wrote: *“I believe it made it worse because it changed less to fewer. Less does not explain it well, but fewer does not fit well in the sentence”*.

3.7.5 Analysis of ChatGPT Machine Revisions

In order to answer Research Question 4 (Can ChatGPT generate correct and useful revisions, and correspondingly what issues arise with using feedback from an unconstrained, generative AI?), we manually review the quality of the ChatGPT machine-revisions while comparing them with the student's original OE responses. Firstly, we noticed that in a substantial amount of cases (26%), the revised text seems to be incorrect, or more incorrect than the original response. For example, one student's original response was: *“Positive slope”* and the GPT revision was *“Robin made it to the destination faster because of the positive slope.”* whereas a correct answer would specify that Angel reached the destination faster. However, in many cases (37%) we found that ChatGPT's revision was correct and helpful for revision. For example, one student responded: *“Angel's line finished first”*, and ChatGPT's revision was: *“Angel's line finished first because it has a steeper slope.”*

We noticed that ChatGPT did not always follow the instructions in our prompt. Interestingly, in about 6 cases (5%), ChatGPT responds with an introductory phrase like, *“the response should be:”*, or *“The correct response to who made it to the destination faster and why is:”*, despite that in the prompt we specifically tell ChatGPT to *“respond only with your updated response”*. We also urge ChatGPT to *“Please change no more than five words of my response...”*, but in a substantial set of revisions (34%), it changes more than five words. For example, one student wrote: *“The time in minutes”*, and ChatGPT's revision was: *“Robin made it to the destination faster than Angel because the blue line has a positive slope, indicating an uphill hike, which means Robin was ascending while Angel was descending.”* It is worth noting that this particular revision is also incorrect.

Conversely, in some cases (13%) ChatGPT changes only one word, or changes nothing besides punctuation. For example, one student wrote: *“When robin reaches 0 elevation.”* and ChatGPT simply changed the period to a comma as such: *“When Robin reached 0 elevation,”*. In another case, ChatGPT only changed one word of a student’s original response: *“Angel’s line ended much earlier.”* was revised to *“Angel’s line ended much sooner.”* Similar to this example, we saw many cases (21%) where ChatGPT’s revision consisted of changing words to respective synonyms or rearranging the words of a sentence. One student wrote: *“The slope is steeper so it indicates less time.”*, and ChatGPT’s revision was: *“The steeper slope suggests less time.”* Finally, in a small set of cases (6%), ChatGPT’s revision was nonsensical. For example, a student responded: *“the x axis”*, and ChatGPT’s revision was: *“Robin made it to the destination faster and why”*.

3.8 Discussion and Limitations

The results of our study give evidence that the RL agent’s AIF mechanism is superior to that from a popular chatbot, ChatGPT, and the non-AI, static feedback mechanism for the control group in terms of encouraging students to improve their short written responses for a mathematical problem solving item. We hypothesize that there are a few reasons why the RL mechanism seems best. Firstly, the RL agent can only make a very particular set of revisions based on the 10 pre-chosen phrases for addition, so we know that the agent is not able to give a student incorrect advice. Feedback from the ChatGPT agent is completely unconstrained as far as what it can change about a student’s revision. We try to constrain ChatGPT as much as possible through our prompt instructions, but as shown in the results, ChatGPT does not always follow the prompt directions and sometimes gives incorrect feedback. We emphasize that this is a major advantage to using an RL agent for feedback as opposed to ChatGPT for not only increased learning gains, but to avoid hindering a student’s learning progress with incorrect information. Further, the RL agent’s revisions are not only constrained to be correct but are customizable to an educator’s preference. This positions teachers as the overall decision-makers of the feedback process, allowing them to exercise their expertise of the subject matter and their students’ understandings.

We further speculate that the task-adaptive nature of the RL agent’s feedback may contribute to the statistically significant difference in learning gains compared to the ChatGPT and non-AIF group, as it makes its revision specific to a student’s response. So generally, the agent should add a phrase that a student did not already include in their own response. Conversely, every student in the control group got the same piece of feedback about the slope of the line, so if a student already included the idea of slope in their response, this feedback may be quite useless for revision. Although ChatGPT had the opportunity to provide task-adaptive feedback, SMEs tended to disagree that the ChatGPT revisions included a novel concept, and we observed that ChatGPT often only changes a single word to a synonym or alters punctuation. The more information feedback contains, the more effective the feedback (Wisniewski et al, 2020) and it is apparent that in aggregate, the RL agent provides more information beyond that which a student had already included in their response.

Additionally, literature shows that student perceptions of feedback are an important factor in determining the impact of feedback on learning (Jonsson, 2013; Rakoczy et al., 2013) and more specifically that students must perceive feedback as helpful to impact downstream work (Jonsson & Panadero, 2018; Van der Kleij & Lipnevich, 2021). As discovered in our analysis of student critiques, many students in the control group commented that they wouldn’t use the

feedback because they already included the information in their response, further supporting our hypothesis that the task-adaptive nature of the RL AIF may explain the difference in downstream revision performance. Although, we note that this contradicts with evidence showing that less individualized and specific feedback forces students to actively engage with feedback, leading to more productive learning (Jonsson & Panadero, 2018).

Further, we found that our conclusions about feedback effect between intervention groups when controlling for OE score does not appear to be the same for students of varying prior knowledge. More specifically, the difference in effect of AIF (RL and ChatGPT groups) compared to the control group shrinks as prior knowledge increases, broadly similar to conclusions previously drawn relating to the moderating effect of prior knowledge (Shute, 2008; Fyfe et al., 2012; Nussbaumer et al., 2014; Razzaq et al., 2020). So, our particular feedback mechanism for revision may not be as useful for students who are already quite comfortable with the subject matter. It is worth noting that we did not observe a negative feedback effect for students of high prior knowledge as shown in Fyfe et al. (2012). In future work, we could build in a mechanism that allows for a different type of feedback for students who exhibit high prior knowledge, although whether there is a particular type of feedback that is more effective for those with high prior knowledge, and what it is an open research question.

In terms of how students act upon feedback, we did see that more students in the ChatGPT and control group chose to not engage in revision at all after answering the OE item compared to those in the RL group. Additionally, for students in the RL and ChatGPT groups, about 9% of students in both simply copied and pasted the machine-revision as their own response. This is certainly not an ideal revision scenario, but we did anticipate some percent of students would be tempted to copy and paste as shown in previous research mentioned. Other recent works examining the effectiveness of AF for revision engagement found that the proportion of students who engage in revision at all is about 0.3-0.5 (Cang et al., 2023; Lee et al., 2021), which is much lower than that of our results (about 0.8-0.9). We hypothesize that prompting students to critique their feedback from an AI peer, eliciting the highest level of cognitive engagement according to the ICAP theory and encouraging social engagement as well, before participating in revision may explain this difference as the aforementioned studies did not include a similar active engagement prompt, and those that did showed increased participation in revision (Matuk & Linn, 2023).

In a real-life use of our intervention, it may be necessary to set integrity guidelines with students in regards directly copying from an AI helper although, it would not be difficult to “catch” students and potentially penalize for this behavior with a simple cosine similarity calculation. Opportunities for academic dishonesty and plagiarism are abundant with tools like ChatGPT easily available to students, and some suggest that such risks can be mitigated by taking a proactive and ethical approach to using these tools (Cotton et al., 2024). However, issues around academic integrity and using AI for writing, coding, and solving problems are complex, for example, views from students themselves are mixed regarding what constitutes cheating when using an AI assistant for writing (Fyfe et al., 2023). Moreover, the role of teachers in the educational integration of AI tools has largely been ignored (Seufert et al., 2021), and little is known about teachers’ knowledge to utilize AI tools in their teaching practices (Kim et al., 2021). Ethical considerations on AI are yet to be explored from the teacher’s perspective (Celik, 2023), so we cannot be sure how teachers may perceive the risk of providing students with an opportunity to directly copy an AI “peer”. We note that this is a limitation of the RL and ChatGPT AIF design.

Regarding students' perceptions of the revision feedback, we did not notice much difference in the sentiment of students' comments across the intervention groups. Generally, students commented positively about the feedback and for all intervention groups, less than 20% of students expressed negative thoughts about the feedback they received. Many students were unhappy with receiving feedback that did not give them any additional information, aligning closely with previous literature noting that students prefer feedback that is helpful and relevant (Dawson et al., 2019), and we mostly saw these types of comments for students in the control group. The importance of feedback that is specific to a students' response can be noted from this finding. Additionally, many students focused on grammatical changes and errors in the feedback they received. We would generally like the students to focus on the content matter of the feedback rather than mechanistic things like spelling and punctuation, so ensuring that feedback is grammatically correct, and that a machine revision includes more than a change in sentence structure may be important to keep students focused on the concept-related ideas in the feedback.

When examining ChatGPT's revisions for feedback, we found many issues. Most notably, ChatGPT gives incorrect feedback about 26% of the time, which could hinder student learning - the opposite of what we hope to achieve with effective AIF. Further, we give ChatGPT strict instructions for how to provide a revision, but in many cases it does not follow our instructions at all. Although many of its revisions were correct and helpful, a few were completely nonsensical. With these substantial problems, we do not feel confident that ChatGPT is ready to be used to provide an example of revision via automatic feedback. We agree with the advice of researchers who warn of well-known issues regarding the use of chatbot AI technologies in educational settings, including generating fake information and potentially hindering students learning (Lo, 2023; Kasneci et al, 2023). While the use of generative models for AIF holds many exciting possibilities, researchers may need to focus on constraining these models such that they can only provide correct and safe feedback. An advantage of using RL is that teachers can play a role in defining what actions – in our case, revisions - the agent takes, mitigating the risk of feedback being incorrect, harmful or biased.

There are many limitations of our study and results. Firstly, we only use one question from a particular subject area to test our feedback mechanism, so we cannot be sure that our results would generalize to different OE items from other subject areas and we collect data from undergraduate students only, so our results may not generalize to K-12 students or adult learners. In future work, we plan to train the agent to revise a broad range of OE response item types in a variety of subject areas and compare the efficacy of the agent across items and subject areas. Further, our sample of SMEs and correspondingly evaluated AIF revisions are small, but nonetheless, having experts evaluate the quality of AI output is a valuable safety precaution to be taken before exposing students to AI technologies (Kasneci et al., 2023). Follow-up work could be conducted with a larger sample size, and expanded questionnaire to further understand how educators and SMEs view the AIF..

Additionally, the performance of the ChatGPT intervention depends somewhat on the prompt we constructed. Although we followed recommendations for prompt engineering, we cannot be certain that our results would be the same for a different prompt. For example, the ChatGPT revision agent was not told to add the same pre-chosen phrases as the RL was able to add, but it was instructed to insert any helpful phrases that were five words or less. In a future study, we could include the phrases that the RL agent is able to add in ChatGPT's prompt and tell ChatGPT that it should only add one of the phrases presented. So, ChatGPT might be less inclined to include incorrect information. Although, we observed that ChatGPT does not always

follow directions, and changing the prompt will not necessarily constrain ChatGPT to only add the correct phrases.

Another limitation is that we chose a specific set of 10 phrases to be used in the RL agent's revisions, but we could have chosen more, less, or a different set of phrases. The phrases we chose to use were designed to include one idea in five words or less, but we could have included a synthesis of multiple ideas for stronger hints, or only keywords for less of a hint. In future work, we could give the agent more revision choices overall with some addition actions including multiple words and phrases, and others including simpler insertions like keywords, and examine which types of revisions seem more or less helpful to students with varying prior knowledge and varying initial response scores.

The RL agent design leaves much flexibility to choose what options the agent has to correct a response. Although, a current flaw in the RL agent's ability to revise is that it is not able to directly connect or synthesize ideas through its revisions. It provides correct information through inserting key phrases, but its corrected responses are not exemplary according to frameworks like Knowledge Integration (KI; Linn et al., 2014) which emphasizes distinguishing among and integrating ideas - both correct and incorrect. Relatedly, although we hope that presenting a new idea through the AI-bot's revision implicitly nudges a student to do this synthesizing themselves in their revision, the AI-bot's revisions do not explicitly encourage students to sort through and connect their original ideas with the new idea that the agent presented. Students may simply add the new idea to their previous response without considering how it fits into their current understanding, while from the constructivist's perspective, learning happens through building upon what one has previously learned.

Inspired by recent advances in RL (Zhou et al., 2024), in future work, we plan to design a multi-turn feedback agent that can ask a series of probing questions after a student's initial response to more directly guide students through the process of integrating their original ideas with new ones presented. For example, if a student responded to the OE item used in this study by saying, "Robin hiked faster because the line is going up" - which is incorrect as the graph shows that Angel hiked faster - the agent could respond with a question like, "Remember that the x-axis represents time spent hiking. Whose line ends with a smaller x-axis value?" such that the student may reconsider their response, given evidence provided by the graph. The student might then respond with, "Angel's line ends first", and the agent could once again ask a probing question to get the student closer to a correct response like, "Does that mean that Angel's hike was a shorter duration?" After a series of conversational turns, the student would be asked to engage in revising their own original response.

References

- Abdelshiheed, M., Hostetter, J. W., Barnes, T., & Chi, M. (2023, June). Leveraging deep reinforcement learning for metacognitive interventions across intelligent tutoring systems. In *International Conference on Artificial Intelligence in Education* (pp. 291-303). Cham: Springer Nature Switzerland.
- Adams, A. M., Wilson, H., Money, J., Palmer-Conn, S., & Fearn, J. (2020). Student engagement with feedback and attainment: the role of academic self-efficacy. *Assessment & Evaluation in Higher Education*, 45(2), 317-329.
- Adiguzel, T., Kaya, M. H., & Cansu, F. K. (2023). Revolutionizing education with AI: Exploring the transformative potential of ChatGPT. *Contemporary Educational Technology*, 15(3), ep429.
- Agarwal, R., Khurana, V., Grover, K., Mohania, M., & Goyal, V. (2022, July). Multi-Relational Graph Transformer for Automatic Short Answer Grading. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (pp. 2001-2012).
- Agarwal, R., Melnick, L., Frosst, N., Zhang, X., Lengerich, B., Caruana, R., & Hinton, G. E. (2021). Neural additive models: Interpretable machine learning with neural nets. *Advances in Neural Information Processing Systems*, 34, 4699-4711.
- Ahadi, A., Singh, A., Bower, M., & Garrett, M. (2022). Text mining in education—A bibliometrics-based systematic review. *Education Sciences*, 12(3), 210.
- Ahmed, F., Shubeck, K., & Hu, X. (2023, July). Chatgpt in the generalized intelligent framework for tutoring. In *Proceedings of the 11th Annual Generalized Intelligent Framework for Tutoring (GIFT) Users Symposium (GIFTSym11)* (p. 109). US Army Combat Capabilities Development Command–Soldier Center.
- Aleedy, M., Atwell, E., & Meshoul, S. (2022). Using AI Chatbots in Education: Recent Advances Challenges and Use Case. *Artificial Intelligence and Sustainable Computing: Proceedings of ICSISCET 2021*, 661-675.
- Anastasopoulos, I., Sheel, S., Pardos, Z., & Bhandari, S. (2023, July). Introducing an Open-source Adaptive Tutoring System to Accelerate Learning Sciences Experimentation. In *Proceedings of the Tenth ACM Conference on Learning@ Scale* (pp. 251-253).
- Anderson, S. E., & Gansneder, B. M. (1995). Using electronic mail surveys and computer-monitored data for studying computer-mediated communication systems. *Social Science Computer Review*, 13(1), 33-46.
- Arneson, A., Wihardini, D., & Wilson, M. (2019). Assessing college-ready data-based reasoning. In *Quantitative Measures of Mathematical Knowledge* (pp. 93-120). Routledge.
- Arnold, K. M., Umanath, S., Thio, K., Reilly, W. B., McDaniel, M. A., & Marsh, E. J. (2017). Understanding the cognitive processes involved in writing to learn. *Journal of Experimental Psychology: Applied*, 23(2), 115–127. <https://doi.org/10.1037/xap0000119>
- Arthurs, N., & Alvero, A. J. (2020). Whose Truth Is the "Ground Truth"? College Admissions Essays and Bias in Word Vector Evaluation Methods. *International Educational Data Mining Society*.
- Bada, S. O., & Olusegun, S. (2015). Constructivism learning theory: A paradigm for teaching and learning. *Journal of Research & Method in Education*, 5(6), 66-70.
- Basu, S., Jacobs, C., & Vanderwende, L. (2013). Powergrading: a clustering approach to amplify human effort for short answer grading. *Transactions of the Association for*

- Computational Linguistics*, 1, 391-402.
- Bertsch, S., Pesta, B. J., Wiscott, R., & McDaniel, M. A. (2007). The generation effect: A meta-analytic review. *Memory & cognition*, 35(2), 201-210.
- Bevilacqua, M., Oketch, K., Qin, R., Stamey, W., Zhang, X., Gan, Y., ... & Abbasi, A. (2023). When Automated Assessment Meets Automated Content Generation: Examining Text Quality in the Era of GPTs. arXiv preprint arXiv:2309.14488.
- Black, P., & Wiliam, D. (1998). Assessment and classroom learning. *Assessment in Education: principles, policy & practice*, 5(1), 7-74.
- Bonthu, S., Rama Sree, S., & Krishna Prasad, M. H. M. (2021). Automated short answer grading using deep learning: A survey. In *Machine Learning and Knowledge Extraction: 5th IFIP TC 5, TC 12, WG 8.4, WG 8.9, WG 12.9 International Cross-Domain Conference, CD-MAKE 2021, Virtual Event, August 17–20, 2021, Proceedings 5* (pp. 61-78). Springer International Publishing.
- Bouchiat, K., Immer, A., Yèche, H., Rätsch, G., & Fortuin, V. (2023). Laplace-Approximated Neural Additive Models: Improving Interpretability with Bayesian Inference. arXiv preprint arXiv:2305.16905.
- Boud, D., & Molloy, E. (2013). *Feedback in higher and professional education: understanding it and doing it well*. Routledge.
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., ... & Amodei, D. (2020). Language models are few-shot learners. *Advances in neural information processing systems*, 33, 1877-1901.
- Bruner, J. (1990). Acts of meaning (Vol. 3). *Harvard university press*.
- Bruner, J. S. (1961). The act of discovery. *Harvard educational review*.
- Buckingham Shum, S., Lim, L. A., Boud, D., Bearman, M., & Dawson, P. (2023). A comparative analysis of the skilled use of automated feedback tools through the lens of teacher feedback literacy. *International Journal of Educational Technology in Higher Education*, 20(1), 40.
- Burrows, S., Gurevych, I., & Stein, B. (2015). The eras and trends of automatic short answer grading. *International Journal of Artificial Intelligence in Education*, 25, 60-117.
- Cai, W., Grossman, J., Lin, Z. J., Sheng, H., Wei, J. T. Z., Williams, J. J., & Goel, S. (2021). Bandit algorithms to personalize educational chatbots. *Machine Learning*, 110(9), 2389-2418.
- Camus, L., & Filighera, A. (2020). Investigating transformers for automatic short answer grading. In *Artificial Intelligence in Education: 21st International Conference, AIED 2020, Ifrane, Morocco, July 6–10, 2020, Proceedings, Part II 21* (pp. 43-48). Springer International Publishing.
- Cang, X., Gnesdilow, D., & Puntambekar, S. (2023). Revisions in Scientific Explanations Using Automated Feedback. In *Proceedings of the 17th International Conference of the Learning Sciences-ICLS 2023*, pp. 170-177. International Society of the Learning Sciences.
- Carless, D. (2019). Feedback loops and the longer-term: towards feedback spirals. *Assessment & Evaluation in Higher Education*, 44(5), 705-714.
- Cavalcanti, A. P., Barbosa, A., Carvalho, R., Freitas, F., Tsai, Y. S., Gašević, D., & Mello, R. F. (2021). Automatic feedback in online learning environments: A systematic literature review. *Computers and Education: Artificial Intelligence*, 2, 100027.
- Celik, I. (2023). Towards Intelligent-TPACK: An empirical study on teachers' professional

- knowledge to ethically integrate artificial intelligence (AI)-based tools into education. *Computers in Human Behavior*, 138, 107468.
- Cer, D., Yang, Y., Kong, S. Y., Hua, N., Limtiaco, N., John, R. S., ... & Kurzweil, R. (2018). Universal sentence encoder. arXiv preprint arXiv:1803.11175.
- Chang, L. H., & Ginter, F. (2024, March). Automatic Short Answer Grading for Finnish with ChatGPT. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 38, No. 21, pp. 23173-23181).
- Chen, D., & Ye, W. (2022, November). Monotonic Neural Additive Models: Pursuing Regulated Machine Learning Models for Credit Scoring. In *Proceedings of the Third ACM International Conference on AI in Finance* (pp. 70-78).
- Chen, S., & Li, L. (2021). Incorporating question information to enhance the performance of automatic short answer grading. In *Knowledge Science, Engineering and Management: 14th International Conference, KSEM 2021, Tokyo, Japan, August 14–16, 2021, Proceedings, Part III 14* (pp. 124-136). Springer International Publishing.
- Chi, M. T., Adams, J., Bogusch, E. B., Bruchok, C., Kang, S., Lancaster, M., ... & Yaghmourian, D. L. (2018). Translating the ICAP theory of cognitive engagement into practice. *Cognitive science*, 42(6), 1777-1832.
- Chi, M. T., De Leeuw, N., Chiu, M. H., & LaVancher, C. (1994). Eliciting self-explanations improves understanding. *Cognitive science*, 18(3), 439-477.
- Christiano, P. F., Leike, J., Brown, T., Martic, M., Legg, S., & Amodei, D. (2017). Deep reinforcement learning from human preferences. *Advances in neural information processing systems*, 30.
- Clariana, R. B. (1999, February). Differential memory effects for immediate and delayed feedback: A delta rule explanation of feedback timing effects. Paper presented at the *Association for Educational Communications and Technology annual convention*, Houston, TX.
- Clariana, R. B., Wagner, D., & Roher Murphy, L. C. (2000). Applying a connectionist description of feedback timing. *Educational Technology Research and Development*, 48(3), 5-22.
- Condor, A. (2020). Exploring automatic short answer grading as a tool to assist in human rating. In *Artificial Intelligence in Education: 21st International Conference, AIED 2020, Ifrane, Morocco, July 6–10, 2020, Proceedings, Part II 21* (pp. 74-79). Springer International Publishing.
- Conneau, A., Kiela, D., Schwenk, H., Barrault, L., & Bordes, A. (2017). Supervised learning of universal sentence representations from natural language inference data. arXiv preprint arXiv:1705.02364.
- Cotton, D. R., Cotton, P. A., & Shipway, J. R. (2024). Chatting and cheating: Ensuring academic integrity in the era of ChatGPT. *Innovations in education and teaching international*, 61(2), 228-239.
- Crawford, L., Lloyd, S., & Knoth, K. (2008). Analysis of student revisions on a state writing test. *Assessment for Effective Intervention*, 33(2), 108-119.
- Dai, W., Lin, J., Jin, H., Li, T., Tsai, Y. S., Gašević, D., & Chen, G. (2023, July). Can large language models provide feedback to students? A case study on ChatGPT. In *2023 IEEE International Conference on Advanced Learning Technologies (ICALT)* (pp. 323-325). IEEE.
- Danielewicz, J., & Elbow, P. (2009). A unilateral grading contract to improve learning and

- teaching. *College Composition and Communication*, 244-268.
- Dawson, P., Henderson, M., Ryan, T., Mahoney, P., Boud, D., Phillips, M., & Molloy, E. (2018). Technology and feedback design. *Learning, design, and technology*.
- Deeva, G., Bogdanova, D., Serral, E., Snoeck, M., & De Weerd, J. (2021). A review of automated feedback systems for learners: Classification framework, challenges and opportunities. *Computers & Education*, 162, 104094.
- Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805.
- Dietterich, T. G. (1998). Approximate statistical tests for comparing supervised classification learning algorithms. *Neural computation*, 10(7), 1895-1923.
- Doroudi, S., Aleven, V., & Brunskill, E. (2019). Where's the reward?. *International Journal of Artificial Intelligence in Education*, 29(4), 568-620.
- Dowell, N., & Kovanovic, V. (2022). Modeling educational discourse with natural language processing. *Education*, 64, 82.
- Dwyer, N., & Marsh, S. (2016, December). How students regard trust in an elearning context. In *2016 14th Annual Conference on Privacy, Security and Trust (PST)* (pp. 682-685). IEEE.
- Ecclestone, K. (2001). 'I know a 2: 1 when I see it': Understanding criteria for degree classifications in franchised university programmes. *Journal of Further and Higher Education*, 25(3), 301-313.
- Ekin, S. (2023). Prompt Engineering For ChatGPT: A Quick Guide To Techniques, Tips, And Best Practices.
- El Fazazi, H., Elgarej, M., Qbadou, M., & Mansouri, K. (2021). Design of an adaptive e-learning system based on multi-agent approach and reinforcement learning. *Engineering, Technology & Applied Science Research*, 11(1), 6637-6644.
- Fahad Mon, B., Wasfi, A., Hayajneh, M., Slim, A., & Abu Ali, N. (2023, September). Reinforcement Learning in Education: A Literature Review. In *Informatics* (Vol. 10, No. 3, p. 74). MDPI.
- Fellows, N. J. (1994). A window into thinking: Using student writing to understand conceptual change in science learning. *Journal of Research in science teaching*, 31(9), 985-1001.
- Finn, J. D., & Zimmer, K. S. (2012). Student Engagement: What Is It? Why Does It Matter? In S. L. Christenson, C. Wylie, & A. L. Reschly (Eds.), *Handbook of Research on Student Engagement* (pp. 97–131). <https://doi.org/10.1007/978-1-4614-2018-7>
- Fonseca, S. C., Pereira, F. D., Oliveira, E. H., Oliveira, D. B., Carvalho, L. S., & Cristea, A. I. (2020). Automatic subject-based contextualisation of programming assignment lists. International Educational Data Mining Society.
- Fredricks, J. A., Blumenfeld, P. C., & Paris, A. H. (2004). School engagement: Potential of the concept, state of the evidence. *Review of Educational Research*, 74(1), 59–109. <https://doi.org/10.3102/00346543074001059>
- Fredricks, J. A., Wang, M.-T., Linn, J. S., Hofkens, T. L., Sung, H., Parr, A., & Allerton, J. (2016). Using qualitative methods to develop a survey measure of math and science engagement. *Learn*, 43, 5–15. <https://doi.org/http://dx.doi.org/10.1016/j.Learninstruc.2016.01.009>
- Fyfe, P. (2023). How to cheat on your final paper: Assigning AI for student writing. *AI & SOCIETY*, 38(4), 1395-1405.
- Fyfe, E. R., & Rittle-Johnson, B. (2016). Feedback both helps and hinders learning: The causal role of prior knowledge. *Journal of Educational Psychology*, 108(1), 82.

- Fyfe, E. R., Rittle-Johnson, B., & DeCaro, M. S. (2012). The effects of feedback during exploratory mathematics problem solving: Prior knowledge matters. *Journal of Educational Psychology*, 104, 1094-1108. doi:10.1037/a0028389
- Gaddipati, S. K., Nair, D., & Plöger, P. G. (2020). Comparative evaluation of pretrained transfer learning models on automatic short answer grading. arXiv preprint arXiv:2009.01303.
- Gerard, L., & Linn, M. C. (2022). Computer-based guidance to support students' revision of their science explanations. *Computers & Education*, 176, Article 104351.
- Gerard, L., Matuk, C., McElhaney, K., & Linn, M. C. (2015). Automated, adaptive guidance for K-12 education. *Educational Research Review*, 15, 41-58.
- Gerard, L. F., Ryoo, K., McElhaney, K. W., Liu, O. L., Rafferty, A. N., & Linn, M. C. (2016). Automated guidance for student inquiry. *Journal of Educational Psychology*, 108(1), 60.
- Gombert, S., Di Mitri, D., Karademir, O., Kubsch, M., Kolbe, H., Tautz, S., ... & Drachsler, H. (2023). Coding energy knowledge in constructed responses with explainable NLP models. *Journal of Computer Assisted Learning*, 39(3), 767-786.
- Goodman, B., & Flaxman, S. (2017). European Union regulations on algorithmic decision-making and a "right to explanation". *AI magazine*, 38(3), 50-57.
- Grassini, S. (2023). Shaping the future of education: exploring the potential and consequences of AI and ChatGPT in educational settings. *Education Sciences*, 13(7), 692.
- Grover, A., & Leskovec, J. (2016, August). node2vec: Scalable feature learning for networks. In *Proceedings of the 22nd ACM SIGKDD international conference on Knowledge discovery and data mining* (pp. 855-864).
- Gunning, D., & Aha, D. (2019). DARPA's explainable artificial intelligence (XAI) program. *AI magazine*, 40(2), 44-58.
- Haller, S., Aldea, A., Seifert, C., & Strisciuglio, N. (2022). Survey on automated short answer grading with deep learning: from word embeddings to transformers. arXiv preprint arXiv:2204.03503.
- Hamilton, B. L. (1977). An empirical investigation of the effects of heterogeneous regression slopes in analysis of covariance. *Educational and Psychological Measurement*, 37(3), 701-712.
- Hamner, B., Morgan, J., Shermis M., Vander Ark, T. (2012). The Hewlett Foundation: Automated Essay Scoring. Kaggle. <https://kaggle.com/competitions/asap-aes>
- Hancock, C. L. (1995). Implementing the assessment standards for school mathematics: Enhancing mathematics learning with open-ended questions. *The Mathematics Teacher*, 88(6), 496-499.
- Harris, Z. S. (1954). Distributional structure. *Word*, 10(2-3), 146-162.
- Hattie, J. (2008). Visible learning: A synthesis of over 800 meta-analyses relating to achievement. Routledge.
- Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of educational research*, 77(1), 81-112.
- Hattie, J., and Zierer, K. (2019). *Visible Learning Insights*. London: Routledge, doi: 10.4324/9781351002226.
- He, P., Gao, J., & Chen, W. (2021). Debertav3: Improving deberta using electra-style pre-training with gradient-disentangled embedding sharing. arXiv preprint arXiv:2111.09543.
- He, P., Liu, X., Gao, J., & Chen, W. (2020). Deberta: Decoding-enhanced bert with disentangled attention. arXiv preprint arXiv:2006.03654.
- Hou, W. J., & Tsao, J. H. (2011). Automatic Assessment of Students' Free-Text Answers with

- Different Levels. *International Journal on Artificial Intelligence Tools*, 20(02), 327-347.
- Holland, P. W., & Rubin, D. B. (2012). On Lord's paradox. In *Principals of modern psychological measurement* (pp. 3-25). Routledge.
- Howell, J. A., Roberts, L. D., & Mancini, V. O. (2018). Learning analytics messages: Impact of grade, sender, comparative information and message style on student affect and academic resilience. *Computers in Human Behavior*, 89, 8-15.
- Huitema, B. (2011). *The analysis of covariance and alternatives: Statistical methods for experiments, quasi-experiments, and single-case studies*. John Wiley & Sons.
- Iraj, H., Fudge, A., Faulkner, M., Pardo, A., & Kovanović, V. (2020, March). Understanding students' engagement with personalised feedback messages. In *Proceedings of the tenth international Conference on learning analytics & knowledge* (pp. 438-447).
- Jacobsen, L. J., & Weber, K. E. (2023). The promises and pitfalls of ChatGPT as a feedback provider in higher education: An exploratory study of prompt engineering and the quality of AI-driven feedback. <https://doi.org/10.31219/osf.io/cr257>
- Jo, W., & Kim, D. (2022). Neural Additive Models for Nowcasting. arXiv preprint arXiv:2205.10020.
- Jonsson, A. (2013). Facilitating productive use of feedback in higher education. *Active learning in higher education*, 14(1), 63-76.
- Jonsson, A., & Panadero, E. (2018). Facilitating Student's Active Engagement with Feedback. *The Cambridge Handbook of Instructional Feedback*. Cambridge University Press.
- Ju, S., Zhou, G., Barnes, T., Chi, M.: Pick the moment: identifying critical pedagogical decisions using long-short term rewards. In: *Proceedings of the 13th International Conference on EDM*, pp. 126–136. EDM, Virtual (2020)
- Jukiewicz, M. (2024). The future of grading programming assignments in education: The role of ChatGPT in automating the assessment and feedback process. *Thinking Skills and Creativity*, 101522.
- Kasneci, E., Seßler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., ... & Kasneci, G. (2023). ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and individual differences*, 103, 102274.
- Kayid, A., Frosst, N., Hinton, G. (2020) Neural additive models Library.
- Kelley, K., & Preacher, K. J. (2012). On effect size. *Psychological methods*, 17(2), 137.
- Kim, S., Jang, Y., Choi, S., Kim, W., Jung, H., Kim, S., & Kim, H. (2021). Analyzing teacher competency with TPACK for K-12 AI education. *KI-Künstliche Intelligenz*, 35 (2), 139–151. <https://doi.org/10.1007/s13218-021-00731-9>
- Klein, P. (2015). Mediators and moderators in individual and collaborative writing to learn. *Journal of Writing Research*, 7, 201–214
- Klein, R., Kyrilov, A., & Tokman, M. (2011, June). Automated assessment of short free-text responses in computer science using latent semantic analysis. In *Proceedings of the 16th annual joint conference on Innovation and technology in computer science education* (pp. 158-162).
- Kluger, A. N., & DeNisi, A. (1996). The effects of feedback interventions on performance: a historical review, a meta-analysis, and a preliminary feedback intervention theory. *Psychological bulletin*, 119(2), 254.
- Kooli, C., & Yusuf, N. (2024). Transforming Educational Assessment: Insights Into the Use of ChatGPT and Large Language Models in Grading. *International Journal of Human-Computer Interaction*, 1-12.

- Koraishi, O. (2023). Teaching English in the age of AI: Embracing ChatGPT to optimize EFL materials and assessment. *Language Education and Technology*, 3(1).
- Kung, F. Y., & Scholer, A. A. (2018). Message framing influences perceptions of feedback (in) directness. *Social Cognition*, 36(6), 626-670.
- Langer, J. A., & Applebee, A. N. (1987). *How Writing Shapes Thinking: A Study of Teaching and Learning*. NCTE Research Report No. 22. National Council of Teachers of English, 1111 Kenyon Rd., Urbana, IL 61801
- Latif, E., & Zhai, X. (2024). Fine-tuning chatgpt for automatic scoring. *Computers and Education: Artificial Intelligence*, 100210.
- Lee, H. S., Gweon, G. H., Lord, T., Paessel, N., Pallant, A., & Pryputniewicz, S. (2021). Machine learning-enabled automated feedback: supporting students' revision of scientific arguments based on data drawn from simulation. *Journal of Science Education and Technology*, 30(2), 168-192.
- Lee, U., Jung, H., Jeon, Y., Sohn, Y., Hwang, W., Moon, J., & Kim, H. (2023). Few-shot is enough: exploring ChatGPT prompt engineering method for automatic question generation in english education. *Education and Information Technologies*, 1-33.
- Lee, G. G., Latif, E., Wu, X., Liu, N., & Zhai, X. (2024). Applying large language models and chain-of-thought for automatic scoring. *Computers and Education: Artificial Intelligence*, 100213.
- Lee, A. V. Y., Luco, A. C., & Tan, S. C. (2023). A Human-centric automated essay scoring and feedback system for the development of ethical reasoning. *Educational Technology & Society*, 26(1), 147-159.
- Levene, H. (1960). Robust tests for equality of variances. *Contributions to probability and statistics*, 278-292.
- Levine, S., Kumar, A., Tucker, G., & Fu, J. (2020). Offline reinforcement learning: Tutorial, review, and perspectives on open problems. arXiv preprint arXiv:2005.01643.
- Levy, K. J. (1980). A Monte Carlo study of analysis of covariance under violations of the assumptions of normality and equal regression slopes. *Educational and Psychological Measurement*, 40(4), 835-840.
- Likert, R. (1932). A technique for the measurement of attitudes. *Archives of psychology*.
- Linn, M. C., Gerard, L., Ryoo, K., McElhaney, K., Liu, O. L., & Rafferty, A. N. (2014). Computer-guided inquiry to improve science learning. *Science*, 344(6180), 155-156.
- Liu, T., Ding, W., Wang, Z., Tang, J., Huang, G. Y., & Liu, Z. (2019). Automatic short answer grading via multiway attention networks. In *Artificial Intelligence in Education: 20th International Conference, AIED 2019, Chicago, IL, USA, June 25-29, 2019, Proceedings, Part II 20* (pp. 169-173). Springer International Publishing.
- Lo, C. K. (2023). What is the impact of ChatGPT on education? A rapid review of the literature. *Education Sciences*, 13(4), 410.
- Luber, M., Thielmann, A., & Säfken, B. (2023). Structural Neural Additive Models: Enhanced Interpretable Machine Learning. arXiv preprint arXiv:2302.09275.
- MacLellan, C. J., & Gupta, A. (2021). Learning Expert Models for Educationally Relevant Tasks Using Reinforcement Learning. *International Educational Data Mining Society*.
- Madnani, N., Burstein, J., Sabatini, J., & O'Reilly, T. (2013). Automated Scoring of Summary-Writing Tasks Designed to Measure Reading Comprehension. *Grantee Submission*.
- Malecka, B., Boud, D., & Carless, D. (2022). Eliciting, processing and enacting feedback:

- mechanisms for embedding student feedback literacy within the curriculum. *Teaching in Higher Education*, 27(7), 908-922.
- Mao, Z., & Lee, I. (2023). Student engagement with written feedback: Critical issues and way forward. *RELC Journal*, 00336882221150811.
- Mariotti, E., Moral, J. M. A., & Gatt, A. (2023). Exploring the balance between interpretability and performance with carefully designed constrainable neural additive models. *Information Fusion*, 101882.
- Matuk, C., & Linn, M. C. (2023). Students' perceptions of the impacts of peer ideas in inquiry learning. *Instructional Science*, 51(1), 65-102.
- Memarian, B., & Doleck, T. (2024). A Scoping Review of Reinforcement Learning in Education. *Computers and Education Open*, 100175.
- Mertens, U., Finn, B., & Lindner, M. A. (2022). Effects of computer-based feedback on lower- and higher-order learning outcomes: A network meta-analysis. *Journal of Educational Psychology*, 114(8), 1743.
- Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. arXiv preprint arXiv:1301.3781.
- Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. *Advances in neural information processing systems*, 26.
- Mislevy, R. J., Almond, R. G., & Lukas, J. F. (2003). A brief introduction to evidence-centered design. ETS Research Report Series, 2003(1), i-29.
- Mohler, M., & Mihalcea, R. (2009, March). Text-to-text semantic similarity for automatic short answer grading. In *Proceedings of the 12th Conference of the European Chapter of the ACL (EACL 2009)* (pp. 567-575).
- Moons, F., Vandervieren, E., & Colpaert, J. (2022). Atomic, reusable feedback: a semi-automated solution for assessing handwritten tasks? A crossover experiment with mathematics teachers. *Computers and Education Open*, 3, 100086.
- Mory, E. H. (2004). Feedback research revisited. In D. Jonassen (Ed.), *Handbook of research on educational communications and technology: A project for the Association for Educational Communications and Technology* (2nd. pp. 745-783). Mahwah, NJ: Lawrence Erlbaum Associates.
- Moslehi, S., Mahjub, H., Farhadian, M., Soltanian, A. R., & Mamani, M. (2022). Interpretable generalized neural additive models for mortality prediction of COVID-19 hospitalized patients in Hamadan, Iran. *BMC Medical Research Methodology*, 22(1), 339.
- Mynlieff, M., Manogaran, A. L., Maurice, M. S., & Eddinger, T. J. (2014). Writing assignments with a metacognitive component enhance learning in a large introductory biology course. *CBE Life Sciences Education*, 13(2), 311–321. <https://doi.org/10.1187/cbe.13-05-0097>
- Narciss, S., & Huth, K. (2006). Fostering achievement and motivation with bug-related tutoring feedback in a computer-based training for written subtraction. *Learning and Instruction*, 16, 310–322. doi:10.1016/j.learninstruc.2006.07.003
- Nicol, D. (2010). From monologue to dialogue: improving written feedback processes in mass higher education. *Assessment & Evaluation in Higher Education*, 35(5), 501-517.
- Nicol, D., Thomson, A., & Breslin, C. (2014). Rethinking feedback practices in higher education: a peer review perspective. *Assessment & evaluation in higher education*, 39(1), 102-122.
- Nussbaumer, D., Schneider, M., & Stern, E. (2014). The influence of feedback on the flexibility

- of strategy choices in algebra problem solving. Proceedings of the Thirty-Sixth Annual Conference of the Cognitive Science Society (pp. 1102-1107). Quebec City, Canada: Cognitive Science Society.
- Nye, B. D., Mee, D., & Core, M. G. (2023). Generative large language models for dialog-based tutoring: An early consideration of opportunities and concerns. In *AIED Workshops*.
- Olejnik, S. F., & Algina, J. (1984). Parametric ANCOVA and the rank transform ANCOVA when the data are conditionally non-normal and heteroscedastic. *Journal of Educational Statistics*, 9(2), 129-149.
- OpenAI. (2023). ChatGPT (Mar 14 version) [Large language model].
<https://chat.openai.com/chat>
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., ... & Lowe, R. (2022). Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35, 27730-27744.
- Pardo, A., Jovanovic, J., Dawson, S., Gašević, D., & Mirriahi, N. (2019). Using learning analytics to scale the provision of personalised feedback. *British Journal of Educational Technology*, 50(1), 128-138.
- Pardos, Z. A., Tang, M., Anastasopoulos, I., Sheel, S. K., & Zhang, E. (2023, April). OATutor: An Open-source Adaptive Tutoring System and Curated Content Library for Learning Sciences Research. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (pp. 1-17).
- Park, H. W., Grover, I., Spaulding, S., Gomez, L., & Breazeal, C. (2019, July). A model-free affective reinforcement learning approach to personalization of an autonomous social robot companion for early literacy education. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 33, No. 01, pp. 687-694).
- Pennington, J., Socher, R., & Manning, C. D. (2014, October). Glove: Global vectors for word representation. In *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)* (pp. 1532-1543).
- Potthoff, R. F. (1965). Some Scheffe-type tests for some Behrens-Fisher-type regression problems. *Journal of the American Statistical Association*, 60(312), 1163-1190.
- Porter, A. C., & Raudenbush, S. W. (1987). Analysis of covariance: Its model and use in psychological research. *Journal of Counseling Psychology*, 34(4), 383.
- Poulton, A., & Eliens, S. (2021, September). Explaining transformer-based models for automatic short answer grading. In *Proceedings of the 5th International Conference on Digital Technology in Education* (pp. 110-116).
- Price, M., Handley, K., Millar, J., & O'donovan, B. (2010). Feedback: all that effort, but what is the effect?. *Assessment & Evaluation in Higher Education*, 35(3), 277-289.
- Qi, H., Wang, Y., Dai, J., Li, J., & Di, X. (2019). Attention-based hybrid model for automatic short answer scoring. In *Simulation Tools and Techniques: 11th International Conference, SIMUtools 2019, Chengdu, China, July 8–10, 2019, Proceedings 11* (pp. 385-394). Springer International Publishing.
- Rakoczy, K., Harks, B., Klieme, E., Blum, W., & Hochweber, J. (2013). Written feedback in mathematics: Mediated by students' perception, moderated by goal orientation. *Learning and Instruction*, 27, 63-73.
- Ray, A. B., Graham, S., Houston, J. D., & Harris, K. R. (2016). Teachers use of writing to support students' learning in middle school: A national survey in the United States. *Reading and Writing*, 29, 1039-1068.

- Razzaq, R., Ostrow, K. S., & Heffernan, N. T. (2020, June). Effect of immediate feedback on math achievement at the high school level. In *International Conference on Artificial Intelligence in Education* (pp. 263-267). Cham: Springer International Publishing.
- Reimers, N., & Gurevych, I. (2019). Sentence-bert: Sentence embeddings using siamese bert-networks. arXiv preprint arXiv:1908.10084.
- Riordan, B., Bichler, S., Bradford, A., Chen, J. K., Wiley, K., Gerard, L., & Linn, M. C. (2020, July). An empirical investigation of neural methods for content scoring of science explanations. In *Proceedings of the fifteenth workshop on innovative use of NLP for building educational applications* (pp. 135-144).
- Rivard, L. O. P. (1994). A review of writing to learn in science: Implications for practice and research. *Journal of research in science teaching*, 31(9), 969-983.
- Roscoe, R. D., Snow, E. L., Allen, L. K., & McNamara, D. S. (2015). Automated detection of essay revising patterns: applications for intelligent feedback in a writing tutor. *Technology, Instruction, Cognition and Learning*, 10, 59-79.
- Rowe, A. D., & Wood, L. N. (2008). Student perceptions and preferences for feedback. *Asian social science*, 4(3), 78-88.
- Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of educational Psychology*, 66(5), 688.
- Rutherford, A. (2011). *ANOVA and ANCOVA: a GLM approach*. John Wiley & Sons.
- Ryoo, K., & Linn, M. C. (2014). Designing guidance for interpreting dynamic visualizations: Generating versus reading explanations. *Journal of Research in Science Teaching*, 51(2), 147-174.
- Schlippe, T., Stierstorfer, Q., Koppel, M. T., & Libbrecht, P. (2022, July). Explainability in Automatic Short Answer Grading. In *International Conference on Artificial Intelligence in Education Technology* (pp. 69-87). Singapore: Springer Nature Singapore.
- Schmucker, R., Pachapurkar, N., Bala, S., Shah, M., & Mitchell, T. (2023, August). Learning to give useful hints: Assistance action evaluation and policy improvements. In *European Conference on Technology Enhanced Learning* (pp. 383-398). Cham: Springer Nature Switzerland.
- Schmucker, R., Xia, M., Azaria, A., & Mitchell, T. (2023). Ruffle&Riley: Towards the Automated Induction of Conversational Tutoring Systems. *arXiv preprint arXiv:2310.01420*.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347.
- Seufert, S., Guggemos, J., & Sailer, M. (2021). Technology-related knowledge, skills, and attitudes of pre-and in-service teachers: The current situation and emerging trends. *Computers in Human Behavior*, 115, Article 106552. <https://doi.org/10.1016/j.chb.2020.106552>
- Shaik, T., Tao, X., Li, Y., Dann, C., McDonald, J., Redmond, P., & Galligan, L. (2022). A review of the trends and challenges in adopting natural language processing methods for education feedback analysis. *IEEE Access*, 10, 56720-56739.
- Shapiro, S. S. and Wilk, M. B. (1965) An analysis of variance test for normality (complete samples). *Biometrika*, 52, 591-611.
- Shute, V. J. (2008). Focus on formative feedback. *Review of educational research*, 78(1), 153-189.
- Siems, J., Schambach, M., Schulze, S., & Otterbach, J. S. (2023). Interpretable Reinforcement

- Learning via Neural Additive Models for Inventory Management. arXiv preprint arXiv:2303.10382.
- Singh, C., Hsu, A. R., Antonello, R., Jain, S., Huth, A. G., Yu, B., & Gao, J. (2023). Explaining black box text modules in natural language with language models. arXiv preprint arXiv:2305.09863.
- Singla, A., Rafferty, A. N., Radanovic, G., & Heffernan, N. T. (2021). Reinforcement learning for education: Opportunities and challenges. arXiv preprint arXiv:2107.08828.
- Steiss, J., Tate, T., Graham, S., Cruz, J., Hebert, M., Wang, J., ... & Olson, C. B. (2024). Comparing the quality of human and ChatGPT feedback of students' writing. *Learning and Instruction*, 91, 101894.
- Sung, C., Dhamecha, T. I., & Mukhi, N. (2019). Improving short answer grading using transformer-based pre-training. In *Artificial Intelligence in Education: 20th International Conference, AIED 2019, Chicago, IL, USA, June 25-29, 2019, Proceedings, Part I* 20 (pp. 469-481). Springer International Publishing.
- Sung, C., Dhamecha, T., Saha, S., Ma, T., Reddy, V., & Arora, R. (2019, November). Pre-training BERT on domain resources for short answer grading. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (pp. 6071-6075).
- Sutton, R. S., & Barto, A. G. (2018). Reinforcement learning: An introduction. MIT press.
- Swart, E. K., Nielen, T. M., & Sikkema-de Jong, M. T. (2019). Supporting learning from text: A meta-analysis on the timing and content of effective feedback. *Educational Research Review*, 28, 100296.
- Tan, H., Wang, C., Duan, Q., Lu, Y., Zhang, H., & Li, R. (2023). Automatic short answer grading by encoding student responses via a graph convolutional network. *Interactive Learning Environments*, 31(3), 1636-1650.
- Tansomboon, C., Gerard, L. F., Vitale, J. M., & Linn, M. C. (2017). Designing automated guidance to promote productive revision of science explanations. *International Journal of Artificial Intelligence in Education*, 27, 729-757.
- Thaker, K., Zhang, L., He, D., & Brusilovsky, P. (2020). Recommending Remedial Readings Using Student Knowledge State. International Educational Data Mining Society.
- Thomas, J. D., & Arnold, R. M. (2011). Giving feedback. *Journal of palliative medicine*, 14(2), 233-239.
- Thomas, D. R., Gupta, S., & Koedinger, K. R. (2023). Comparative analysis of learnersourced human-graded and ai-generated responses for autograding online tutor lessons. In *International Conference on Artificial Intelligence in Education* (pp. 714-719). Cham: Springer Nature Switzerland.
- Tornqvist, M., Mahamud, M., Guzman, E. M., & Farazouli, A. (2023, July). ExASAG: Explainable Framework for Automatic Short Answer Grading. In *Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2023)* (pp. 361-371).
- Tulu, C. N., Ozkaya, O., & Orhan, U. (2021). Automatic short answer grading with semspace sense vectors and malstm. *IEEE Access*, 9, 19270-19280.
- Utkin, L., & Konstantinov, A. (2022). An extension of the neural additive model for uncertainty explanation of machine learning survival models. In *Cyber-Physical Systems: Intelligent Models and Algorithms* (pp. 3-13). Cham: Springer International Publishing.

- Van Breukelen, Gerard JP. "ANCOVA versus change from baseline had more power in randomized studies and more bias in nonrandomized studies." *Journal of clinical epidemiology* 59.9 (2006): 920-925.
- Van der Kleij, F. M., & Lipnevich, A. A. (2021). Student perceptions of assessment feedback: A critical scoping review and call for research. *Educational Assessment, Evaluation and Accountability*, 33, 345-373.
- Van Riesen, S. A., Gijlers, H., Anjewierden, A. A., and de Jong, T. (2019). The influence of prior knowledge on the effectiveness of guided experiment design. *Interact. Learn. Environ.* 2019, 1–17. doi: 10.1080/10494820.2019.1631193
- VanLehn, K. (2011). The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational psychologist*, 46(4), 197-221.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in neural information processing systems*, 30.
- Wainer, H. (1993). Measurement problems. *Journal of educational measurement*, 30(1), 1-21.
- Waltman, K., Kahn, A., & Koency, G. (1998). Alternative Approaches to Scoring: The Effects of Using Different Scoring Methods on the Validity of Scores from a Performance Assessment. *CSE Technical Report 488*.
- Wang, T., Funayama, H., Ouchi, H., & Inui, K. (2021). Data augmentation by rubrics for short answer grading. *Journal of Natural Language Processing*, 28(1), 183-205.
- Wilson, M. (2023). *Constructing measures: An item response modeling approach*. Taylor & Francis.
- Wilson, M., Bathia, S., Morell, L., Gochyyev, P., Koo, B. W., & Smith, R. (2022). Seeking a better balance between efficiency and interpretability: Comparing the likert response format with the Guttman response format. *Psychological Methods*.
- Wilson, M., & Sloane, K. (2000). From principles to practice: An embedded assessment system. *Applied measurement in education*, 13(2), 181-208.
- Winstone, N., Boud, D., Dawson, P., & Heron, M. (2022). From feedback-as-information to feedback-as-process: a linguistic analysis of the feedback literature. *Assessment & Evaluation in Higher Education*, 47(2), 213-230.
- Winstone, N. E., Nash, R. A., Parker, M., & Rowntree, J. (2017). Supporting learners' agentic engagement with feedback: A systematic review and a taxonomy of recipience processes. *Educational psychologist*, 52(1), 17-37.
- Wisniewski, B., Zierer, K., & Hattie, J. (2020). The power of feedback revisited: A meta-analysis of educational feedback research. *Frontiers in psychology*, 10, 487662.
- Xiao, Y., Zingle, G., Jia, Q., Akbar, S., Song, Y., Dong, M., ... & Gehringer, E. (2020). Problem Detection in Peer Assessments between Subjects by Effective Transfer Learning and Active Learning. International Educational Data Mining Society.
- Xu, F., Uszkoreit, H., Du, Y., Fan, W., Zhao, D., & Zhu, J. (2019). Explainable AI: A brief survey on history, research areas, approaches and challenges. In *Natural Language Processing and Chinese Computing: 8th CCF International Conference, NLPCC 2019, Dunhuang, China, October 9–14, 2019, Proceedings, Part II* 8 (pp. 563-574). Springer International Publishing.
- Xu, S., Bu, Z., Chaudhari, P., & Barnett, I. J. (2022). Sparse neural additive model: Interpretable deep learning with feature selection via group sparsity. arXiv preprint arXiv:2202.12482.
- Yang, T. C., Chen, M. C., & Chen, S. Y. (2018). The influences of self-regulated learning support and prior knowledge on improving learning performance. *Computers &*

- Education*, 126, 37-52.
- Yang, X., Huang, Y., Zhuang, F., Zhang, L., & Yu, S. (2018). Automatic Chinese short answer grading with deep autoencoder. In *Artificial Intelligence in Education: 19th International Conference, AIED 2018, London, UK, June 27–30, 2018, Proceedings, Part II* 19 (pp. 399-404). Springer International Publishing.
- Zeng, Z., Li, X., Gasevic, D., & Chen, G. (2022, July). Do Deep Neural Nets Display Human-like Attention in Short Answer Scoring?. In *Proceedings of the 2022 conference of the north American chapter of the Association for Computational Linguistics: Human language technologies* (pp. 191-205).
- Zhang, Y., Shah, R., & Chi, M. (2016). Deep Learning+ Student Modeling+ Clustering: A Recipe for Effective Automatic Short Answer Grading. *International Educational Data Mining Society*.
- Zhang, Z. V., & Hyland, K. (2022). Fostering student engagement with feedback: An integrated approach. *Assessing Writing*, 51, 100586.
- Zhou, Y., Zanette, A., Pan, J., Levine, S., & Kumar, A. (2024). ArCHer: Training Language Model Agents via Hierarchical Multi-Turn RL. *arXiv preprint arXiv:2402.19446*.
- Zhu, M., Lee, H. S., Wang, T., Liu, O. L., Belur, V., & Pallant, A. (2017). Investigating the impact of automated feedback on students' scientific argumentation. *International Journal of Science Education*, 39(12), 1648-1668.
- Zhu, M., Liu, O. L., & Lee, H. S. (2020). The effect of automated feedback on revision behavior and learning gains in formative assessment of scientific argument writing. *Computers & Education*, 143, 103668.
- Zhu, X., Wu, H., & Zhang, L. (2022). Automatic Short-Answer Grading via BERT-Based Deep Neural Networks. *IEEE Transactions on Learning Technologies*, 15(3), 364-375.

Appendix

Screenshots of the Survey in OATutor

OATutor (v1.5.1) Demographic Questions

Please answer the following demographic questions.

What gender do you identify as?

☐ Prefer not to say

☐ Non binary

☐ Female

☐ Male

SUBMIT

What is your age?

- ☐ 31 or older
- ☐ 25 – 30
- ☐ 22 – 25
- ☐ 18 – 21
- ☐ 17 or younger

SUBMIT**What is your ethnicity?**

- ☐ Multiple ethnicities / Other
- ☐ White / Caucasian
- ☐ Hispanic
- ☐ Black or African American
- ☐ Asian / Pacific Islander
- ☐ American Indian or Alaskan Native

In which of the following courses are you enrolled?

- ☐ CSC 110
- ☐ STA 4173
- ☐ STA 4102
- ☐ MTH 4420
- ☐ STA 2023 at UCF
- ☐ STA 2023 at SCF
- ☐ STA 1001
- ☐ Not Applicable or None of the Above

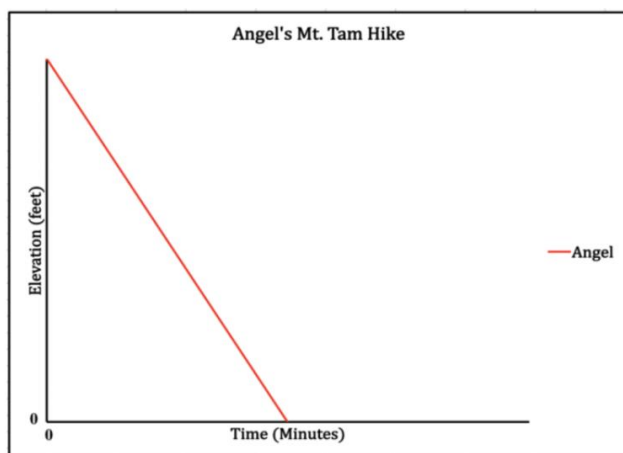
SUBMIT**NEXT PROBLEM**

Elements of an Equation

Angel is hiking down to the bottom of Mt. Tam, with an elevation of 2000 feet (ft).

She hikes at a constant rate of speed on her way down.

The graph below depicts Angel's hike (as a red line) from the top of Mt. Tam, 2000 ft., down to the bottom.



Each symbol in the linear equation $y = mx + b$ represents a “real-life” component of Angel's 2000 ft hike to the bottom of Mt. Tam.

What does the symbol y represent?

- ☐ Total distance of Angel's hike.
- ☐ Angel's elevation after a given number of minutes.
- ☐ Number of minutes.
- ☐ Starting elevation of Angel's hike.
- ☐ Angel's change in elevation in one minute.

SUBMIT

Each symbol in the linear equation $y = mx + b$ represents a “real-life” component of Angel's 2000 ft hike to the bottom of Mt. Tam.

What does the symbol m represent?

- ☐ Total distance of Angel's hike.
- ☐ Angel's elevation after a given number of minutes.
- ☐ Number of minutes.
- ☐ Starting elevation of Angel's hike.
- ☐ Angel's change in elevation in one minute.

SUBMIT

Each symbol in the linear equation $y = mx + b$ represents a “real-life” component of Angel's 2000 ft hike to the bottom of Mt. Tam.

What does the symbol x represent?

- ☐ Total distance of Angel's hike.
- ☐ Angel's elevation after a given number of minutes.
- ☐ Number of minutes.
- ☐ Starting elevation of Angel's hike.
- ☐ Angel's change in elevation in one minute.

SUBMIT

Each symbol in the linear equation $y = mx + b$ represents a “real-life” component of Angel's 2000 ft hike to the bottom of Mt. Tam.

What does the symbol b represent?

- ☐ Total distance of Angel's hike.
- ☐ Angel's elevation after a given number of minutes.
- ☐ Number of minutes.
- ☐ Starting elevation of Angel's hike.
- ☐ Angel's change in elevation in one minute.

SUBMIT

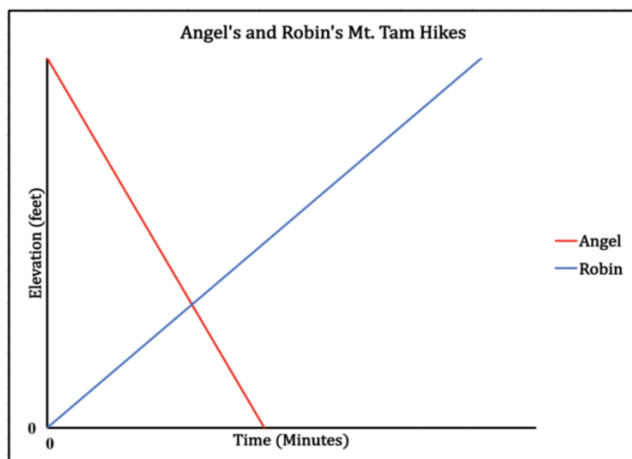
NEXT PROBLEM

OATutor (v1.5.1)

Hiking 1

Crossing Paths

Angel and Robin begin hiking at the same time, both at a constant rate of speed. Angel starts from the top of Mt. Tam, and Robin starts from the bottom. The red line shows Angel's downhill hike while the blue line shows Robin's uphill hike.



Who made it to their destination faster?

☐ Robin

☒ Angel

SUBMIT



What feature(s) of the graph led you to make your selection above?

We are training an Artificial Intelligence bot to change your response so that it is more correct! See how the bot revised your response below, then proceed to the next question.

it was because hiking downwards is faster than going up, a steeper line

It was because hiking downwards is faster than going up.

We need your help!

The bot is still in training so it doesn't write in perfect English, and sometimes messes up.

What did the bot add to, or remove from your response, and why does this make your response better or worse?

SUBMIT

Revise your own response.

Please keep in mind the AI bot's changes, and revise your own original response below.

SUBMIT

